

Scale and conformal symmetry in four-dimensional operational quantum field theory

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Abstract

We prove that scale symmetry implies local conformal symmetry for a class of four-dimensional unitary, positive-energy quantum field theories with a Poincaré- and scale-invariant vacuum. The class has a discrete, bounded-below spectrum of scaling dimensions with finite multiplicities, and each original field has finite scaling support. The original fields and their adjoints have compatible affiliated realizations in a causally commuting bounded local net, and a physical local dilatation current generates scale transformations through its local Ward identities. The operational framework admits sharply localized reconstructed fields after their joint products and compatible affiliated realizations in the same net have been established. Under these hypotheses, the stress tensor admits a symmetric, conserved, traceless improvement with unchanged translation and Lorentz charges. The corresponding conformal currents satisfy unbroken local Ward identities on the physical field algebra. The conclusion is local; integration to a global conformal action on the whole bounded net remains a separate question.

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1 Introduction

Scale symmetry constrains correlation functions under uniform changes of length. Conformal symmetry also includes special conformal transformations and their local Ward identities. The implication from the former to the latter is therefore a statement about the local operator content of a theory, not only about the action of a global charge.

For a symmetric conserved stress tensor in four dimensions, a conserved dilatation current has the form

$$D_\mu = x^\nu T_{\mu\nu} - V_\mu, \quad T^\mu{}_\mu = \partial^\mu V_\mu. \quad (1.1)$$

The problem is to improve T to a traceless tensor without changing its affine charges or leaving the original physical field algebra. A sufficient condition is a local scalar L with $T^\mu{}_\mu = \square L$, but the general improvement problem allows tensor potentials. Accordingly, a proof must derive the scalar condition if it uses one; it cannot impose it as part of the desired conclusion.

Stress-tensor improvements that preserve the Poincaré generators were a central feature of the work of Callan, Coleman and Jackiw [12]. The issue here is to construct the required scalar within the original physical field algebra.

1.1 The result and its scope

We work in a vacuum representation of a local, unitary, positive-energy theory on $\mathbb{R}^{1,3}$. The vacuum is invariant under Poincaré transformations and continuous dilatations. The original point-field labels form an algebraic direct sum

$$\mathcal{F} = \bigoplus_{\Delta \in \Sigma} \mathcal{F}_\Delta,$$

where $\mathcal{F}_\Delta$ consists of homogeneous dilation eigenfields of dimension Δ . The real dimension spectrum Σ is discrete, with no finite accumulation, bounded below, and has finite multiplicities in the Lorentz-component convention of Assumption 2.1. Thus each original field is a finite linear combination of homogeneous fields: it has *finite scaling support*. Both T and the virial field V in (1.1) are physical local fields, with the corresponding compact-flux Ward identities.

The operational data also include a fixed covariant, causally commuting net of bounded local algebras, with compatible affiliated realizations of the original fields and their adjoints. A further finite homogeneous field family belongs to this physical theory only after its localization in arbitrarily small regions, all finite mixed products, common invariant domain and original-net affiliation have been established. The required domain contains the original vacuum-word domain $\mathcal{D}_{\text{old}}$, constructed in Proposition 2.2. Thus membership is a conclusion to be proved for the improvement scalar. Finite scaling support restricts field labels; arbitrary finite word lengths and joint Schwartz test kernels remain available.

We use *physical field algebra* in this joint sense: polynomial insertions from the original fields and any finite family admitted together with them under Assumption 2.1. Products of separately constructed families require their mutual reconstruction. Proposition 8.11 will extend the same improvement and Ward action to each such admitted family. The framework requires neither Haag duality nor a split property, a phase-space estimate, or field-operator bounds in terms of energy.

Theorem 1.1. *Suppose Assumption 2.1 holds, including the algebraic direct-sum decomposition with finite scaling support for every original field. Then the physical stress tensor admits a local symmetric conserved improvement with the same translation and Lorentz Ward charges and with zero trace. More precisely, let $T^{(4)}$ and $V^{(3)}$ be the Hermitian dimension-four and dimension-three components selected by the affine Ward identities. There is a Hermitian physical scalar field L of dimension two such that*

$$\partial^\mu V_\mu^{(3)} = \square L, \quad t_{\mu\nu} = T_{\mu\nu}^{(4)} + \frac{1}{3}(\partial_\mu \partial_\nu - \eta_{\mu\nu} \square)L \quad (1.2)$$

is symmetric, conserved and traceless. The replacement $T \mapsto T^{(4)}$ also preserves the affine charges. The currents $X^\nu t_{\mu\nu}$ for conformal Killing vector fields X have unbroken local Ward identities on the physical field algebra, extending its Poincaré and dilation actions.

The preserved physical data are the Hilbert space, bounded local net, original polynomial fields on $\mathcal{D}_{\text{old}}$, and their translation and Lorentz charges. The common domain is enlarged by the reconstructed vacuum words. This does not identify new affiliated extensions with previously chosen closed operators on every vector of their larger domains. The conclusion is a local conformal Ward action; integration to a global conformal action on the whole bounded net is a separate question.

The field-space and bounded-net hypotheses delimit the class in Theorem 1.1. In particular, weak commutation of unbounded fields alone does not supply the bounded locality used by the proof. In a gauge description, the construction takes place in the chosen gauge-invariant local algebras.

The full free scalar theory. The massless real scalar in its four-dimensional vacuum Fock representation gives a concrete example. Use its full bounded local net and the Wick monomials in the scalar and its derivatives as original fields. Their scaling grades are finite-dimensional and integer-valued. The joint-domain and affiliation verification is given in the example of Section 2.2.

With $Q = :\phi^2:$, the canonical stress tensor and virial field are

$$T_{\mu\nu}^{\text{can}} = :\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} (\partial\phi)^2:, \quad V_\mu = -\frac{1}{2} \partial_\mu Q.$$

The free equation gives $(T^{\text{can}})^\mu{}_\mu = - :(\partial\phi)^2: = \partial^\mu V_\mu = \square(-Q/2)$. The canonical commutation relations give the Poincaré and dilation Ward actions on compact Cauchy slabs. Here the scalar is already present, $L = -Q/2$, and (1.2) is the familiar improvement $t_{\mu\nu} = T_{\mu\nu}^{\text{can}} + (\eta_{\mu\nu} \square - \partial_\mu \partial_\nu)Q/6$.

The local-current hypothesis is substantive. Restricting instead to shift-invariant Wick polynomials removes Q and $V_\mu = -:\phi \partial_\mu \phi:$; the available dimension-three linear fields $\partial_\mu \partial_\nu \phi$ contain no Lorentz vector, and products of differentiated fields start at dimension four. This is the field-algebra issue in the free-scalar and dual noncompact two-form examples discussed by Dymarsky and Zhiboedov [14], Nakayama [35]. A different operational theory must also specify its bounded net and intrinsic field membership: deleting a name from the original list alone does not exclude a field reconstructible in the same net.

1.2 History and the local-field question

The two-dimensional history runs through Zamolodchikov's c -theorem and Polchinski's enhancement result under suitable hypotheses [37, 42]. In four dimensions, local renormalization-group consistency conditions give a substantial perturbative result. The flow relevant to the trace includes flavor-current contributions, and positivity in the perturbative regime forces that flow to vanish at a scale-invariant theory. Perturbative limit cycles of ordinary beta functions can therefore describe conformal theories [19, 25]. The required positivity is not established by those consistency identities

at arbitrary coupling. Nakayama reviews the distinctions among scale currents, improvements, unitarity and the hypotheses of enhancement arguments [35].

The four-dimensional a -theorem of Komargodski and Schwimmer uses anomaly matching and a positive dilaton dispersion relation for flows between conformal endpoints [26]. Luty, Polchinski and Rattazzi developed this method to establish conformal asymptotics in perturbatively controlled regimes and to constrain nonperturbative scale-invariant theories [30]. Their general argument leaves a passage from on-shell dilaton constraints to an off-shell local trace identity. Dymarsky, Komargodski, Schwimmer and Theisen sharpened the vanishing-amplitude argument, retaining infrared-regulation hypotheses and the need to justify the inference from trivial probe scattering to a removable local interaction [15].

That inference is sensitive to contact and semi-local terms. Bzowski and Skenderis showed how semi-local contributions can carry the scale anomaly even when on-shell dilaton amplitudes vanish [11]. Dymarsky, Farnsworth, Komargodski, Luty and Prilepina analyzed the generalized-free-trace possibility: without a suitable dimension-two scalar the anomaly rules it out; when such a scalar exists, either an improvement is conformal or an improved trace can be chosen that is not generalized free [16]. The latter alternative retains an improvement problem.

Earlier arguments also aim directly at a local scalar. Yonekura proposes the inverse-wave-operator construction $\square^{-1}T^\mu{}_\mu$ and states the crossing and momentum-space analyticity assumptions used to establish its locality [41]. Sachs gives a Weyl-gauging and consistency argument for a restricted class of metric-couplable Lagrangian theories with a gauge-invariant virial current [38]. Here the required limited momentum continuation is derived inside primitive causal tubes, and the improvement is reconstructed in the prescribed physical theory. No asymptotic scattering states or independently assumed crossing relation enter the argument.

Null-cut modular methods have a substantial earlier development. Faulkner, Leigh, Parrikar and Wang related first-order deformations of half-space modular Hamiltonians to horizon stress-tensor integrals [18]. Casini, Testé and Torroba studied null-plane modular Hamiltonians and their half-sided modular relations [13]. The bounded-cut argument below establishes the flux limits, domains and commuting transverse measure needed under Assumption 2.1.

The remaining issue is an operator one. Intrinsic recovery of point fields from local algebras under polynomial energy bounds was studied by Fredenhagen and Hertel [20]; those bounds are not assumed here. Dividing the trace’s vacuum polarization by invariant mass squared produces a scalar two-point wave. Even if that wave has the right dimension and localization, it is not yet a physical local operator. One must construct its mixed products at every finite length, establish their positivity and a compatible common domain, and prove affiliation in the original algebras. Section 8 is devoted to this passage.

1.3 Structure of the proof

1. *Bounded null cuts.* Wedge modular data and half-sided modular inclusions give positive generators for null cuts. A bounded-profile Ward identity identifies their linear response with the physical stress tensor. It produces a commuting positive transverse measure and its quadratic moment. The spectral comparison used here is stronger than form order; its proof uses modular strip analyticity.
2. *Removal of the critical scalar tail.* The available scalar comparison is sharpened by ruling out its sole possible obstruction, a critical tail. A causal stress/virial dispersion comparison between the two null horizons isolates that tail as an order-zero Hankel leakage coefficient. A logarithmic test leaves a positive multiple of the scalar leakage norm as the leading term,

forcing the tail to vanish.

3. *From charge forms to actual modular action.* The improved scalar comparison first gives local forms with one scalar insertion. Their quadratic charges are identified with positive self-adjoint null moments on a specified domain. Future and past cone comparisons force the residual dilation charge to vanish. A KMS argument then identifies the actual cone modular groups, and half-sided inclusions give the actual time-Möbius action on line diamonds.
4. *Extraction and identification of the scalar wave.* Actual diamond modular transformations extract a dimension-two scalar wave S from the physical virial field, with real localization and bounded approximants in the original net. The formal trace wave l need not initially equal S . Hilbert products through length four give a conformal representation on pairs. Its radial positivity and normalized current fluxes force the residual $l - S$ to vanish.
5. *Reconstruction and improvement.* Bounded approximants give analytic products at every finite length. Ambient differential identities control pure scalar products; a causal wave equation and characteristic Gaussian tests propagate temperedness and spectral support through mixed orderings with homogeneous original slots. Finite multilinearity then extends the scalar and Hilbert-valued boundaries to every fixed list of original fields, with the same joint test kernels. The resulting products give a common invariant domain, and exchange with the original bounded commutants gives affiliated realizations. Only after this reconstruction is the improvement in (1.2) an operator identity in the physical algebra.

The intermediate results separate tools that may be useful beyond the present theorem: a bounded-cut modular Ward formula, a two-horizon critical-tail test, and a reconstruction argument that distinguishes a local vacuum wave from a field of the original theory. Their hypotheses and domain restrictions are stated at the points of use.

Three group actions serve different purposes in this argument. Their relation is summarized here to distinguish the two-point comparison from the actual local operations used in the construction. The space Π is the closed sum of the scalar l , conserved-vector J , and traceless-tensor t vacuum-wave channels. The pair space $\mathcal{H}_{\leq 2}$ is spanned by the vacuum and the one- and two-insertion tube vectors for L, S, J, t .

Action	Space	Role
Auxiliary conformal action	One-field space Π	Fixed by the primary two-point kernels; supplies localization and spectral comparisons. The null moments are later shown to act this way on Π (Theorem 5.7).
Actual time-Möbius action	Original $\mathcal{H}$, for each time direction	Uses cone and diamond modular groups. It transports bounded operations in line diamonds and extracts S (Theorem 5.10).
Conformal action on pairs	$\mathcal{H}_{\leq 2}$	Constructed from short-product Ward identities and four-point positivity. It supplies the radial estimates proving $l = S$ (Proposition 7.7).

Section 2 fixes the field-space conventions and modular inputs. Sections 3 and 4 establish the null-cut and dispersion arguments. Sections 5 and 6 identify actual modular symmetries and extract the scalar vacuum wave. Section 7 proves its equality with the trace polarization. Section 8 constructs the field and completes the local Ward identities.

The following notation records the successive constructions.

Symbol	Role in the proof
l	Trace scalar vacuum wave, defined in (2.8).
S	Scalar vacuum wave extracted from the virial field, with physical real localization (Proposition 6.4) and later bounded approximants (Proposition 6.6).
ℓ	Residual wave $l - S$, written $L - S$ inside short products; Theorem 7.11 proves $\ell = 0$.
L	Initially a scalar insertion symbol, then an entry in analytic products. Its physical field realization is obtained only after Theorem 8.1, the common-domain construction, and Proposition 8.8.

2 Operational framework, vacuum waves, and null cuts

We use the metric $\eta = \text{diag}(1, -1, -1, -1)$ and $U(x) = e^{iP \cdot x}$. Scalar products are antilinear in their first argument. Our dilation convention is

$$U_D(s)P_\mu U_D(-s) = e^s P_\mu, \quad U_D(s)O(x)U_D(-s) = e^{s \cdot d} O(e^s x)$$

for a field of dimension d . The distinction between a field, its vacuum wave, and a bounded local operation will be maintained throughout.

2.1 Original physical data and the vacuum-generated domain

The hypotheses below specify the original fields, their bounded local net, and their Ward currents. We also fix the domain required when a new field is admitted to that theory. Proposition 2.2 constructs this domain from the original vacuum distributions and proves its compatibility with the given affiliated realizations.

Assumption 2.1 (Operational physical theory). The physical Hilbert space $\mathcal{H}$ is generated from a vacuum Ω by the original physical fields and their adjoints. The fields, in finite-component Lorentz multiplets, and their ordered products are operator-valued Wightman distributions on a common invariant polynomial domain, which we call the initial realization domain $\mathcal{E}$. They have positive translation spectrum, Poincaré covariance, and locality, with the usual grading when fermions occur. The vacuum is invariant under Poincaré transformations and continuous dilations; uniqueness of the invariant vector is not assumed.

Let $\mathcal{D}_{\text{fact}} \subset \mathcal{E}$ be the algebraic span of Ω and all finite original vacuum words with compactly supported factor tests. To specify the vacuum-generated domain, for every finite ordered list $\mathbf{A} = (A_1, \dots, A_n)$ of original fields and adjoints, write $\Psi_{\mathbf{A}}^{\text{old}}(h)$ for the continuous extension of its original factor-test vacuum map to joint kernels $h \in \mathcal{S}((\mathbb{R}^4)^n)$. These kernels include $C_c^\infty((\mathbb{R}^4)^n)$ and finite sums of factorized tests. Set

$$\mathcal{D}_{\text{old}} = \text{span}\{\Omega, \Psi_{\mathbf{A}}^{\text{old}}(h) : n = 1, 2, \dots, h \in \mathcal{S}((\mathbb{R}^4)^n)\}.$$

The existence of these extensions and their insertion actions is proved in Proposition 2.2; it is not an additional assumption. This is an algebraic span: it has no Hilbert or graph completion and no bound on word length or tensor rank. The domain whose containment is required below is precisely $\mathcal{D}_{\text{old}}$, not an independently larger choice of $\mathcal{E}$.

Let $\mathcal{F}$ be the complex vector space of original point-field labels, modulo equality of their operator-valued distributions on $\mathcal{D}_{\text{fact}}$; Proposition 2.2 makes this equivalently equality on $\mathcal{D}_{\text{old}}$. It is the

algebraic direct sum

$$\mathcal{F} = \bigoplus_{\Delta \in \Sigma} \mathcal{F}_\Delta, \quad A = \sum_{\Delta \in S_A} A_\Delta, \quad S_A \subset \Sigma \text{ finite},$$

where $\Sigma \subset \mathbb{R}$ is a discrete set of real dimensions, bounded below and locally finite: $\Sigma \cap [a, b]$ is finite for every finite interval $[a, b]$. Each $\mathcal{F}_\Delta$ is a finite-dimensional Lorentz-stable space of finite-component fields satisfying

$$U_D(s)A_\Delta(x)U_D(-s) = e^{s\Delta}A_\Delta(e^s x)$$

as operator-valued distributions on $\mathcal{D}_{\text{fact}}$, and hence on $\mathcal{D}_{\text{old}}$ by that proposition. Multiplicity counts independent component fields after including all Lorentz component indices and imposing linear field identities; equivalently, each grade contains finitely many finite-component Lorentz multiplets. Homogeneous components are themselves original physical fields. Adjunction preserves each real grade, with conjugate Lorentz components, and covariant distributional differentiation sends $\mathcal{F}_\Delta$ into $\mathcal{F}_{\Delta+1}$, with a missing grade interpreted as zero. Lorentz transformations preserve the grades and statistics; translations act on spacetime arguments. No completion by infinitely many scaling components is taken.

There is a fixed isotone net $O \mapsto \mathcal{M}(O)$ of weakly closed algebras of physical bounded operations, covariant under the same physical Poincaré and dilation representations. Algebras at spacelike separation commute, with the fixed statistics twist in the graded case. Each smeared original physical field has a closed affiliated realization in its localization algebra, extending its realization on $\mathcal{E}$, whose adjoint also extends the field adjoint there. The corresponding extensions on $\mathcal{D}_{\text{old}}$ are derived in Proposition 2.2. For any other open region R ,

$$\mathcal{M}(R) = \bigvee_{\overline{O} \subset R} \mathcal{M}(O),$$

where O ranges over double cones. Physical point-field membership in $\mathcal{A}_{\text{phys}}$ is intrinsic to this fixed net, not to an enumeration of symbols. An operationally reconstructed finite family of homogeneous finite-component point fields of real scaling dimensions is admitted, together with its finite linear span, once its localization supports can shrink inside arbitrarily small double cones; all joint field distributions of the family with itself and with every fixed finite list of original fields and adjoints have been established at every finite length on a compatible common invariant domain containing $\mathcal{D}_{\text{old}}$, with the required adjoints and the preceding covariance, spectrum, and locality/statistics properties; and compatible smeared realizations affiliated with the original localization algebras have been established, with their adjoints extending the field adjoints on that domain. Separately constructed fields are not combined into a common polynomial field algebra without their mutual joint distributions, common invariant domain with the required adjoints, and original-net affiliations having been established. The criterion asserts no candidate's existence or analytic properties and admits no point field merely by convergence of an infinite nonhomogeneous scaling sum.

The original physical fields include a symmetric conserved stress tensor $T_{\mu\nu}$ with the translation and orbital Lorentz Ward identities, and a local virial field V_μ such that

$$D_\mu = x^\nu T_{\mu\nu} - V_\mu, \quad \partial^\mu D_\mu = 0, \quad T^\mu{}_\mu = \partial^\mu V_\mu.$$

The Ward identities of D generate the stated dilations. Charge Ward identities are understood through compact Cauchy slabs surrounding an insertion and extend to polynomial products by the derivation rule. The identities displayed above are flat-space field identities; the usual contact terms in time-ordered distributions are allowed.

Finite scaling support concerns field labels. It leaves all finite product lengths and all joint Schwartz and compact smooth kernels available, without a finite tensor-rank requirement. It restricts neither Hilbert completions and generator spectra nor weak closures of bounded algebras. The bounded-locality and original-net affiliation clauses are separate hypotheses: neither weak commutation of field restrictions nor affiliation in a larger relatively local net supplies them. No Haag duality, split property, phase-space estimate, scattering theory, or conformal symmetry is assumed.

Proposition 2.2 (Original joint-test domain). *For every fixed finite ordered list of original fields and adjoints, the vacuum distribution has a unique tempered extension and its vacuum map has a continuous Hilbert-valued extension to all joint Schwartz kernels. The space $\mathcal{D}_{\text{old}}$ is Hilbert-dense and invariant under old field insertions and continuous joint polynomial insertions, with their prescribed adjoints. Finite field-label identities, covariance, locality, and the compact-slab Ward identities pass to this domain with their stated support conditions.*

For a compactly smeared original field $B(f)$, each retained closed affiliated realization and its Hilbert adjoint agree on $\mathcal{D}_{\text{old}}$ with these insertion actions. This assertion also holds for an originally supplied Schwartz realization. It does not require $\mathcal{D}_{\text{old}}$ to be a graph core for that realization.

Proof. Tempered distributions and finite field-label expansions. For every fixed ordered list of homogeneous original fields $A_i \in \mathcal{F}_{\Delta_i}$, its already given vacuum distribution is tempered. To see this, remove the common translation coordinate. In the remaining relative coordinates ξ , dilation invariance of the vacuum gives

$$W(\lambda\xi) = \lambda^{-\sum_i \Delta_i} W(\xi), \quad \lambda > 0,$$

in the distributional pullback sense. A compact neighborhood of the origin is controlled by the local finite order of the given distribution. On a fixed compact annulus it is bounded by finitely many test-function seminorms. Dilating that annulus transfers the bound to every outer dyadic annulus with a fixed power determined by the dimensions and the seminorm order. Summation against a sufficiently high Schwartz seminorm proves temperedness. The zero- and one-point cases are immediate.

Adjoints are homogeneous at the same real grade. For any fixed list of original fields $A_i = \sum_{\Delta \in S_i} A_{i,\Delta}$, multilinearity on $\mathcal{D}_{\text{fact}}$ first gives the component expansion on compact factor tests. Compact tensor-test density and continuity of the given joint distributions then give, for the same compact joint kernel φ ,

$$W(A_1, \dots, A_n)[\varphi] = \sum_{(\Delta_1, \dots, \Delta_n) \in S_1 \times \dots \times S_n} W(A_{1,\Delta_1}, \dots, A_{n,\Delta_n})[\varphi]. \quad (2.1)$$

The finite sum of homogeneous tempered extensions defines the unique tempered extension of the left side, so the same identity holds for every joint Schwartz kernel. It agrees with any originally specified Schwartz distribution by its stated continuity and compact-test density. Its constants and seminorm orders may depend on the fixed list; no single total scaling dimension is assigned to a nonhomogeneous sum.

Hilbert-valued words and insertion actions. For a fixed old list $\mathbf{A}$, write $\Psi_{\mathbf{A}}^0(u)$ for the original vacuum vector of a finite sum u of compact factor tests. The doubled tempered distribution gives a continuous Schwartz seminorm p and a constant C such that

$$\|\Psi_{\mathbf{A}}^0(u) - \Psi_{\mathbf{A}}^0(v)\|^2 = W(\mathbf{A}^*, \mathbf{A})[(u-v)^* \otimes (u-v)] \leq Cp(u-v)^2.$$

Here the star reverses and adjoints the field list and conjugates and reverses the kernel. Compact finite tensor tests are dense in the joint Schwartz space. The estimate therefore constructs the unique continuous Hilbert-valued extension $\Psi_{\mathbf{A}}^{\text{old}}$ used in the definition of $\mathcal{D}_{\text{old}}$; polarization passes all its Gram identities to the limit. Any originally specified Schwartz vacuum-word map agrees with this extension by its stated continuity and the same density. Consequently the original vacuum-generation assumption also makes $\mathcal{D}_{\text{fact}}$ Hilbert-dense. This argument concerns only original products whose existence is assumed; it supplies no product containing a newly constructed field.

For a fixed old word $\mathbf{A}$ and a fixed finite old prefix $\mathbf{B}$, the same doubled vacuum-distribution estimate applies to the longer word. Consequently, if $h_j \rightarrow h$ in the joint Schwartz topology, then for a fixed prefix kernel g ,

$$\Psi_{\mathbf{A}}^{\text{old}}(h_j) \longrightarrow \Psi_{\mathbf{A}}^{\text{old}}(h), \quad \Psi_{\mathbf{B},\mathbf{A}}^{\text{old}}(g \otimes h_j) \longrightarrow \Psi_{\mathbf{B},\mathbf{A}}^{\text{old}}(g \otimes h)$$

in Hilbert norm. The estimates are continuous jointly in g, h on their Schwartz spaces. Reversed adjoint kernels obey the same bounds. Compact joint kernels can be approximated, after a finite partition into product patches, by finite sums of factor tests with the prescribed local supports; Schwartz kernels admit finite tensor approximations in the Schwartz topology. Concatenation, reversal, and differentiation are continuous in these test spaces.

Left concatenation of old test words is well-defined on their vacuum vectors. Indeed, if a finite old test polynomial P represents zero, positivity and Cauchy–Schwarz applied to P and Q^*QP give $W(P^*Q^*QP) = 0$ for every finite old test polynomial Q . This is the squared norm of the proposed vector for QP . Thus the null vectors form a left ideal, including cancellations between different words, and the same vacuum pairing places adjoint insertion inside the Hilbert adjoint. Concatenation therefore makes $\mathcal{D}_{\text{old}}$ invariant under the old field and polynomial insertions. It is Hilbert-dense because it contains the original factorized vacuum-polynomial vectors that generate $\mathcal{H}$.

Agreement with the retained realizations. Let $T_B(f)$ be a chosen closed affiliated realization of an original field with compact test f , and choose finite sums of compact tensor tests $h_j \rightarrow h$ as above. On the vectors in $\mathcal{D}_{\text{fact}} \subset \mathcal{E}$ the original extension clause gives

$$T_B(f)\Psi_{\mathbf{A}}^{\text{old}}(h_j) = \Psi_{B,\mathbf{A}}^{\text{old}}(f \otimes h_j).$$

Both sides converge in Hilbert norm by the two displayed continuity statements. Closedness of $T_B(f)$ therefore proves the same equality for h . Applying the argument to the closed Hilbert adjoint $T_B(f)^*$ gives its equality with the insertion of $B^\dagger(f)$ there. Iterating this one-field statement proves the corresponding actions of every finite factorized prefix and its reversed adjoint on $\mathcal{D}_{\text{old}}$. Joint polynomial insertions are the continuous concatenations already defined; this argument does not assign an affiliated closure to an arbitrary joint smear.

If an original Schwartz smearing $B(f)$ and its closed affiliated realization were already supplied, the same conclusion holds for that realization. To identify its action on a fixed vector of $\mathcal{D}_{\text{fact}}$, approximate f by compact tests in Schwartz topology. The longer doubled correlations give Hilbert convergence of the inserted vectors, while the original Schwartz distributional continuity gives the same limiting matrix elements against $\mathcal{D}_{\text{fact}}$. Density identifies the vectors, also for the adjoint. The preceding h_j closed-graph argument then applies unchanged. This uses only an originally supplied Schwartz action and its stated continuity; it asserts no pre-existing realization for a smearing that was not originally supplied.

Identities and local Ward actions. The same paired vector and inserted-vector limits extend every finite field-label identity from $\mathcal{D}_{\text{fact}}$ to $\mathcal{D}_{\text{old}}$. The converse follows from $\mathcal{D}_{\text{fact}} \subset \mathcal{D}_{\text{old}}$, so the two equality conventions for $\mathcal{F}$ agree. The homogeneous dilation law and the adjoint and derivative

grade laws extend in the same way, because the transformed, reversed, and differentiated kernels remain in the stated test spaces. Thus none of these identities on joint vectors was used to construct those vectors.

The original covariance and flat field identities pass from the factorized vectors to joint vectors by the same longer-word continuity, with transformed or differentiated kernels. Locality and compact-slab Ward identities pass under their stated support conditions. For a general Schwartz spectator vector, keep the localized slab and insertion tests fixed and approximate only the spectator kernel in Schwartz topology. If a localized insertion polynomial itself has a compact joint kernel, use the support-preserving compact approximants. Thus the operational identities used below hold on the dense invariant vacuum-generated domain $\mathcal{D}_{\text{old}}$, with the original vacuum correlations, Hilbert space, and bounded net unchanged. The factor approximations just used converge in the graph of each specified $T_B(f)$ on these joint vectors; Hilbert density alone was not used for that conclusion. No claim is made that $\mathcal{D}_{\text{old}}$ is a graph core for a chosen realization, that the closure of its field restriction equals that realization, or that a newly constructed field acts on arbitrary additional vectors of $\mathcal{E}$. $\square$

We will use (2.1) as a finite-label permanence principle for later distributions. Every term uses the same joint kernel; continuity, common spectral cones, and linear distributional identities pass through the finite sum. Positivity and descent through null polynomial vectors require their full summed forms and will be checked where they arise.

The dimension set is countable, and each grade has a finite component basis. Countable dense test families for the resulting countable collection of finite words, together with the Hilbert-valued continuity just proved, show that $\mathcal{H}$ is separable. This supplies the separability used in the direct-integral descriptions below.

The spectrum condition now gives tube analyticity, uniqueness from an open boundary patch, and Reeh–Schlieder for these original distributions. In particular, a separated-point field identity valid on a mutually spacelike open patch extends to ordered vacuum distributions by tube uniqueness. This assertion does not assign differentiated Ward identities to a particular choice of time ordering.

2.2 Example: the full free scalar theory

The massless real scalar in its four-dimensional vacuum Fock representation gives a concrete example. Take the full scalar net $\mathcal{M}(O) = \{e^{i\phi(f)} : f \in C_c^\infty(O, \mathbb{R})\}''$ and, as original fields, finite linear combinations of Wick monomials $:\prod_{j=1}^n \partial^{\alpha_j} \phi:$. Their dimensions are the nonnegative integers $n + \sum_j |\alpha_j|$, with finitely many component monomials at each grade. The Weyl relations give bounded locality, and Wick’s construction supplies all joint products and their invariant domain [10, Theorem 5.7 and its following remark]. Affiliation has a concrete justification here. On the common domain invariant under Wick fields and Weyl operators [9, Section 2.2, equation (11)], a Wick smearing commutes with exterior Weyl operators. Its closed graph retains this commutation and domain invariance. Double-cone duality of the free massless scalar [23, Section 3, Corollary 2] identifies the generated exterior algebra with $\mathcal{M}(O)'$, proving affiliation. The formal adjoint gives the required Hilbert-adjoint extension. This is a property of the example, not a duality hypothesis of the theorem.

2.3 Selecting the current weights

We may, and henceforth do, replace T and V by Hermitian homogeneous fields of dimensions four and three. We justify both the selection and preservation of the charges. First take the even

components of the spacetime currents. The prescribed unitary affine derivations preserve parity, so the odd-current Ward contributions vanish by parity separation on parity-homogeneous insertions; the flat field identities split by parity as well. For an even current J with a real slab test, polynomial adjunction on $\mathcal{D}_{\text{old}}$ gives

$$(i[J, A])^\dagger = i[J^\dagger, A^\dagger].$$

The prescribed unitary derivation preserves adjoints, and its Ward identity also applies to $A^\dagger$ with the same support. Hence $J^\dagger$ and $(J + J^\dagger)/2$ have the same Ward action as J . Applying this to the translation and orbital currents of T and to $D = xT - V$ shows that the Hermitian parts preserve all required charges. The real metric, coordinate, and derivative coefficients also preserve the flat field identities under adjunction. Relabel these even Hermitian representatives as T, V . The adjoint grade law preserves their decompositions. Write the finite sums

$$T = \sum_{d \in S_T} T^{(d)}, \quad V = \sum_{e \in S_V} V^{(e)},$$

with every missing component interpreted as zero. Directness of the grading, Lorentz covariance, and the derivative grade law split symmetry, Hermiticity, conservation, and the trace identity. Thus, for every $d \in S_T \cup \{e + 1 : e \in S_V\}$,

$$T_{\mu\nu}^{(d)} = T_{\nu\mu}^{(d)}, \quad \partial^\mu T_{\mu\nu}^{(d)} = 0, \quad (T^{(d)})^\mu{}_\mu = \partial^\mu V_\mu^{(d-1)},$$

and the components can be taken Hermitian. In particular each geometric current

$$D_\mu^{(d)} = x^\nu T_{\mu\nu}^{(d)} - V_\mu^{(d-1)}$$

is conserved.

Fix a homogeneous insertion O of dimension α and homogeneous spectators. A compact-slab translation commutator made from $T^{(d)}$ has weight $\alpha + d - 3$: the slab integral removes three powers of length. An orbital Lorentz commutator from $xT^{(d)}$ has weight $\alpha + d - 4$, as does a dilation commutator from the pair $D^{(d)}$. Conservation of these component currents permits the slabs to be rescaled while keeping the insertion inside. After testing the homogeneous spectators, these are identities between ordinary scalar distributions. The prescribed translation, Lorentz, and dilation actions have weights $\alpha + 1$, α , and α , respectively. After testing the remaining variables, the Ward identities are finite scalar sums of real exponentials in the logarithmic scale. Their linear independence selects the $d = 4$ translation and orbital contributions and the $d = 4$ dilation pair $xT^{(4)} - V^{(3)}$; the other corresponding Ward contributions vanish. This is a statement about charges, not the vanishing of the nonselected fields.

Finite multilinearity now extends these identities to every fixed list of original fields and adjoints under the same test kernels, and the retained derivation rule extends them to every finite polynomial word. The selected components retain conservation, the trace identity, and all required Ward actions. Replacing the original stress tensor by $T^{(4)}$ adds a local symmetric conserved tensor with zero translation and Lorentz charges. This is an admissible charge-preserving change of stress tensor; it does not presume a scalar improvement. The field decompositions here are identities on $\mathcal{D}_{\text{old}}$; no equality between a chosen closed realization of a sum and a sum of chosen closed realizations is needed. Each selected component has its own affiliated realization by Assumption 2.1.

2.4 Modular input and a spectral comparison lemma

For a standard real subspace $X \subset \mathcal{H}$, write

$$\mathfrak{s}_X(x + iy) = x - iy \quad (x, y \in X), \quad \mathfrak{s}_X = J_X \Delta_X^{1/2}, \quad S_X = -\frac{1}{2\pi} \log \Delta_X.$$

Here standard means $\overline{X + iX} = \mathcal{H}$ and $X \cap iX = \{0\}$. The symplectic complement is $X' = \{\xi : \text{Im} \langle \xi, x \rangle = 0 \text{ for } x \in X\}$. For a standard algebra $(\mathcal{M}, \Omega)$ its associated real space is $X_{\mathcal{M}} = \{A\Omega : A = A^* \in \mathcal{M}\}$. Statistics twists in complementary spaces and algebras are understood in all statements involving odd fields.

Every original polynomial word in this subsection has the finite component expansion described above, with its joint test kernel unchanged. Thus the preceding temperedness and finite-component covariance apply to every word used in the polynomial argument.

The Wightman Bisognano–Wichmann theorem identifies the wedge polynomial Tomita graph with $J_W e^{-\pi B}$, where B generates the wedge boost; hence $\Delta_W^{it} = e^{-2\pi it B}$ and the opposite wedge has the complementary real space [3]. The published theorem assumes a unique translation-invariant vacuum. We explain its boundary calculation and core argument in the version needed here, followed by the passage to bounded algebras. First, uniqueness of the vacuum is unnecessary for this graph identification. The reflection itself must be constructed before the core argument. Extend the covariance of the vacuum distributions along the connected complex Lorentz group. On a real Jost patch, locality reverses the order of the fields with the prescribed statistics signs; analytic uniqueness extends the resulting distributional equality. Combine it with the Hermitian-conjugation identity for vacuum distributions. If F and G are smeared polynomial words and $\xi_F = F\Omega$, it says

$$\langle \xi_F^{\natural}, \xi_G^{\natural} \rangle = \langle \xi_G, \xi_F \rangle.$$

Here $\xi_F^{\natural}$ is the reflected-adjoint word: conjugate its test functions, reflect their arguments by $j_W(u, v, y) = (-u, -v, y)$, and replace the components by their adjoints with the finite-component matrices of the imaginary wedge boost at rapidity $i\pi$ and the fixed statistics twist. Apply this operation to each factor in the *original word order*; the order reversal above occurs in the vacuum distributions used to compute scalar products. This is precisely the finite-component reflection boundary calculation in the polynomial BW proof. It uses covariance, the spectrum condition, locality, and Hermitian conjugation, with no cluster limit or condition on the dimension of the invariant subspace.

The displayed identity shows that the conjugate-linear word map preserves norms and descends through all zero-norm polynomial relations. Reflected test functions again range over the full test space, so its range is dense. It therefore extends to an antiunitary J_W ; reflection twice, with the normal-statistics phases, gives $J_W^2 = 1$. Now use polynomial vacuum vectors whose first point and successive gaps lie in an open wedge-ordered patch. Imaginary boost through rapidity π takes their ordered distributions through the positive-spectrum tubes, and locality at the spacelike boundary reverses the fields. Tube uniqueness makes their complex span Hilbert-dense. Adjoin the adjoint vectors and all real boost translates. The resulting span is boost-invariant and its adjunction agrees with $J_W e^{-\pi B}$. To check that it is a graph core, let $\mathcal{G}$ be its closed graph and smooth by $e^{-\epsilon B^2}$. This smoothing preserves $\mathcal{G}$, and both $e^{-\epsilon B^2}$ and $e^{-\pi B} e^{-\epsilon B^2}$ are bounded. Hilbert approximation by the indicated span consequently approximates every smoothed vector in graph norm. Finally $e^{-\epsilon B^2} \xi \rightarrow \xi$ in the graph norm of $e^{-\pi B}$ for ξ in its domain. This proves graph totality without a clustering argument.

Second, the polynomial and bounded-algebra graphs coincide under Assumption 2.1. Start with finite sums of factorized smeared words. If A is a closed affiliated realization of such a polynomial with $\Omega \in \text{Dom } A \cap \text{Dom } A^*$, its bounded polar cutoffs $A_m = A \mathbf{1}_{[0, m]}(|A|)$ satisfy

$$A_m \Omega \longrightarrow A \Omega, \quad A_m^* \Omega \longrightarrow A^* \Omega.$$

For the second convergence use $A_m^* \Omega = \mathbf{1}_{[0, m]}(|A|) A^* \Omega$. Use natural-domain products and sums of the chosen closed affiliated single-field realizations. Their domains contain $\mathcal{D}_{\text{old}}$, and their adjoints

contain the corresponding adjoint polynomials there, so they are closable. Their natural domains and graphs are invariant under commutant unitaries; the closures remain affiliated. These are the extensions to which the same polar-cutoff argument applies. For an admitted compact joint kernel, first use a finite partition into product patches inside the localization region, then approximate each part by finite sums of tensor tests in its smooth test-function topology. The reversed adjoint kernels of these approximants converge to the reversed adjoint kernel in the same topology. The old Hilbert-valued joint-distribution bounds proved above pass both the vacuum and adjoint-vacuum vectors to the limit. Closedness of the bounded-algebra Tomita graph therefore gives the same graph inclusion for the joint word. This vector-level limit does not assert a closed affiliated realization for an arbitrary joint smear. Let $X_{\text{pol}}(R)$ be the closed real span of vacuum vectors of finite original polynomial expressions that are Hermitian on $\mathcal{D}_{\text{old}}$ and compactly localized in R , with admitted compact joint kernels included by the graph limits just proved. This is an auxiliary vacuum-word graph space; it neither restricts the original test-function spaces nor redefines $\mathcal{D}_{\text{old}}$. Consequently,

$$X_{\text{pol}}(W) \subset X_{\mathcal{M}(W)} \subset X'_{\mathcal{M}(W')} \subset X_{\text{pol}}(W')' = X_{\text{pol}}(W). \quad (2.2)$$

This proves equality and the bounded wedge modular formula. The inclusion in the middle is bounded locality, not a claim about commuting unbounded restrictions. To pass from graph equality to algebra wedge duality, use the modular-invariant-subalgebra theorem: a subalgebra invariant under the modular group of a faithful normal state admits a state-preserving conditional expectation [39]; see also [1, Theorems 3.5 and 5.2]. If the subalgebra is $\mathcal{N}$ and E is this expectation, its GNS implementation is the vector identity

$$E(x)\Omega = P_{\mathcal{N}\Omega}x\Omega.$$

Apply it to the local inclusion of the opposite, twist-corrected wedge algebra in $\mathcal{M}(W)'$. The larger algebra's modular group is the geometric opposite boost and preserves the smaller algebra. The latter has cyclic vacuum, so the implementing projection is the identity. The expectation is therefore the identity by separation, proving equality of these algebras. Double-cone duality is not used.

We use the following exact forms of the modular translation theorems [2, 5]; for the positive-translation statement we use the explicit formulation and proof in [6, Theorem II.5.2]. If e^{icb} fixes Ω , $b \geq 0$, and $\text{Ad}(e^{icb})\mathcal{M} \subset \mathcal{M}$ for $c \geq 0$, where Ω is cyclic and separating for $\mathcal{M}$, then

$$e^{isS_{\mathcal{M}}}be^{-isS_{\mathcal{M}}} = e^sb, \quad J_{\mathcal{M}}bJ_{\mathcal{M}} = b. \quad (2.3)$$

If $\mathcal{N} \subset \mathcal{M}$ have the same cyclic separating vector and $\text{Ad}(e^{isS_{\mathcal{M}}})\mathcal{N} \subset \mathcal{N}$ for $s \geq 0$, the half-sided modular inclusion theorem gives

$$\begin{aligned} H = \overline{S_{\mathcal{M}} - S_{\mathcal{N}}} \geq 0, \quad H\Omega = 0, \quad e^{isS_{\mathcal{M}}}He^{-isS_{\mathcal{M}}} &= e^sH, \\ J_{\mathcal{M}}HJ_{\mathcal{M}} = H, \quad \mathcal{N} = \text{Ad}(e^{iH})\mathcal{M}, \quad e^{isS_{\mathcal{N}}} &= e^{i(1-e^s)H}e^{isS_{\mathcal{M}}}. \end{aligned} \quad (2.4)$$

The closure in the first line is the self-adjoint closure on the common logarithm domain; subsequent sums of affine-group generators will have this meaning. The applicable inclusion result is [2, Theorem 2.1], specialized to a vector state. It requires no cyclicity for $\mathcal{N}' \cap \mathcal{M}$. Since $J_{\mathcal{M}}$ is antiunitary, its unitary-group formula is $J_{\mathcal{M}}e^{itH}J_{\mathcal{M}} = e^{-itH}$, consistently with the generator formula above. These results also hold for standard real subspaces; one may apply the algebra result to their CCR Weyl algebras and restrict the vacuum modular objects to the one-particle space.

For later form comparisons, $X \subset X_1$ implies $\mathfrak{s}_X \subset \mathfrak{s}_{X_1}$. Thus the closed form of Δ_X restricts that of Δ_{X_1} and $\Delta_X \geq \Delta_{X_1}$ in form order. The resolvent integral for the operator-monotone logarithm

gives $S_X \leq S_{X_1}$ on common operator domains. This statement by itself is weaker than the next lemma. Koot's spectral characterization of inclusions of standard pairs uses the modular-strip and half-plane mechanism [28, Theorem 3.1, Lemma 3.4 and Theorem 3.5]. We prove the contraction version between two possibly different standard spaces that is needed for the embedded comparison channels.

Lemma 2.3 (Strip comparison). *Let X_i be standard real spaces with normalized modular generators B_i , and let $K_i \geq 0$ be self-adjoint with $e^{itB_i} K_i e^{-itB_i} = e^t K_i$. Suppose $j : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ is a contraction intertwining the boost groups and*

$$j e^{iK_1} X_1 \subset e^{iK_2} X_2.$$

Then, for every $R \geq 0$,

$$\mathbf{1}_{(R, \infty)}(K_2) j \mathbf{1}_{[0, R]}(K_1) = 0. \quad (2.5)$$

In particular, for $s > 0$ and $\xi \in \text{Dom } K_1^s$, $j\xi \in \text{Dom } K_2^s$ and $\|K_2^s j\xi\| \leq \|K_1^s \xi\|$.

Proof. We first prove that, for every $r > 0$,

$$e^{rK_2} j e^{-rK_1} \text{ is everywhere defined and contractive.} \quad (2.6)$$

Its exponential growth on a high target band will then exclude that band from the image of a low source band.

Set $A = e^{-iK_2} j e^{iK_1}$. It maps X_1 into X_2 , so $\mathfrak{s}_2 A \xi = A \mathfrak{s}_1 \xi$ on $\text{Dom } \mathfrak{s}_1 = \text{Dom } \Delta_1^{1/2}$. Equivalently,

$$\Delta_2^{1/2} A \xi = J_2 A J_1 \Delta_1^{1/2} \xi \quad (\xi \in \text{Dom } \Delta_1^{1/2}).$$

Thus $\Delta_2^{1/2} A \Delta_1^{-1/2}$, initially on $\text{Dom } \Delta_1^{-1/2}$, has the contractive bounded extension $J_2 A J_1$. The operators

$$F(z) = \Delta_2^{iz} A \Delta_1^{-iz}, \quad -\frac{1}{2} < \text{Im } z < 0,$$

have contractive boundary values: the real boundary multiplies A on either side by a unitary, and the other boundary is $\Delta_2^{it} J_2 A J_1 \Delta_1^{-it}$. The analytic-extension theorem [2, Theorem 2.3], followed by three lines, gives the contractive analytic continuation to the strip. Equivalently, sandwich between bounded spectral intervals of the modular logarithms, apply three lines to their scalar matrix coefficients, and remove the intervals. The displayed Tomita relation supplies the full source domain needed at the second boundary. Write $\tilde{F}(\zeta) = F(z)$ for $\zeta = e^{-2\pi z}$, which maps the strip bijectively onto the upper half-plane. Boost covariance and the intertwining property of j identify its positive real boundary value as $e^{-i\zeta K_2} j e^{i\zeta K_1}$. To justify the interior expression despite its potentially unbounded first factor, take ξ and η in bounded spectral bands of K_1 and K_2 , respectively. The scalar function

$$\left\langle e^{i\bar{\zeta} K_2} \eta, j e^{i\zeta K_1} \xi \right\rangle$$

is entire and agrees with $\left\langle \eta, \tilde{F}(\zeta) \xi \right\rangle$ on a positive real interval. Boundary uniqueness, or reflection of their zero difference across that interval, identifies them throughout the upper half-plane. At $\zeta = ir$, $r > 0$, contractivity therefore gives

$$\left| \left\langle e^{rK_2} \eta, j e^{-rK_1} \xi \right\rangle \right| \leq \|\eta\| \|\xi\|.$$

Bounded spectral vectors are a core for e^{rK_2} . The adjoint-domain criterion implies $j e^{-rK_1} \xi \in \text{Dom } e^{rK_2}$ and identifies its image with $\tilde{F}(ir) \xi$. Finally approximate any source vector by its K_1 spectral truncations. Boundedness of e^{-rK_1} and $\tilde{F}(ir)$, followed by closedness of e^{rK_2} , proves that

$e^{rK_2}je^{-rK_1}$ is everywhere defined and contractive. If $\xi = \mathbf{1}_{[0,R]}(K_1)\xi$, apply this extension to $e^{rK_1}\xi$ to obtain

$$\|e^{rK_2}j\xi\| \leq e^{rR} \|\xi\|.$$

Projection onto $K_2 \geq R + \epsilon$ and $r \rightarrow \infty$ shows that this projection annihilates $j\xi$. Taking $\epsilon \downarrow 0$ proves (2.5). The resulting tail inequality is

$$\|\mathbf{1}_{(a,\infty)}(K_2)j\xi\| \leq \|\mathbf{1}_{(a,\infty)}(K_1)\xi\|.$$

Integrate its square against $2sa^{2s-1}da$. Truncation followed by monotone convergence proves both the domain assertion and the norm bound. $\square$

2.5 Timelike vacuum channels

Fix future null vectors $n, \bar{n}$ with $n \cdot \bar{n} = 1$ and orthonormal transverse axes e_i , $i = 1, 2$, with $e_i \cdot e_j = -\delta_{ij}$. Write

$$\begin{aligned} x &= vn + u\bar{n} + y, & p &= an + b\bar{n} + k, & p \cdot x &= au + bv - \mathbf{k} \cdot \mathbf{y}, \\ & & 2ab &= M^2 + |\mathbf{k}|^2. \end{aligned} \tag{2.7}$$

The variables $a, b, \mathbf{k}$ also denote their commuting momentum operators. Coordinate subscripts on covariant tensors mean contraction with the coordinate basis vectors. The boost B acts on spacetime by $v\partial_v - u\partial_u$, and null rotations G_i by $u\partial_i + y_i\partial_v$.

Disintegrate the positive two-point measure of a dimension- d multiplet on the open timelike cone. Lorentz transformations and dilations act transitively there, so relative to d^4p its vacuum amplitude at rest has norm proportional to M^{d-2} . The proportionality is a positive finite-dimensional rotation-invariant matrix; transport to other momenta uses the finite Lorentz matrices and unitary fiber transport. There is no independent positive singular measure on an interior mass shell, since dilation moves every positive mass shell and fixes the homogeneous measure class on their union.

A possible contribution on the nonzero light cone is normalized instead by $d^4p \theta(p^0) \delta(p^2)$. Its amplitude on a fixed null ray would grow as energy to the power $d - 1$. A boost along that ray has only the boost weights of the finite Lorentz multiplet; positivity therefore excludes the contribution if $d - 1$ exceeds their maximum. The maxima are one for a vector and two for a symmetric rank-two tensor, whereas $d - 1$ is respectively two and three for V and T . A zero-momentum atom for a positive-dimensional field is also impossible: its norm is invariant under unitary dilation but would be multiplied by e^{sd} . Thus both $V\Omega$ and $T\Omega$ have purely timelike spectrum.

Let $\Theta = T^\mu{}_\mu$. At this stage define only vacuum waves,

$$l = -\Theta/M^2, \quad J_\mu = V_\mu - ip_\mu l, \quad t_{\mu\nu} = T_{\mu\nu} - \frac{1}{3}(-\eta_{\mu\nu}M^2 + p_\mu p_\nu)l. \tag{2.8}$$

The local Ward identities imply $p^\mu J_\mu = 0$, $p^\mu t_{\mu\nu} = 0$, and $\eta^{\mu\nu} t_{\mu\nu} = 0$. At rest their rotation types are respectively spin zero, one, and two, so they are orthogonal. Their rest norms are constant, proportional to M , and proportional to M^2 . In particular l is a well-defined dimension-two wave; division by M^2 has introduced no infrared singularity in its two-point measure.

These channels have the two-point data of a scalar primary of dimension two, a conserved vector primary of dimension three, and a conserved symmetric traceless primary of dimension four. This follows directly from the uniqueness of a positive rotation-invariant metric on each of the three rest representations and homogeneous Lorentz transport. Equivalently their position-space kernels are the positive-energy boundary values proportional to

$$\frac{1}{(x^2)^2}, \quad \frac{I_{\mu\nu}(x)}{(x^2)^3}, \quad \frac{I_{\mu\rho}I_{\nu\sigma} + I_{\mu\sigma}I_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma}/2}{2(x^2)^4}, \quad I_{\mu\nu} = \eta_{\mu\nu} - \frac{2x_\mu x_\nu}{x^2}.$$

Their overall constants include the prescribed spectral boundary normalization. Let $\Pi \subset \mathcal{H}$ be the Hilbert sum of these wave channels, omitting zero-norm channels. Kernel covariance defines an auxiliary unitary conformal action on Π : transform tube point vectors with their indicated tensor matrices and weights, observe preservation of their scalar products, and extend from their dense span. This is a comparison representation, not yet a symmetry of the physical theory. Its real localization spaces can equivalently be constructed as those of the corresponding positive generalized free fields.

The following rest axes are adapted to the fixed null direction:

$$e_z^{(n)} = \frac{M}{b}n - \frac{p}{M}, \quad e_i^{(n)} = e_i + \frac{k_i}{b}n. \quad (2.9)$$

They are orthogonal to p , have squared norm -1 , and are mutually orthogonal. Define the constant spin labels j_A and s_{AB} by the components of $J/(iM)$ and t/M^2 in this tetrad, and put $s_z = s_{zz}$. Then

$$V_v = ibc_V, \quad T_{vv} = b^2c_T, \quad c_V = l + j_z, \quad c_T = l/3 + s_z. \quad (2.10)$$

The phases refer to real derivatives of Hermitian fields. Expansion of (2.9) shows that every stress component is a linear combination of constant labels with polynomial coefficients in a, b, k, M , allowing only powers of b in the denominators. Its norm is bounded by a constant times $w_\alpha w_\beta$, where

$$w_u = a, \quad w_i = \sqrt{ab}, \quad w_v = b. \quad (2.11)$$

For example $M, |k| \leq \sqrt{2ab}$, and the apparent $1/b$ factors in the rest axes acquire a compensating M or k from the rest amplitude.

2.6 Scalar comparison copies

For each label in the following table, some tangential derivative of $M^\nu d_*$ is the vacuum wave of a physical differential field:

d_*	ν
c_V, c_T	0
l, j_z, s_z	2
j_i, s_{zi}	1
$(s_{ij})_{\perp \text{TF}}$	2

(2.12)

Here the derivative has multiplier b^m for some integer $m \geq 0$, up to a fixed phase. To see the physical content of this claim, set $q_i = be_i + k_i n$, a vector whose contractions are differential combinations with polynomial coefficients. The identities

$$M^2 l = -\Theta, \quad bM^2 j_z = M^2 V_v / i + b\Theta, \quad b^2 M^2 s_z = M^2 T_{vv} + b^2 \Theta / 3$$

prove the scalar and longitudinal entries. Since $p \cdot q_i = 0$,

$$V(q_i) = ibMj_i, \quad T(n, q_i) = b^2 Ms_{zi}, \quad (T(q_i, q_j))_{\perp \text{TF}} = b^2 M^2 (s_{ij})_{\perp \text{TF}}.$$

The scalar improvement in (2.8) drops from the first mixed tensor contraction and from the transverse traceless part. Finally c_V, c_T come directly from (2.10). All factors M^2, b, k_i in these identities are polynomial momentum multipliers; no inverse differential operator has been declared local. The scalar reality phase may be chosen as $-i$ on the j_i, s_{zi} copies and as 1 on the other entries. This accounts for the factor i in a real tangential derivative.

Suppose a region R is invariant under positive v translations, as are the cuts used below. Derivatives in v can be removed in the real closure of its wave space. Choose real $\rho \in C_c^\infty(0, 1)$ with integral one, and set $\rho_R(v) = R^{-1}\rho(v/R)$. For a test function f supported in the region, the wave of $(1 - \rho_{R^*v})^m f$ has multiplier $(1 - \widehat{\rho}(Rb))^m$ and converges to that of f by dominated convergence, since $b > 0$ almost everywhere. Each such test has its first m v -moments zero. Its m -fold primitive is therefore compactly supported; choosing the primitive to vanish at $v = -\infty$ keeps its support in the positive- v hull of the original support and its positive translates. Thus it is the m th derivative of a real test in the same region. With the fixed derivative phase, the scalar generalized free wave space of dimension $2 + \nu$ on each label embeds in $X(R)$.

For $W = \{u < 0, v > 0\}$ this copy has precisely its own scalar wedge Tomita data. Indeed, its embedding intertwines boosts and the wedge reflection, including the derivative phase; hence its projection reduces $J_W e^{-\pi B}$, and the fixed real space is the scalar one. This does not say that its curved-cut real space exhausts physical localization.

In light-front variables the measure is, up to one common fixed normalization,

$$d^4p = \frac{db}{b} d^2k M dM, \quad b > 0, M > 0. \quad (2.13)$$

For a scalar copy of dimension $2 + \nu$, put $Y = i\partial_k$ and define

$$H_\nu = b(|Y|^2 + h_\nu), \quad h_\nu = -\partial_M^2 - M^{-1}\partial_M + \nu^2/M^2. \quad (2.14)$$

The operator h_ν is the regular nonnegative Hankel realization on $L^2(\mathbb{R}_+, M dM)$: the Hankel transform with kernel $J_\nu(Mr)$ conjugates it to multiplication by r^2 . In particular, the logarithmic solution at zero is excluded when $\nu = 0$.

The associated null conformal map is

$$(u, y, v) \mapsto (\gamma u, \gamma y, v + c\gamma|y|^2), \quad \gamma = (1 - 2cu)^{-1}, \quad c > 0. \quad (2.15)$$

It maps W diffeomorphically to

$$-\frac{1}{2c} < u < 0, \quad v > \frac{|y|^2}{2u + 1/c}.$$

Two-point conformal covariance therefore maps the scalar wedge real space onto the scalar real space of this region. Its unitary implementer is e^{icH_ν} . For verification, apply (2.14) to $M^\nu e^{ip \cdot x}$. The direct expression is

$$H_\nu(M^\nu e^{ip \cdot x}) = \left(2u^2a - 2uk \cdot y + b|y|^2 - 2iu(\nu + 2)\right) M^\nu e^{ip \cdot x}.$$

For complex tube coordinates, the squares in this formula are analytic bilinear squares. Multiplication by i gives the infinitesimal coordinate action

$$2u^2\partial_u + 2uy_i\partial_i + |y|^2\partial_v + 2u(2 + \nu),$$

which integrates to (2.15) with the scalar weight. This calculation is valid on tube point vectors: their imaginary future displacement gives exponential damping in b and Gaussian damping in k, M at fixed b ; at $M = 0$ their radial behavior is M^ν times an even smooth function, the regular Hankel boundary condition. Applying any power of $b(|Y|^2 + h_\nu)$ retains an integrable polynomial times this damping. The factors of b cancel the inverse Gaussian variances as $b \downarrow 0$. Consequently differentiation occurs in the operator domain. Uniqueness of the unitary evolution identifies this realization with the kernel-covariance implementer.

2.7 The physical null-cut generators

For a real transverse profile f , define

$$W_f = \text{int} \left\{ u < 0 : v + \frac{|y - z|^2}{-2u} \geq f(z) \text{ for every } z \in \mathbb{R}^2 \right\}. \quad (2.16)$$

Equivalently, put

$$\mathcal{L}_d f(y) = \sup_z \left(f(z) - \frac{|y - z|^2}{2d} \right); \quad W_f = \{u < 0, v > \mathcal{L}_{-u} f(y)\}^\circ.$$

The allowed profiles initially are nonnegative bounded smooth functions and nonnegative circular parabolas of positive curvature. We subsequently allow constants of either sign, and sums of a bounded smooth function with a positive-curvature circular parabola. All bounded smooth profiles used in estimates have the stated derivatives bounded. Write $p_d(y) = |y - d|^2$ and $W = W_0$.

For a nonnegative profile, $W_f \subset W$ has nonempty open interior and an open spacelike complement. Reeh–Schlieder and the test-space density above make the vacuum vectors of finite factorized polynomials from one small double cone dense. Their affiliated realizations have bounded polar cutoffs in the algebra, so Ω is cyclic for $\mathcal{M}(W_f)$. Locality and cyclicity in the complement make it separating. The boost sends W_f to $W_{e^s f}$, which is contained in W_f for $s \geq 0$. Every hypothesis of (2.4) is therefore met, and it defines $H_f \geq 0$ with

$$\begin{aligned} S(W_f) &= B - H_f, & \mathcal{M}(W_f) &= \text{Ad}(e^{iH_f})\mathcal{M}(W), \\ e^{isS(W_f)} &= e^{i(1-e^s)H_f} e^{isB}, & H_{\lambda f} &= \lambda H_f \quad (\lambda > 0). \end{aligned} \quad (2.17)$$

The last assertion follows by conjugating the modular difference by a boost and applying its affine covariance.

Translations e^{icb} act inward on both W and W_f for $c \geq 0$. Equation (2.3) says that both modular groups scale b by the same factor; their quotient in (2.17) consequently commutes with the whole b -group. Hence H_f and b strongly commute. A constant translation gives

$$H_{f+c} = H_f + cb. \quad (2.18)$$

Use this identity to define signed profiles bounded below; it agrees with their modular difference and is independent of the positive shift chosen. Inclusion and modular-log order then give, for bounded g ,

$$-\|g\|_\infty b \leq H_g \leq \|g\|_\infty b. \quad (2.19)$$

The inequalities can first be tested on vectors with b in a compact band, smooth for the relevant affine-group representations. On such a band they assert a bounded self-adjoint extension, and exhaustion of the bands gives the displayed decomposable-operator inequality.

Proposition 2.4 (Parabolic generators). *On $\{b > 0\}$ there is a boost-independent internal Hilbert space $\mathcal{K}$ and a nonnegative self-adjoint operator h on it such that*

$$\mathcal{H}_{b>0} \simeq L^2(\mathbb{R}_+, db/b) \otimes L^2(\mathbb{R}^2, d^2k) \otimes \mathcal{K}, \quad H_{p_d} = b(|Y - d|^2 + h), \quad Y = i\partial_k. \quad (2.20)$$

Here $G = bY$, $P_i = -k_i$, and $(e^{isB}\psi)(b, k) = \psi(e^s b, k)$. The operators H_{p_d} vanish on $\ker b$. Moreover $J_W b J_W = b$, $J_W Y J_W = Y$, and $J_W h J_W = h$.

Proof. Put $H = H_{p_0}$. Completion of the square in (2.16) shows that $W_{p_0/(2R)}$ is the portion, with $u < 0$, of the future cone with tip $(u, v, y) = (-R, 0, 0)$. The null rotations about this tip have generators $G_i + RP_i$. They preserve the algebra and the vacuum, so commute with its modular group $e^{i(1-e^s)H/(2R)}e^{isB}$. Set $\tau = (1 - e^s)/(2R)$. Boost conjugation of the rotation group then gives, at the level of Weyl elements, the transformation

$$((1 - 2R\tau)\beta, R\beta) \mapsto (\beta, R\beta)$$

for coefficients of $(G, P_\perp)$; its central coefficient b is fixed. For a fixed sufficiently small nonzero τ , vary R over an open positive interval. Two distinct values give linearly independent coefficient pairs. Multiplying the corresponding Weyl elements, with the same central Weyl phase on both sides, yields the full shear

$$\text{Ad}(e^{i\tau H})G_i = G_i, \quad \text{Ad}(e^{i\tau H})P_i = P_i + 2\tau G_i. \quad (2.21)$$

The group law extends it to every real τ .

Positive energy confines the nonzero momentum spectrum on $\ker b$ to one null ray. A small spatial rotation takes this ray to a disjoint ray; a vector supported there would be orthogonal to its arbitrarily small rotations, contradicting strong continuity. Thus $\ker b$ is precisely the zero-momentum subspace. On it $P_i = 0$, and (2.21) forces $G_i = 0$. Repeating in all null directions makes every connected Lorentz generator vanish there. In particular $B = 0$, and $e^{isB}He^{-isB} = e^sH$ forces $H = 0$ on this subspace.

For $b > 0$, the null rotations and transverse translations form the Weyl pair $G_i = bY_i$, $P_i = -k_i$. Their regular representation is the Schrödinger representation with a multiplicity space; the boost action on $\log b$ trivializes that multiplicity over b . This gives the representation in (2.20). The operator $b|Y|^2$ implements exactly the shear (2.21). Consequently $e^{-itb|Y|^2}e^{itH}$ commutes with the transverse Weyl representation and with b . It is a one-parameter unitary group on the multiplicity space, fiber by fiber. Boost homogeneity identifies its generator as bh with h independent of b . Since $H \geq 0$, its expression $b(|Y|^2 + h)$ implies $h \geq 0$: localize the Y variable near zero and test an arbitrary spectral interval of h .

Transverse translation covariance replaces Y by $Y - d$, proving the formula for p_d . The modular-conjugation relation for H and b , together with geometric conjugation of the null rotations, gives the three statements involving J_W . $\square$

The use of a regular Weyl representation in this proof is the multiplicity form of the Stone-von Neumann theorem, or equivalently translation imprimitivity. No assertion that the internal space consists only of one-field waves is made. In the channels of Π , M and the spin labels are internal variables.

Proposition 2.5 (Bounded cuts commute with the parabolic data). *For bounded smooth g ,*

$$H_g = bh_g, \quad \|h_g\| \leq \|g\|_\infty,$$

where h_g is independent of b and strongly commutes with Y, h .

Proof. The formula (2.20) and constant shifts make all parabolic generators commute. Consequently $\text{Ad}(e^{itH_{p_d}})\mathcal{M}(W_q) = \mathcal{M}(W_{q+tp_d})$ for every positive-curvature circular parabola q and $t \geq 0$. To pass to a bounded g we need an algebra-generation argument. For $x_0 = (u_0, v_0, y_0)$, $u_0 < 0$, its null shadow is

$$q_{x_0}(z) = v_0 + \frac{|z - y_0|^2}{-2u_0}, \quad W_{q_{x_0}} = I^+(x_0) \cap \{u < 0\}.$$

Given a double cone compactly contained in W_g , choose x_0 slightly timelike before its lower tip, still inside W_g . Its shadow strictly dominates g and its future cone contains the double cone. Conversely, $q \geq g$ implies $W_q \subset W_g$. Inner generation therefore proves

$$\mathcal{M}(W_g) = \bigvee_{q \geq g} \mathcal{M}(W_q). \quad (2.22)$$

For W_{g+tp_d} the interior-shadow construction supplies a generating subfamily of dominating parabolas of curvature strictly greater than t . Indeed lower curvature cannot dominate $g + tp_d$ at transverse infinity, while equality corresponds to the boundary value $u_0 = -1/(2t)$ and cannot be the shadow of an interior point. Thus the join defining $\mathcal{M}(W_{g+tp_d})$ may be restricted to all dominating parabolas of curvature greater than t . Subtraction of tp_d bijects this restricted family with the positive-curvature parabolas dominating g . Arbitrary dominating parabolas may have curvature exactly t ; they are not needed for the generating family. Taking joins yields

$$\text{Ad}(e^{itH_{p_d}})\mathcal{M}(W_g) = \mathcal{M}(W_{g+tp_d}).$$

Conjugate the modular group and use $\text{Ad}(e^{itH_{p_d}})B = B - tH_{p_d}$. Applying the same identity to λg and λt , then dividing by λ by boost homogeneity, gives

$$H_{g+tp_d} = tH_{p_d} + e^{i\lambda tH_{p_d}}H_g e^{-i\lambda tH_{p_d}} \quad (\lambda > 0). \quad (2.23)$$

There is no need to differentiate an unspecified unbounded sum. On a band $\epsilon \leq b \leq L$, H_g is bounded by $L \|g\|_\infty$. The second term in (2.23) is independent of λ , and its strong limit as $\lambda \downarrow 0$ is H_g . It follows that H_g commutes with the full group of every H_{p_d} . Exhausting the b bands proves strong commutation globally. In the joint calculus of the commuting parabolic family,

$$H_{p_d} - H_{p_0} = b(-2d \cdot Y + |d|^2).$$

These differences recover Y on $b > 0$, and $H_{p_0}/b - |Y|^2$ recovers h . Hence H_g commutes with Y, h . Finally (2.19) gives the bounded operator $h_g = H_g/b$, and boost homogeneity makes it independent of b . $\square$

We next justify the reflected localization needed for lower bounds. The wedge conjugation reflects (u, v, y) to $(-u, -v, y)$. Even if its action on an arbitrarily chosen bounded double-cone algebra has not been postulated, it produces a net compatible with the original one in spacelike separation. Indeed every bounded operation in a double cone belongs to all containing wedge algebras; their images under J_W are the corresponding reflected, twist-corrected wedge algebras by (2.2) and Poincaré covariance. Two compact spacelike separated double cones admit separating complementary wedges. To see this, write their causal tips as a_i, b_i . A future null linear functional separating b_1 from a_2 , and a second one separating b_2 from a_1 , define the two wedge boundaries; their strict inequalities place the cones in opposite wedges. Locality of these wedge algebras gives the claim.

In particular, the reflected region $J_W W_{-g}$ is spacelike to W_g . This also follows directly by minimizing the sum of the two quadratic penalties in (2.16); for $s, t > 0$ its minimum is $|y_- - y_+|^2/(2(s+t))$. Thus

$$J_W X(W_{-g}) \subset X(W_g)', \quad H_g \geq -H_{-g}. \quad (2.24)$$

For the second implication use modular-log order, $J_W B J_W = -B$, and $J_W H_g J_W = H_g$. The latter follows from the positive-cut conjugation relation and the constant-shift formula.

Lemma 2.6 (Taylor comparison). *If g is bounded and smooth with bounded Hessian, then*

$$|h_g - g(Y)| \leq Ch, \quad C = \frac{1}{2} \sup_y \|g''(y)\|. \quad (2.25)$$

In particular $\|(h_g - g(Y))\psi\| \leq C \|h\psi\|$ for $\psi \in \text{Dom } h$.

Proof. For every transverse center z ,

$$g(y) \leq g(z) + \nabla g(z) \cdot (y - z) + C|y - z|^2.$$

The right side is a signed positive-curvature parabola if $C > 0$. Inclusion, modular-log order, and (2.20) give

$$h_g \leq g(z) + \nabla g(z) \cdot (Y - z) + C|Y - z|^2 + Ch.$$

Applying the same argument to $-g$ and then (2.24) gives the lower inequality with both occurrences of C negated. For this fixed g , all operators involved commute strongly by Proposition 2.5. Impose the inequalities first for a countable dense set of z and then, on each joint spectral fiber, let z tend to the value of Y . They become $-Ch \leq h_g - g(Y) \leq Ch$ in a commuting spectral calculus, which proves both assertions. If $C = 0$, boundedness makes g constant and the result is (2.18). $\square$

Proposition 2.7 (Scalar spectral comparison). *On any scalar copy from (2.12), with its embedding into the ambient internal space understood, one has*

$$\mathbf{1}_{(\lambda, \infty)}(h) \mathbf{1}_{[0, \lambda]}(h_\nu) = 0, \quad \|h^s \psi\| \leq \|h_\nu^s \psi\| \quad (s > 0). \quad (2.26)$$

The norm assertion includes the corresponding inclusion of domains. It does not assert that h leaves the scalar copy invariant. For bounded profiles $f \leq g$ one also has

$$\mathbf{1}_{(\lambda, \infty)}(h_f) \mathbf{1}_{(-\infty, \lambda]}(h_g) = 0. \quad (2.27)$$

Proof. Let j_ν embed the scalar comparison copy. The derivative-removal argument in Section 2.6 gives its real wave localization inside the physical cut space. By (2.15) and (2.17),

$$j_\nu e^{icH_\nu} X_\nu(W) \subset e^{icH_{p_0}} X(W) \quad (c > 0).$$

The embedding intertwines the flat wedge boosts, so Lemma 2.3 applies. It also intertwines the external variables b, Y . To see the latter point independently of the internal operator h , a null rotation N_β fixes n, b, M and sends k to $k + b\beta$ and e_i to $e_i + \beta_i n$. The adapted axes in (2.9) therefore satisfy $N_\beta e_A^{(n)}(p) = e_A^{(n)}(N_\beta p)$ for $A = z, 1, 2$. The constant labels and their fixed derivative phases transform accordingly. Thus j_ν intertwines the null-rotation groups generated by $G = bY$, as well as b ; disintegration on $b > 0$ gives the claimed Y intertwining. This uses the Poincaré action on the comparison waves and does not require h to preserve their individual channels. Both sides are decomposable in b, Y , and j_ν intertwines these variables. In the common spectral fibers their generators are $b(|Y|^2 + h_\nu)$ and $b(|Y|^2 + h)$. Subtracting the common scalar $b|Y|^2$ and dividing by $b > 0$ in the band-inclusion conclusion gives the first assertion in (2.26). More formally, apply the whole-generator conclusion for rational band endpoints, localize (b, Y) to shrinking measurable rectangles, and use monotone convergence of spectral projections. The power bound follows by integrating spectral tails as in Lemma 2.3.

If $f \leq g$, the cut inclusion is $W_g \subset W_f$. Apply the same lemma with source g , target f , and the identity embedding. If needed add the same constant to make both profiles nonnegative. After disintegrating b , removing this common constant gives (2.27). This is spectral order obtained from an analytic strip, rather than from ordinary quadratic-form order. $\square$

2.8 Normal displacement and the required form domain

Lemma 2.8 (Normal form bound). *Let g be bounded and smooth with Lipschitz constant C_1 , and write $H_g(-\delta) = e^{-i\delta a} H_g e^{i\delta a}$. For $\delta, \lambda > 0$,*

$$\pm(H_g - H_g(-\delta)) \leq \delta \left(\frac{a}{\lambda} + \frac{\lambda C_1^2}{2} b \right) \quad (2.28)$$

as forms on a joint momentum-smooth core. The commutator form with a extends continuously to $\text{Dom } \sqrt{1+a+b}$. If $\psi \in \bigcap_{m \geq 1} \text{Dom}(1+a+b)^m$, then $H_g \psi \in \text{Dom } \sqrt{a}$.

Proof. The quadratic sup-convolution obeys $\mathcal{L}_{d+\delta} f = \mathcal{L}_d(\mathcal{L}_\delta f)$. Completing the square in the Lipschitz estimate gives

$$\mathcal{L}_\delta(\lambda g)(y) \leq \lambda g(y) + \frac{\delta \lambda^2 C_1^2}{2}.$$

Hence the cut with profile $\lambda g + \delta \lambda^2 C_1^2/2$, translated by $u = -\delta$, lies inside $W_{\lambda g}$. The translated boost is $B - \delta a$. Modular-log order thus gives

$$B - \delta a - \lambda H_g(-\delta) - \frac{\delta \lambda^2 C_1^2}{2} b \leq B - \lambda H_g.$$

This is the upper sign of (2.28). Apply J_W , which fixes a, b, H_g and reverses the translation parameter by antiunitarity, and then translate by $-\delta$; this gives the lower sign. One may initially use vectors smooth for translations and the boost, then remove the boost smoothing in the resulting form bound.

Put $E = 1 + a + b$ and $Q_\lambda = a/\lambda + \lambda C_1^2 b/2$. Differentiating at $\delta = 0$ on momentum-smooth vectors defines the Hermitian form

$$c_g(\phi, \psi) = i(\langle a\phi, H_g \psi \rangle - \langle H_g \phi, a\psi \rangle).$$

Differentiation is legitimate using $H_g = b h_g$, bounded h_g , and translation smoothness of both vectors; it does not presume that H_g preserves $\text{Dom } a$. The two inequalities give $|c_g(\psi, \psi)| \leq \langle \psi, Q_\lambda \psi \rangle$. Polarization, or representation of this Hermitian form after $Q_\lambda^{1/2}$, gives

$$|c_g(\phi, \psi)| \leq \|Q_\lambda^{1/2} \phi\| \|Q_\lambda^{1/2} \psi\|. \quad (2.29)$$

For fixed λ this proves continuity in the $E^{1/2}$ form norm.

For the domain assertion set $\chi = H_g \psi$. The b bound and strong b commutation imply $\chi \in \text{Dom } \sqrt{1+b}$. On a joint spectral core for a, b ,

$$\langle a\phi, \chi \rangle = \langle H_g \phi, a\psi \rangle - i c_g(\phi, \psi), \quad |\langle H_g \phi, a\psi \rangle| \leq \|g\|_\infty \|\sqrt{b} \phi\| \|\sqrt{b} a\psi\|.$$

Both terms are continuous in $\|\sqrt{E} \phi\|$. Adding $\langle (1+b)\phi, \chi \rangle$ consequently gives

$$|\langle E\phi, \chi \rangle| \leq C_\psi \|\sqrt{E} \phi\|.$$

The spectral theorem characterizes this inequality as $\chi \in \text{Dom } \sqrt{E}$: test with spectral truncations of χ and let their endpoints tend to infinity. Thus $H_g \psi \in \text{Dom } \sqrt{a}$, as required. $\square$

When opposite localization of conjugated bounded operations is needed, one may therefore use the reflected, twist-corrected net just described. It is spacelike compatible with the original net. Conjugating a chosen original affiliated realization by J_W gives an affiliated realization in the reflected algebra, with its polynomial restriction on $J_W \mathcal{D}_{\text{old}}$. On generated factorized vacuum words, its vacuum and adjoint-vacuum data are the reflected-adjoint word data from the wedge boundary calculation; admitted compact joint word data follow by the graph limits above. No equality of $J_W \mathcal{D}_{\text{old}}$ with $\mathcal{D}_{\text{old}}$ is used. The construction can also be repeated with past cuts of the original net. Section 3 will identify their bounded-profile generators from the physical current and thereby compare the resulting moments.

3 The bounded null-cut Ward identity

The distinction between a modular generator and a stress-tensor flux is essential here. The preceding construction supplies the generators H_g without identifying their action on physical fields. We now make that identification for bounded profiles. Throughout this section g is real and belongs to $C_b^\infty(\mathbb{R}^2)$, meaning that every derivative is bounded. All estimates involve only finitely many of these bounds. We write $h_1 = \mathbf{1}$ for the constant profile 1; a numerical subscript ν in h_ν always denotes the radial Hankel operator of Section 2.

The proof has two distinct outputs. First we identify matrix elements of H_g with a stress-tensor flux and extend that identity to physical polynomial tests. Then we use its linear dependence on g , together with the spectral order already proved, to construct a commuting transverse measure. Its quadratic moment is the input to Section 4. The identification concerns bounded profiles throughout; the unbounded parabolic generator retains its separate role as a comparison operator.

Put

$$j_\alpha = vT_{\alpha v} - uT_{\alpha u}, \quad \partial^\alpha j_\alpha = 0, \quad \mathcal{C}_\epsilon(\phi, Z) = \langle \phi, Z \rangle - \epsilon \langle Z, \phi \rangle. \quad (3.1)$$

The last expression is a real bilinear form, rather than a complex bilinear pairing. Complex versions below are obtained by polarization. The stress tensor is even. Thus only the case in which the two local tests have equal total parity contributes; ϵ is their interchange sign.

3.1 Bounded local tests and differentiation

Let A and C be bounded Hermitian operators, localized in compact double cones strictly inside W and its opposite, respectively. We first smooth them by compactly supported convolution under the Poincaré group. A slightly larger double cone still contains each support. Such smoothing gives bounded commutators of every prescribed finite order with the infinitesimal Poincaré action, and the vacuum vectors and their products are smooth translation vectors. On the opposite side one may use the twist-corrected reflected net described in Section 2.

The first task is to keep both tests separated while varying them with H_g . This requires containment of the conjugated algebras for both signs of the flow parameter. The bound $|H_g| \leq \|g\|_\infty b$ alone supplies a domain estimate and does not supply that containment.

There are $\alpha, c > 0$ such that the support of A lies strictly inside the truncated cut with profile $\alpha|y|^2 + c$; the reflected assertion holds for C . Write $d = \|g\|_\infty$. For $t > 0$, the positive-profile half-sided inclusion and the constant-shift identity give

$$\text{Ad}(e^{itH_g})\mathcal{M}(W) = \mathcal{M}(W_{tg}) \supset \mathcal{M}(W_{td}) = \text{Ad}(e^{itdb})\mathcal{M}(W).$$

Conjugating this inclusion by e^{-itH_g} , and using commutation with b , yields

$$\text{Ad}(e^{-itH_g})\mathcal{M}(W) \subset \mathcal{M}(W_{-td}).$$

For positive t the same upper containment follows from $tg \geq -td$. These are algebra inclusions obtained from the cuts; they do not follow merely from an operator-form bound. Since H_g commutes with the parabolic generators and b , conjugation of $\mathcal{M}(W_{\alpha p_0+c})$ therefore lies in $\mathcal{M}(W_{\alpha p_0+c-|s|d})$. Choosing $|s|d < c/2$ shows that

$$A_s = \text{Ad}(e^{isH_g})(A), \quad C_s = \text{Ad}(e^{isH_g})(C)$$

remain in fixed, strictly separated truncated regions. Reflection gives the assertion for C_s . No identity $H_{-g} = -H_g$ is used. Hence A_s and C_t graded commute for all sufficiently small s, t .

Set $Z = CA\Omega$. Since $|H_g| \leq \|g\|_\infty b$, these smooth vectors lie in $\text{Dom } H_g$. Differentiating vectors, rather than unbounded operator products, gives

$$Z_{C'} = \epsilon A(iH_g C\Omega), \quad Z_{A'} = C(iH_g A\Omega), \quad Z_{C'} + Z_{A'} = iH_g Z. \quad (3.2)$$

For example, $C_s A\Omega = \epsilon A C_s \Omega$ in a whole interval about zero, so the first derivative follows from strong vector differentiation. The product rule then follows by subtracting and adding $C_s A\Omega$ in $C_s A_s \Omega$. The normal form estimate of Lemma 2.8 and the bounded translation commutators give

$$Z_{A'}, Z_{C'} \in \text{Dom}(1+a)^{1/2}. \quad (3.3)$$

For example, Lemma 2.8 puts $H_g A\Omega$ in $\text{Dom}(1+a)^{1/2}$. The bounded commutator of C with the normal translation generator makes C bounded on $\text{Dom}(1+a)$ with its graph norm; interpolation with its Hilbert-space bound gives the same property on $\text{Dom}(1+a)^{1/2}$. This proves the assertion for $Z_{A'}$, and the roles of A, C give the other one. In particular no invariance of $\text{Dom } a$ under H_g is being used.

These are the only normal-energy domains needed for the narrow-strip term below. The boost calculation uses only $\text{Dom } B \cap \text{Dom } H_g$. Indeed the affine unitary relation $e^{itB} e^{isH_g} = e^{ie^t s H_g} e^{itB}$, differentiated in t on this intersection, gives

$$B e^{isH_g} \xi = e^{isH_g} (B + sH_g) \xi.$$

Thus the H_g unitary group preserves this intersection and is continuous in its B graph norm. Differentiating the corresponding matrix identity, and moving B onto the fixed first vector in one difference quotient, gives

$$i(\langle B\xi, H_g \eta \rangle - \langle H_g \xi, B\eta \rangle) = \langle \xi, H_g \eta \rangle, \quad \xi, \eta \in \text{Dom } B \cap \text{Dom } H_g. \quad (3.4)$$

Our smoothed vacuum vectors belong to this domain. No domain for $BH_g \eta$ is required.

Decrease c and relabel it so that the fixed containing truncations for all sufficiently small s have profiles $\alpha p_0 + c$. We choose a time step whose two half-slabs remain separated from the opposite truncated regions. The geometry can be read directly from the cut formula. Put $R_0 = 1/(2\alpha)$. Then

$$W_{\alpha p_0+c} = \left\{ -R_0 < u < 0 : \quad v > c + \frac{|y|^2}{2(R_0 + u)} \right\}.$$

For $u = -d < 0$, the reflected-cut separation inequality says that

$$v > -c - \frac{|y|^2}{2(R_0 + d)}$$

is spacelike to the reflected opposite truncation. The margin satisfies

$$d + c + \frac{|y|^2}{2(R_0 + d)} \geq c_0 \sqrt{1 + d^2 + |y|^2} \quad (d \geq 0) \quad (3.5)$$

for some $c_0 > 0$: $d + c$ controls $R_0 + d$, and the arithmetic–geometric mean controls $|y|$. Reflection gives the corresponding inequality for $u > 0$.

Choose a positive smooth width $w(u, y)$ comparable to $\sqrt{1 + u^2 + |y|^2}$, with bounded first derivatives and homogeneous of degree one outside a compact set. If σ is a fixed smooth step, zero below -1 and one above 1 , set $\rho = \sigma((u + v)/(\varepsilon w(u, y)))$. For sufficiently small ε , its transition slab is spacelike, and (3.5) makes its negative and positive halves spacelike to the respective opposite truncated regions. Near $u = 0$ we also require separation from both fixed compact tests A, C . Their compact normal-coordinate margins give this gap on bounded transverse sets; at large $|y|$, their quadratic spacelike separation dominates the slab width. Shrinking ε supplies the required central gap. This last assertion concerns the compact tests, not both entire truncated regions.

Take $\chi \in C_c^\infty(\mathbb{R} \times \mathbb{R}^2)$, real and equal to one near zero, and write

$$\chi_R(u, y) = \chi(u/R, y/R), \quad F_R = \chi_R d\rho, \quad j_F \Omega = \int F \cdot j(x) \Omega d^4x.$$

The full flux vectors satisfy

$$\sup_{R \geq R_0} \|j_{F_R} \Omega\| < \infty, \quad j_{F_R} \Omega \rightharpoonup 0 \quad (R \rightarrow \infty). \quad (3.6)$$

Indeed conservation allows replacement of ρ by an entirely scaled step agreeing with it near $\text{supp } d\chi_R$. The difference of the fluxes is the integral of $d\chi_R(\rho - \rho_R)$ against j and vanishes. The resulting compact flux has scale degree zero, so its vector is a unitary dilation of one fixed vector. Its spectral mass is driven to zero under this dilation. The stress-tensor vacuum channel has no zero-momentum part, proving weak convergence; the norm is unchanged. This argument uses neither clustering nor a norm bound on the field.

The physical Ward identities are assumed for original polynomials. The following two mixed functionals extend them in exactly the form needed for differentiation. In each, the boost acts on the fixed smoothed test.

Let ϑ be smooth, equal to one on $(-\infty, -2]$ and zero on $[-1, \infty)$, and put

$$\theta_{-, \delta}(u) = \vartheta(u/\delta), \quad \theta_{+, \delta}(u) = \theta_{-, \delta}(-u), \quad j_\pm = j_{\theta_{\pm, \delta} F_R}.$$

Lemma 3.1 (Mixed boost Ward identities). *For the bounded smoothed opposite tests above, for sufficiently large R , sufficiently small $\delta > 0$, and s in a neighborhood of zero,*

$$\begin{aligned} \langle C_s \Omega, BA \Omega \rangle &= \mathcal{C}_\epsilon(j_- \Omega, C_s A \Omega), \\ \langle BC \Omega, A_s \Omega \rangle &= -\mathcal{C}_\epsilon(j_+ \Omega, C A_s \Omega). \end{aligned} \quad (3.7)$$

Proof. Start with a Hermitian finite algebraic sum C_0 of factorized original words compactly localized in the opposite wedge. Its closed affiliated realization was constructed in Section 2. The boost Ward identity applies to its original polynomial restriction; affiliation and opposite locality allow the compact current flux around C_0 to pass the bounded operator A .

Choose the partition for the flux around A once. Its time-zero causal shadow is compactly contained in the positive spatial halfspace, so a small slab neighborhood of that shadow lies compactly inside W and is spacelike to every opposite-wedge polynomial. Denote the resulting fixed compact current vacuum vector by η_A . For each C_0 , its own time-zero causal shadow is a compact subset of the negative halfspace. Thin or deform the slab only near that shadow, leaving the A partition and the homogeneous shape at infinity unchanged. Conservation and (3.6), applied to each deformation, transfer the opposite flux to this same η_A . The signs of the commutator give

$$\langle BC_0 \Omega, A \Omega \rangle = \langle C_0 \Omega, BA \Omega \rangle = \mathcal{C}_\epsilon(\eta_A, C_0 A \Omega).$$

In particular the current side obeys

$$|\mathcal{C}_\epsilon(\eta_A, C_0 A \Omega)| \leq 2 \|\eta_A\| \|A\| \|C_0 \Omega\|,$$

since $C_0 A \Omega = \epsilon A C_0 \Omega$. No uniform bound on the changing flux near C_0 is needed. The boost has been moved to the fixed vector $A \Omega$ before taking any Hilbert limit.

Hermitian factorized polynomial vacuum vectors are real dense in the opposite wedge standard subspace by the Tomita-graph argument in Section 2. The displayed bounded functional therefore extends to $C_s \Omega$. For the fixed family of truncated localizations, the compact spatial partition can be replaced by the negative half-slab: C_s is spacelike there, and the commutator vanishes outside the fixed causal intersection of A . This gives the first identity in (3.7). Interchanging the fixed side gives the second. Its minus sign comes from placing the boost on the bra; the opposite flux contributes $\epsilon \langle C A_s \Omega, j_+ \Omega \rangle - \langle j_+ \Omega, C A_s \Omega \rangle$. The operators A_s, C_s need only lie in their fixed truncated algebras; compact localization of these varied operators is not required. $\square$

Differentiate (3.7) in the real parameter s and add. Equation (3.4) identifies the left side as

$$i(\langle BC \Omega, H_g A \Omega \rangle - \langle H_g C \Omega, B A \Omega \rangle) = \langle C \Omega, H_g A \Omega \rangle.$$

The right side is $\mathcal{C}_\epsilon(j_- \Omega, Z_{C'}) - \mathcal{C}_\epsilon(j_+ \Omega, Z_{A'})$. Substituting (3.2) and writing the two half-slabs as the full slab minus its central strip gives

$$\begin{aligned} \langle C \Omega, H_g A \Omega \rangle &= \mathcal{C}_\epsilon(j_{\theta_-, \delta F_R} \Omega, i H_g Z) - \mathcal{C}_\epsilon(j_{F_R} \Omega, Z_{A'}) \\ &\quad + \mathcal{C}_\epsilon(j_{(1-\theta_-, \delta - \theta_+, \delta) F_R} \Omega, Z_{A'}). \end{aligned} \quad (3.8)$$

All current entries here are actual smeared vacuum vectors. This calculation extends a bounded vector functional; it does not postulate a stress-tensor operator commutator with an arbitrary bounded operation.

3.2 The three cutoff estimates

There are three different error mechanisms in (3.8). The narrow strip disappears weakly after one inverse half-power of normal energy. The normal face is compared in norm with the multiplication operator $bg(Y)$, which yields the physical plane flux. The distant cap is controlled after dilation by a mass-dyad sum. Keeping these mechanisms separate specifies which vector domains are needed and which bounds must be uniform in the normal width.

Here and below $a = (M^2 + |k|^2)/(2b)$, $b > 0$. Constants in the one-field norm can be suppressed, since there are only finitely many polarizations. The two useful versions of the measure are

$$da \, db \, d^2 k \quad (2ab \geq |k|^2), \quad \frac{db}{2b} \, d^2 k \, dm, \quad m = M^2 \geq 0. \quad (3.9)$$

All inequalities below are componentwise inequalities in the finite rest-spin spaces, summed at the end.

We make the dependence on test functions precise. For a compact smooth function $q(u, y, v)$ put

$$p_m(q) = \sum_{|\beta| \leq m} \left\| (1 + |u| + |y| + |v|)^m \partial^\beta q \right\|_{L^1}.$$

In estimates involving a step, this seminorm is applied to its v derivative and to its products with the indicated compact functions of (u, y) , not to the step itself. At a fixed R , all constants denoted

$C_{R,N}$ are bounded by a polynomial in the corresponding p_{4N+20} seminorms, the derivatives of χ_R on their compact support, and the derivatives of the fixed step ϑ . After rescaling a cap by R , use the same seminorms of its fixed smooth shapes; the constants are then independent of R and of the sharp normal width. Factors u , v , or a coordinate derivative appearing in the current are included in these seminorms. This convention allows uniformity for compact families of local tests and for all the smoothing operations used below.

First consider the last term in (3.8). Its normal support has width $O(\delta)$. Fourier transformation gives a factor δ times a rapidly decreasing function of $(\delta a, b, k)$; the part containing $uT_{\alpha u}$ has an additional δ . The polarization weights for indices u, i, v are $a, \sqrt{ab}, b$. Consequently, after applying $(1+a)^{-1/2}$, the square norm is bounded by a constant times

$$\int_0^\infty \{\delta^2(1+a) + \delta^4(1+a)^3\}(1+\delta a)^{-6} da \leq C_R. \quad (3.10)$$

The omitted (b, k) integrals are convergent rapid-decay integrals; extending their domain only enlarges this bound. On every compact momentum set the weighted Fourier coefficients tend to zero. Uniform boundedness and compact-momentum density therefore give weak convergence to zero. By (3.3),

$$\lim_{\delta \downarrow 0} \mathcal{C}_\epsilon(j_{(1-\theta_{-, \delta} - \theta_{+, \delta})F_R} \Omega, Z_{A'}) = 0. \quad (3.11)$$

Thus the strip estimate uses exactly the half-energy domain established before the differentiation. We turn to the first term, whose sharp normal boundary will survive as the Ward flux.

Next integrate the first term of (3.8) by parts. Moving iH_g to the first entry means applying $-iH_g$ there. Before that application, conservation replaces the flux by

$$B_{R,\delta} - E_{R,\delta} = \int (-\theta'_{-, \delta}) \chi_R \rho j_v \Omega d^4x - \int \theta_{-, \delta} \rho [(\partial_u \chi_R) j_v - (\partial_i \chi_R) j_i] \Omega d^4x. \quad (3.12)$$

For this integration first introduce a smooth v cutoff of length S . At fixed R, δ , let $S \rightarrow \infty$ before making any other limit. At $b > 0$, the transforms of ρ and $v\rho$ have respectively the bounds b^{-1} and b^{-2} times rapid large- b symbols. This follows by differentiating the compact function $\partial_v \rho$; the possible distributions supported at $b = 0$ are killed by the displayed stress-tensor powers of b . The error containing the derivative of the v cutoff, after multiplication by b , is bounded by

$$C_{R,\delta,N} b(abS + a^2)(1 + Sb + a + |k|)^{-N}.$$

Its square integral in (3.9) is $O(S^{-3})$ for N large: set $B = Sb$, and enlarge the cone to the full positive (a, b) quadrant. Since $\|H_g \xi\| \leq \|g\|_\infty \|b\xi\|$, this removes the cutoff in the required matrix element. The remaining terms converge after multiplication by b , using normal Fourier decay at fixed δ .

We now estimate the normal face $B_{R,\delta}$. Its decomposition requires only the scalar comparison channels with $\nu = 0, 1$:

$$T_{vv} = b^2 c_T, \quad T_{iv}/b = M s_{zi} - k_i c_T,$$

while conservation expresses T_{uv} as a linear combination of $(ab, |k|^2)c_T$ and $k_i M s_{zi}$. Before a radial derivative, the coefficient of the vT_{vv} term has order one. That of the uT_{uv} term has order $\delta(M^2 + |k|^2)/b$; the mixed term $\delta|k|M/b$ is bounded by the same expression. Differentiating in $m = M^2$ costs δ/b , and

$$h_\nu(M^\nu f(m)) = -4M^\nu((1+\nu)f'(m) + mf''(m)). \quad (3.13)$$

For example, a mixed $\nu = 1$ coefficient has the form $\delta k_i M f_\delta(m, b, k)/b$, up to fixed constants and finite sums. Rescaling its normal test by $u = \delta s$ gives

$$|\partial_m^j f_\delta| \leq C_{R,N} (\delta/b)^j (1 + b + |k| + \delta a)^{-N}, \quad j = 0, 1, 2.$$

Its radial image is

$$bh_1 \left(\frac{\delta k_i M}{b} f_\delta \right) = -4\delta k_i M \left(2\partial_m f_\delta + m\partial_m^2 f_\delta \right).$$

The potentially singular factor is now $\delta|k|M/b \leq \delta a$; also $\delta m/b \leq 2\delta a$. These factors are absorbed by the rapid decay, leaving an overall factor δ . The $\nu = 0$ terms follow from the same radial identity. Consequently, for every N , each coefficient of $bh_\nu B_{R,\delta}$ obeys

$$|bh_\nu B_{R,\delta}(b, k, M)| \leq C_{R,N} \delta (1 + b + |k| + \delta a)^{-N}. \quad (3.14)$$

The spectral comparison (2.26), applied to these channels, and the Taylor bound (2.25) imply

$$\|(H_g - bg(Y))B_{R,\delta}\| \leq C_R \|g''\|_\infty \sqrt{\delta}. \quad (3.15)$$

In particular the estimate has no unrecorded lower cutoff in b . Indeed the square integral of the right side of (3.14) is bounded by

$$\begin{aligned} C\delta^2 \int \frac{db}{b} d^2k \int_0^\infty \left(1 + b + |k| + \frac{\delta(m + |k|^2)}{2b} \right)^{-2N} dm &\leq C_N \delta \int db d^2k (1 + b + |k|)^{1-2N} \\ &\leq C'_N \delta. \end{aligned}$$

For the principal part use $w = 1 + b + M^2/b$, which commutes with $g(Y)$. For any fixed $q > 1/2$, the vectors $w^{-q} b B_{R,\delta}$ are norm bounded and converge to their plane limits. To see this, the coefficients after multiplication by b are bounded by

$$C_{R,N} b (1 + \delta a) (1 + b + |k| + \delta a)^{-N};$$

the m integral against w^{-2q} is finite, and the remaining b, k integrals converge. Dominated convergence applies, and the term with u tends to zero. Since a translation-smooth Z belongs to $\text{Dom } w^q$ for every fixed q , this proves convergence in the pairing. Fourier inversion in k inserts $g(y)$. Finally ib acts as ∂_v on $T_{vv}\Omega$; integration by parts leaves

$$\int \chi_R(0, y) g(y) (\rho(0, y, v) + v\partial_v \rho(0, y, v)) \mathcal{C}_\epsilon(T_{vv}(0, y, v)\Omega, Z) dv d^2y. \quad (3.16)$$

The term with $\partial_v \rho$ vanishes by the spacelike gap.

The face comparison has now removed the internal operator h from the surviving flux. Its error tends to zero with δ at each fixed R , including the region $b \downarrow 0$. To complete the argument we need a different estimate for the distant cap: its bound must remain valid after the normal step has become sharp.

It remains to control the cap $E_{R,\delta}$ uniformly as the normal step becomes sharp. Dilate its variables by R . Its smooth shapes then range over a fixed bounded set of the seminorms specified above, and its current smearing has scale degree zero. In these coordinates the generator becomes

$$R^{-1} b h_{g_R}, \quad g_R(y) = g(Ry). \quad (3.17)$$

Normal integration by parts gives a factor $(1 + a)^{-1}$ from a half-space step, and $(1 + a)^{-2}$ if its coefficient contains u . These bounds, as well as their logarithmic a derivatives up to order two, are

uniform in the smoothing width. For example, split the normal integral into $|u| \leq (1+a)^{-1}$ and its complement. The first part has the claimed length, or its square with an extra u ; on the second part integrate by parts. The derivative of the step has uniformly bounded total variation, and its support has $|u| = O(\delta/R)$. Repeating this argument proves the stated symbol bounds. Tangential integrations by parts give arbitrary rapid decay in b, k , with the same b^{-2}, b^{-1} factors from $v\rho, \rho$.

Let F_v, F_i denote the resulting cap coefficients before multiplication by b . The polarization weights give, for any N ,

$$|F_v| \leq C_N(1+a)^{-1}\tau_N(b, k), \quad |F_i| \leq C_N \frac{\sqrt{a/b}}{1+a} \tau_N(b, k), \quad \tau_N = (1+b+|k|)^{-N}. \quad (3.18)$$

The same inequalities hold after up to two logarithmic M derivatives, with a different C_N . This last assertion can also be checked directly in each tetrad coefficient: every occurrence of a is $(M^2 + |k|^2)/(2b)$, so $M\partial_M a \leq 2a$.

Use a smooth dyadic partition $\eta_{\text{lo}}(M) + \sum_{j \geq 0} \eta_j(M) = 1$ with η_{lo} supported in $M \leq 2$ and η_j supported in $N_j/2 \leq M \leq 2N_j$, $N_j = 2^j \geq 1$, and uniformly bounded logarithmic derivatives. If $F_j = \eta_j F$ and $F_{\text{lo}} = \eta_{\text{lo}} F$, then

$$\|bF_{\text{lo}}\| \leq C, \quad \|bF_j\| \leq C, \quad \|bhF_j\| \leq CN_j^{-2} \quad (j \geq 0). \quad (3.19)$$

Here the last expression uses the high-mass pieces only. For clarity, the potentially worst integral in the first two bounds is

$$\int_{M \asymp N} \frac{dm}{2b} \frac{ab}{(1+a)^2} = \frac{1}{2} \int_{M \asymp N} \frac{a}{(1+a)^2} dm \leq Cb \quad (N \geq 1),$$

followed by the convergent (b, k) integral with τ_N^2 . For $M \leq 2$ the integrand is simply bounded and the m interval has bounded length. Thus the factor $b^{-1/2}$ in (3.18) is harmless. To prove the h bound, decompose into the finite pure-label channels of (2.12). On $M \asymp N_j$, the identity (3.13) and the logarithmic derivative bounds give a factor N_j^{-2} , including when derivatives hit η_j . Apply (2.26) to each channel and sum their norms. No commutation of h with invariant mass is asserted or used.

Write $h_{g_R} = g_R(Y) + D_R$. The bounded-profile estimate and (2.25) give simultaneously

$$\|D_R \xi\| \leq 2 \|g\|_\infty \|\xi\|, \quad \|D_R \xi\| \leq \frac{1}{2} R^2 \|g''\|_\infty \|h\xi\|.$$

The second is an operator-norm estimate on $\text{Dom } h$, since D_R strongly commutes with h . Thus the high dyads of the remainder in (3.17) have norms at most

$$\frac{C_g}{R} \min(1, R^2/N_j^2),$$

and their sum, including the low-mass part, is

$$O_g((1 + \log R)/R). \quad (3.20)$$

Indeed, the dyads with $N_j \leq R$ contribute at most C_g/R each, and there are $O(1 + \log R)$ of them. The remaining bound is a geometric series in R^2/N_j^2 , whose sum is bounded independently of R . This gives the stated rate without any orthogonality assumption on the images of the dyads.

This estimate allows h to mix masses: the summation is a triangle inequality after applying D_R , not an orthogonal decomposition of its image. The principal part $R^{-1}bg_R(Y)$ does preserve mass. Pair its j th dyad against the correspondingly rescaled Z_R . Translation smoothness gives, for every fixed integer r ,

$$\|1_{M \asymp N_j} Z_R\| \leq C_r(Z)(1 + N_j/R)^{-r}.$$

Consequently its pairing is bounded by

$$\frac{C_g C_r(Z)}{R} \left(1 + \sum_{j \geq 0} (1 + N_j/R)^{-r} \right) \leq C_{g,r,Z} \frac{1 + \log R}{R}, \quad r \geq 1. \quad (3.21)$$

Equations (3.20)–(3.21) prove cap vanishing in precisely the order required by (3.8).

3.3 The Ward formula and polynomial tests

The order of limits is

$$S \longrightarrow \infty, \quad \delta \downarrow 0 \text{ at fixed } R, \quad R \longrightarrow \infty. \quad (3.22)$$

The strip term vanishes by (3.11), the cap by (3.20)–(3.21), and the full-flux term by (3.6). The normal face gives (3.16). In particular its dependence on g is linear before the last limit. This proves linearity of H_g as a form on these local vectors for all profiles under consideration. The estimate

$$|\langle \xi, H_g \eta \rangle| \leq \|g\|_\infty \|b^{1/2} \xi\| \|b^{1/2} \eta\|$$

extends this equality to the whole form domain. The needed graph density follows by applying a small compact translation convolution to Reeh–Schlieder approximants in any fixed double cone: convolution is bounded from Hilbert norm to the graph norm of $b^{1/2}$, and its approximate identity tends to the identity in that graph norm.

Proposition 3.2 (Bounded-profile Ward formula). *The map $g \mapsto H_g = b h_g$ is real linear on $C_b^\infty(\mathbb{R}^2)$. For g compactly supported or Schwartz, and the bounded smooth opposite tests above,*

$$\langle C\Omega, H_g A\Omega \rangle = \int_{\mathbb{R}^2} \int_{\mathbb{R}} g(y) \rho(v) \mathcal{C}_\epsilon(T_{vv}(0, y, v)\Omega, C A\Omega) \, dv \, d^2 y. \quad (3.23)$$

Here ρ is any smooth step whose derivative lies in the v gap between the two causal intersections with the plane. The expression is a vacuum-vector distribution paired with the indicated smooth vector, not an unsmeared operator integral.

To replace the original plane step by this $\rho(v)$, interpolate inside the spacelike gap and use locality. The half-step vacuum wave is continuous after an inverse polynomial energy weight: T_{vv} supplies b^2 , the step costs b^{-1} , and tangential smearing supplies rapid Fourier decay. This is also the weighted convergence proved in the normal-face calculation. It justifies the plane restriction and the removal of χ_R in (3.23).

Proposition 3.3 (Polynomial extension). *Equation (3.23) holds for compactly smeared physical polynomial tests, including their adjoints, on the common field domain $\mathcal{D}_{\text{old}}$. The convergence to it is uniform when the smearing tests vary over a compact smooth parameter family.*

The extension has three separate passages, with uniformity for fixed word labels and lengths and a compact smooth family of their tests. For finite sums of factorized polynomials, polar truncation first replaces each polynomial by a bounded operation in its original localization algebra. Group convolution then gives the translation derivatives needed in the Ward formula. The rapid truncation error absorbs the polynomial growth of the norms of the other bounded factors; only after that limit do we remove the convolution. Finally, test-space density and the original joint-distribution bounds extend the identity to admitted compact joint kernels.

Proof. First let A be a Hermitian finite algebraic sum of factorized physical polynomial words on the invariant field domain $\mathcal{D}_{\text{old}}$. Sums and products of its affiliated single-field realizations, followed by closure, give a closed affiliated operator T such that $T|_{\mathcal{D}_{\text{old}}} = A$ and $T^*|_{\mathcal{D}_{\text{old}}} = A$. The product is closable because its adjoint contains the reversed adjoint product on the dense domain $\mathcal{D}_{\text{old}}$. Write $T = U|T|$, let $E_L = 1_{[0,L]}(|T|)$, and set

$$T_L = TE_L, \quad A_L = \frac{1}{2}(T_L + T_L^*).$$

Then A_L is bounded Hermitian in the same local algebra, with $\|A_L\| \leq L$. For $\psi \in \mathcal{D}_{\text{old}}$ and every nonnegative integer m , the spectral-tail bounds give

$$\begin{aligned} \|(T - T_L)\psi\| &\leq L^{-2m} \| |T|^{2m+1}\psi \|, \\ \|(T^* - T_L^*)\psi\| &\leq L^{-2m} \| |T|^{2m}T^*\psi \|. \end{aligned} \tag{3.24}$$

Both right sides are finite polynomial-domain norms. Indeed $\mathcal{D}_{\text{old}}$ is invariant under T, T^* , and powers of T^*T and TT^* there agree with the corresponding iterated physical polynomials. Hence $A_L\psi - A\psi = O(L^{-N})$ for every prescribed N . This proves the same assertion for any fixed finite family of polynomial states and moments needed in the argument. No essential self-adjointness is required.

One must not differentiate the sharp spectral projection E_L . Instead first convolve A_L under the spacetime group, with a fixed compact smooth kernel f . Group derivatives then fall on f . The estimate (3.24) is uniform for the transformed polynomial tests on the compact support of f , since its right sides are continuous norms of smeared polynomial vectors. Thus the convolved vacuum vectors converge in every fixed translation Sobolev norm. Apply the same construction to C separately. Their localization regions remain spacelike. For example,

$$C_L A_L \Omega - C A \Omega = C_L (A_L - A) \Omega + (C_L - C) A \Omega,$$

whose first term is bounded by L times a faster-than-any-power tail and whose second is another polynomial-state tail. The same argument after convolution gives convergence of products in the finite Sobolev norms used above. To make the product convergence explicit, let α_r denote the Poincaré action and write $A_L(r) = \alpha_r(A_L)$, $A(r) = \alpha_r(A)$, with analogous notation for C . Uniformly for r, s in fixed compact group sets,

$$\|C_L(s)A_L(r)\Omega - C(s)A(r)\Omega\| \leq LC_N L^{-N} + C'_N L^{-N}.$$

For the second term apply the polar-tail estimate to the polynomial state $A(r)\Omega$; its iterated $C(s)$ -polynomial norms are uniformly bounded on these compact sets. For a product of the two convolved operators, a common translation is left translation of both group variables. Integration by parts consequently places each prescribed translation derivative on the convolution kernels. The resulting finite sum of compact kernel integrals has the same tail estimate, with new constants. This proves the required Sobolev convergence without differentiating either polar projection.

For these factorized polynomials, the left side of (3.23) converges by its $b^{1/2}$ form bound, and the right side by the inverse-energy estimates for the half-step wave. Finally remove the convolution on the original smooth field tests. All constants are finite test seminorms, so this last limit and compact-parameter uniformity follow from distributional continuity.

It remains to pass to the admitted compact joint kernels. For an old word A , write $\Psi_A(\varphi_A)$ for its vacuum vector smeared with a joint kernel φ_A . First partition a compact support into finitely many product patches inside the required localization region. On each patch, finite sums of tensor

tests are dense in the smooth test-function topology. Choose such kernels $\varphi_{A,j} \rightarrow \varphi_A$ with their supports in those patches; their reversed adjoint kernels converge as well. Do the same for the kernel φ_C of C . Polarization gives the factorized identity for arbitrary complex word tests, so it applies to these approximants. This is a test-space approximation, not a finite-rank requirement on the given kernels.

For every fixed nonnegative integer k , the original joint-distribution bounds imply

$$\|(1 + P^0)^k(\Psi_A(\varphi_{A,j}) - \Psi_A(\varphi_A))\| \rightarrow 0,$$

and analogously for C and for the concatenated old word with kernel $\varphi_{C,j} \otimes \varphi_{A,j} \rightarrow \varphi_C \otimes \varphi_A$. Indeed, a collective time-translation derivative of a vacuum word is the same word with the corresponding derivative of its joint test. The doubled original vacuum distributions therefore bound each displayed graph norm by finitely many joint test seminorms. Positive translation spectrum makes these norms control the $b^{1/2}$ graph norm and every fixed translation-Sobolev norm used above. The left side of (3.23) is continuous by its $b^{1/2}$ form bound; the right side is continuous by the inverse-energy half-step bound applied to the concatenated old vacuum word. Taking this kernel limit after the polar and convolution limits proves the stated Ward identity for the joint kernels.

On a compact smooth parameter family the same finitely many seminorms are uniformly bounded. The tensor approximations can be chosen uniformly in those seminorms, using a finite cover of the compact family and a smooth partition in its parameter. The preceding graph and flux bounds then give the asserted compact-parameter uniformity. This final passage does not assert a closed affiliated operator for an arbitrary joint smear. $\square$

3.4 The commuting measure and its second moment

The analytic part of the argument has established the bounded Ward identity and its polynomial extension. We now use the resulting linearity to organize all bounded profiles simultaneously. Positivity will give a measure, but normalization requires a separate tightness argument, and its quadratic moment requires comparison with the actual parabolic cuts.

Linearity upgrades the spectral order of bounded profiles to commutativity. To record the argument, put $A = h_g$, $B = h_f$ and $\varepsilon = \|f\|_\infty / s$. The profile comparison of Section 2 gives the spectral brackets

$$A - \varepsilon \mathbf{1} \leq_{\text{sp}} A + B/s \leq_{\text{sp}} A + \varepsilon \mathbf{1}.$$

If $r < t$ and $2\varepsilon < t - r$, the low A subspace $1_{(-\infty, r]}(A)\mathcal{H}$ is contained in the spectral subspace of $A + B/s$ below $r + \varepsilon$, while the high A subspace $1_{[t, \infty)}(A)\mathcal{H}$ is contained in its spectral subspace above $t - \varepsilon$. Thus

$$1_{(-\infty, r]}(A)(A + B/s)1_{[t, \infty)}(A) = 0.$$

The same matrix of A is zero, so the matrix of B is zero. Varying r, t proves that B commutes with every spectral projection of A . This argument uses the spectral-support comparison (2.27), not merely ordinary operator order.

Proposition 3.4 (Internal measure and quadratic moment). *On the $b > 0$ representation there is a normalized positive operator-valued Borel measure μ on $\mathbb{R}^2$, with mutually commuting values on the internal multiplicity space, such that*

$$H_g = b \int_{\mathbb{R}^2} g(Y + z) \mu(dz), \quad \mu(\mathbb{R}^2) = \mathbf{1}. \quad (3.25)$$

Its values commute with h and with the external variables b, Y . Its first moment is zero as a polarized integral on $\text{Dom } h^{1/2}$. Its trace second moment is the increasing closed-form limit of compactly truncated moments and defines a positive self-adjoint operator κ satisfying

$$\int z \mu(dz) = 0, \quad \kappa = \int |z|^2 \mu(dz) \leq h, \quad K_n = b(|Y|^2 + \kappa). \quad (3.26)$$

The second-moment inequality is a quadratic-form inequality and, since the operators strongly commute, has its usual joint spectral meaning. No assertion that μ is projection-valued, or that $\kappa = h$, is made.

Proof. The map $g \mapsto h_g$ is positive and contractive in the supremum norm, and $h_1 = \mathbf{1}$. It commutes strongly with Y, h, b and is boost independent. In the Y representation write $h_g = M_{F_g}$, where F_g is an essentially bounded internal-operator-valued function. For $\tau_t g(z) = g(z + t)$, transverse covariance gives

$$F_{\tau_t g}(y) = F_g(y + t) \quad \text{almost everywhere,} \quad \|F_g(\cdot + t) - F_g\|_{L^\infty} \leq \|\tau_t g - g\|_\infty.$$

The right side tends to zero. Weak-operator convolution with a smooth approximate identity therefore converges to F_g in essential-supremum norm. The convolutions are norm continuous; their pairwise essential-supremum bounds are consequently actual supremum bounds. Their uniform limit selects a unique norm-continuous representative $\tilde{F}_g$. Covariance now holds pointwise, so

$$\mathcal{L}(g) = \tilde{F}_g(0), \quad \tilde{F}_g(y) = \mathcal{L}(\tau_y g)$$

defines one positive contractive internal functional. Linearity, positivity, and the commuting relations pass to these representatives. For inequalities involving the unbounded h , first restrict to a bounded spectral band of h , evaluate the resulting continuous inequality at zero, and then exhaust the bands.

The Riesz representation theorem applied to scalar matrix elements of $\mathcal{L}$ on compact tests, and polarized, gives a positive operator-valued measure μ . Commutativity of all h_g and commutation with h pass to its Borel values by bounded monotone approximation. This proves (3.25) first for compactly supported smooth profiles.

It remains to rule out mass at infinity, which is not automatic from a representation on compact tests. Choose a radial $\chi_0 \in C_c^\infty(\mathbb{R}^2)$, $0 \leq \chi_0 \leq 1$, equal to one near zero and nonincreasing along rays. Set $\chi_{0,R}(z) = \chi_0(z/R)$. The Taylor estimate at the $Y = 0$ fiber gives, on $\text{Dom } h^{1/2}$,

$$0 \leq \mathbf{1} - \mu(\chi_{0,R}) \leq CR^{-2}h. \quad (3.27)$$

For $\mathcal{E}_R = \mathbf{1} - \mu(\chi_{0,R})$, positivity gives $0 \leq \mathcal{E}_R \leq \mathbf{1}$, hence $\|\mathcal{E}_R \psi\|^2 \leq \langle \psi, \mathcal{E}_R \psi \rangle$. Equation (3.27) therefore gives strong convergence to zero first on $\text{Dom } h^{1/2}$ and then, by density and the contraction bound, on the whole internal space. Monotone convergence and density imply $\mu(\chi_{0,R}) \uparrow \mathbf{1}$ strongly. This proves normalization and excludes escaping mass. It also extends the representation from compact tests to bounded smooth profiles. Indeed positivity and linearity at the $Y = 0$ fiber give

$$|\mathcal{L}(g) - \mu(g\chi_{0,R})| \leq \|g\|_\infty (\mathbf{1} - \mu(\chi_{0,R})).$$

The right side tends strongly to zero; it controls the omitted scalar matrix elements, and polarization completes the extension.

For the precise second-moment constant, put $q_R(z) = |z|^2 \chi_{0,R}(z)$. These bounded smooth profiles increase to $|z|^2$ and satisfy $q_R \leq p_0$. Direct cut inclusion and log order give $h_{q_R} \leq |Y|^2 + h$. At $Y = 0$ this is

$$0 \leq \mu(q_R) \leq h.$$

Monotone convergence defines the densely defined closed second moment form κ , dominated by h . All its bounded spectral approximants commute with h , so κ is affiliated with the same commutative algebra. This proves $\kappa \leq h$ with coefficient one, rather than a Hessian-dependent constant.

Finally let $e \in \mathbb{R}^2$ and $a_R(z) = (e \cdot z)\chi_{0,R}(z)$. Then $a_R(0) = 0$ and $\|a_R''\|_\infty \leq C|e|/R$, so

$$|\mu(a_R)| \leq C|e|R^{-1}h.$$

The second-moment bound makes the first moment integrable on the h form domain: for $\psi \in \text{Dom } h^{1/2}$,

$$\int |z| \langle \psi, \mu(dz)\psi \rangle \leq \|\psi\| \|h^{1/2}\psi\|.$$

Dominated convergence and polarization now give $\int e \cdot z \mu(dz) = 0$. This proves (3.26). Covariance of the construction under changes of transverse axes and dilations follows from covariance of the original cut generators; boost independence has already removed the choice of boost coordinate. $\square$

The measure and its moment have been constructed using one horizon. For the comparison between two horizons we also need them to be independent of whether the bounded tests were taken in the original or the reflected net. The physical Ward formula supplies precisely that identification.

Corollary 3.5 (Reflected agreement). *Repeat the construction with past cuts of the original physical net, or with the twist-corrected reflected net, and reflect back to the same future orientation. The bounded profile functionals, the measure μ , and the quadratic moment K_n agree.*

Proof. Wightman wedge reflection gives the same polynomial vacuum data and the same physical stress-tensor flux on the right side of (3.23). Proposition 3.3 therefore identifies the two bounded-profile forms on opposite polynomial vacuum vectors. Their $b^{1/2}$ graph density and the uniform form bound identify the operators. The compact-profile measure and its monotone moments are consequently identical. This argument identifies the bounded functionals and their quadratic moment; it does not identify the full parabolic modular generator with K_n at this stage. $\square$

4 Two horizons and the scalar spectral comparison

The comparison needed for the trace polarization is stronger than the order-two comparison furnished by its physical differential field. We prove that the quadratic moment of the bounded-profile generators satisfies an order-zero comparison on the vacuum wave l . No local field with vacuum wave l is used in this Section.

The argument has two parts. First, the same causal stress-tensor commutator is evaluated from two null directions. This relates their bounded-profile matrices after all local subtraction terms have been removed. Second, the difference between Hankel orders two and zero is shown to have only one possible tail. A logarithmically long radial test makes the two-horizon identity detect the squared norm of that tail, and positivity forces it to vanish.

We recall precisely the inputs from Sections 2 and 3. On the internal Hilbert space at fixed light-front coordinates, $h \geq 0$ is the parabolic operator and $h_\nu = -\partial_M^2 - M^{-1}\partial_M + \nu^2 M^{-2}$ has its regular Hankel realization in $L^2(\mathbb{R}_+, M dM)$. The comparison (2.26) applies with

constant polarization label	ν	(4.1)
$c_V = l + j_z, \quad c_T = l/3 + s_z$	0	
l, j_z, s_z	2	
j_i, s_{zi}	1.	

The bounded Ward formula (3.23), including its polynomial extension, and the moment representation give

$$H_g = b \int g(Y + z') \mu(dz'), \quad \mu(1) = \mathbf{1}, \quad \mu(z'_i) = 0, \quad \kappa := \mu(|z'|^2) \leq h. \quad (4.2)$$

Here μ is a positive commutative operator-valued measure; its values commute with h , Y and b . It need not be projection-valued. The construction is covariant under dilatations and rotations, independent of the boost coordinate, and agrees with the reflected construction. Consequently, on every copy in (4.1),

$$\mathbf{1}_{(s,\infty)}(\kappa) \mathbf{1}_{[0,s]}(h_\nu) = 0, \quad \|\kappa^t \psi\| \leq \|h_\nu^t \psi\| \quad (t > 0). \quad (4.3)$$

Indeed κ and h commute, so $\kappa \leq h$ combines with the spectral comparison for h . All internal formulas below are Hilbert space formulas or distributions tested on compact output mass intervals; they do not presume integral kernels for μ .

In particular, the comparison in (4.3) retains the embedding of each scalar copy in the ambient internal space. It bounds the spectrum of its image; it does not say that κ preserves that copy. Reduction will be proved only after the charge identities are available.

4.1 The plane matrix and its causal continuation

The bounded Ward identity initially supplies a localized plane matrix. To compare the two horizons we need its causal continuation and a fixed-normal restriction in the particular three-point channel used here. We establish these two operations before carrying out any Cauchy subtraction.

Fix future null vectors $n, \bar{n}$ with $n \cdot \bar{n} = 1$ and use the transverse frame of Section 2. Write

$$\begin{aligned} p &= an + b\bar{n} + w, & p^2 &= M_o^2 > 0, \\ k &= a_x n + q, & p - k &= (a - a_x)n + b\bar{n} + r, & w &= q + r, \\ X &= -k^2 = |q|^2, & z &= -(p - k)^2 = |r|^2 - 2b(a - a_x). \end{aligned} \quad (4.4)$$

The comparison point is $a_x = a$, where both virtualities are strictly spacelike: $X = |q|^2 > 0$ and $z = |r|^2 > 0$.

Lemma 4.1 (The full-plane matrix). *On the spacelike-transfer part of $n \cdot k = 0$, the full T_{nn} matrix integrated along n is the matrix of $b \int g(Y + z') \mu(dz')$. Initially the assertion is a joint distributional identity, averaged over the null direction n and tested in the normal displacement u of the plane. For the stress-tensor three-point matrix with timelike spin-one output used below, the angular average can be removed: n may be fixed. The displacement u remains a distributional variable, whose Fourier transform supplies a_x .*

Proof. Translate the plane normally by u , test u with a compact smooth function, and choose $\hat{g}$ smooth with compact support in a transverse annulus. Average n over a small smooth patch, taking a smooth transverse frame and a sphere normalization of n . Opposite wedge tests can be chosen uniformly for these small changes. Insert a cutoff in the longitudinal coordinate v and split its matrix by the separating step used in (3.23). One term is that Ward formula; the remaining term is an all-plane vacuum matrix. The latter vanishes at spacelike transfer by the vacuum spectrum. The operator on the Ward side at normal coordinate u is $U_a(u) H_g U_a(-u)$.

The removal of the v cutoff has an independent distributional meaning. In momentum space it inserts $2\pi\delta(n \cdot k)$ after averaging over n . On its support, $k = a_x n + q$ with q in the chosen annulus.

The derivative of $n \cdot k$ in an angular direction of n has the nonzero transverse component q . The coarea formula therefore turns the averaged delta insertion into a smooth multiplier. The Fourier transform of the normal test is Schwartz. Derivatives of the moving coordinates cost polynomial factors, absorbed by this decay and the compact annular test. Thus the cutoff multipliers converge in the Schwartz seminorms needed to pair with any of the tempered external field distributions. For an arbitrary fixed list of original fields, these are the finite sums of homogeneous-list distributions established in Section 2; no single scaling dimension is assigned to the whole list.

The identity on opposite wedge tests extends to arbitrary external translations by positive-energy tube uniqueness. On the operator side, $|H_g| \leq \|g\|_\infty b$ supplies the polynomial energy bound: insert positive-energy exponential damping in the two external vectors, obtain the holomorphic matrix elements, and remove the damping as distributional boundary values. Equality on the open wedge patch then determines the boundary distribution everywhere. This establishes the identity jointly with the normal parameters. The fixed-normal pullback needed here is justified in Lemma 4.2. $\square$

Lemma 4.2 (Fixed-spacelike-virtuality dispersion). *Test $[T_{\alpha\beta}(x), T_{\rho\sigma}(0)]\Omega$ against the spin-one output wave, with output measure d^4p on the open timelike cone. At fixed $X > 0$, the contracted matrix in (4.4) is the jump of polynomially bounded functions holomorphic in the upper and lower z half-planes. The jump is the first Wightman order and is supported in $z \leq 0$. At fixed $z > 0$, the corresponding X jump is the reversed Wightman order with the same sign. The two gap values come from one causal splitting of this commutator.*

Proof. Retain compact timelike output tests throughout the construction. Testing against a j label amounts to testing with V and smooth transverse projectors on that compact set. Lorentz and scale covariance trivialize the open timelike orbit, leaving only the finite rest-rotation representation. They identify the resulting distributions in simultaneous moving coordinates. Fixed- p formulas below refer to this covariant disintegration; their estimates are uniform on compact positive output mass ranges.

Local commutativity supports the relative-position commutator in $\bar{V}_+ \cup \bar{V}_-$. Off the origin choose future and past cutoffs constant near these two cones on the unit sphere. Finite distribution order allows their products to be extended through the origin by subtracting a sufficiently long Taylor jet of the test function. This is the cone-supported distribution extension used in causal splitting; see [17, Section 5.1]. Restore the total commutator by a distribution supported at the origin. This gives tempered cone-supported distributions whose difference is the commutator. Average the splitting over rest rotations and transport it along the timelike Lorentz and scale orbit. Its local ambiguity is a finite sum of derivatives of a delta function. With the Fourier convention $D(p, x) = \int e^{ikx} D(p, k) dk$, the resulting boundary values D^+, D^- are holomorphic in the upper and lower primitive tubes, respectively, with polynomial bounds in momentum and inverse distance to the tube boundary [7, Section 1, pp. 155–156].

Here is a direct path from those tubes to the required half-planes. In the rest frame of p , set

$$k^0 = \frac{M_o^2 - X + z}{2M_o}, \quad R^2 = (k^0)^2 + X, \quad \mathbf{k} = R\omega, \quad \omega \in S^2 \text{ real.} \quad (4.5)$$

Write $k^0 = u + iv$, $R = s + it$. Direct calculation gives

$$v^2 - t^2 = \frac{2Xv^2}{u^2 + v^2 + X + |(u + iv)^2 + X|} > 0, \quad |v| - |t| \geq \frac{X|v|}{2(u^2 + v^2 + X)}. \quad (4.6)$$

Thus (4.5) lies strictly in the relevant tube whenever $\text{Im } z \neq 0$. On compact positive X, M_o ranges the distance is bounded below by a constant times $|\text{Im } z|/(1 + |z|)^2$. The tube bounds therefore give polynomial bounds in $|z| + |\text{Im } z|^{-1}$.

The square root in (4.5) introduces no extra branch. A tensor with the fixed finite collection of external indices has only finitely many rotation harmonics. A harmonic of degree s is R^s times a Cartesian harmonic polynomial in ω ; its remaining scalar coefficient is even and analytic in R . At $R = 0$, where the path is still inside a tube, analyticity in the Cartesian momentum components proves regularity of that coefficient. Divide by R^s and reconstruct with Cartesian harmonic polynomials. The scalar coefficients are single-valued functions of the two invariants. They can consequently be evaluated at the actual moving light-front momentum in (4.4), even though that complex momentum itself has null imaginary part. The branch points $k^0 = \pm i\sqrt{X}$ are nonreal; none belongs to the real locus on which a distributional restriction is taken.

We also verify that the real boundary obtained this way is the required restriction. For fixed real ω , $dR/dk^0 = k^0/\sqrt{(k^0)^2 + X}$ has absolute value less than one. The real tangent is strictly timelike. Integration by parts against a compact test interval therefore gives the uniform decay on the dual cone needed for the Fourier–Laplace boundary limit. The harmonic reconstruction gives the moving-direction restriction. There is a useful equivalent noncharacteristic check. The tube boundary distributions have ordinary wavefront covectors in the corresponding causal cones: compact localization in momentum convolves the cone-supported tempered inverse Fourier transform with a Schwartz function, giving rapid decay in directions outside that cone. For the full finite tensor family, rest-rotation covariance gives equations $L_{ij}D = \rho_{ij}D$, where L_{ij} is the angular rotation vector field and ρ_{ij} a constant matrix on the tensor indices. The principal symbols of these equations force every singular covector to annihilate the angular orbit. Since real spacelike k has nonzero spatial part, its annihilator is exactly $\text{span}\{p, k\}$. If $\xi = Ap + Bk$ annihilates the tangent n in (4.4), then $Ab = 0$, so $\xi = Bk$. As $k^2 = -X < 0$, no nonzero such covector is causal. The pullback along $a_x \mapsto a_x n + q$ is therefore noncharacteristic. On compact positive X and timelike output ranges the separation of its conormal bundle from these wavefront cones persists in a small angular patch of n . The restricted distribution thus depends smoothly on that patch, removing the angular average in Lemma 4.1. The normal displacement u is still tested, or Fourier transformed; no point evaluation of an arbitrary distribution in u is needed. This is a restriction statement for this covariant three-point matrix, not for arbitrary null-plane field distributions.

In the first Wightman order the vacuum leg has momentum $p - k$; in the reverse order it has momentum k . At $X > 0$ the reverse order vanishes by the spectrum condition, and the first can occur only for $z \leq 0$. The two holomorphic boundary values glue in the positive z gap. If instead $z > 0$ is fixed and X varies, upper-half-plane continuation corresponds to the lower original tube; its jump is minus the commutator, hence the reverse Wightman order with a positive sign. Both gap values are values of the same split commutator at two spacelike virtualities. This proves the comparison directly from causal tubes, without a crossing assumption. $\square$

4.2 Subtractions and the two-horizon identity

Causal continuation identifies the two spacelike gap values, but a dispersion formula determines each only up to a polynomial. The next step tracks the dependence of these polynomials on the transverse variables. The useful conclusion is their separated form: each ambiguity is polynomial in at least one of the two variables.

Write $[\psi]_{j_A}$ for the output distribution tested against the constant spin-one label in the indicated horizon frame, and put $\mu_n(q) = \int e^{iq \cdot z'} \mu(dz')$ in that frame. Application of Lemma 4.1 to the first jump gives, after testing it with $\phi(z)$, a fixed nonzero common constant times

$$\left[\mu_n(q) (M^2 + |r|^2)^2 c_T^{(\bar{n})}(M, r) \phi(-M^2) \right]_{j_A} (M_o). \quad (4.7)$$

The normalization is worth checking. The plane insertion is $b\mu_n(q)e^{iq \cdot Y}$, conjugated by $U_a(u)$, and the input polarization is $T_{\bar{n}\bar{n}} = a_i^2 c_T^{(\bar{n})}$ with $a_i = (M^2 + |r|^2)/(2b)$. The normal Fourier measure is $da_x = dz/(2b)$. Thus the jump with respect to dz contains $2b^2 a_i^2 = (M^2 + |r|^2)^2/2$; no boost factor remains. The opposite channel has exactly the same constant. These identities can first be tested on external wave packets, so (4.7) means bounded operator application to an internal M dM wave and testing on compact output masses.

Lemma 4.3 (The subtraction ambiguities). *There are continuous $L^2(M \text{ d}M)$ -valued regularizations $\chi_{\text{reg}}^n(r, M)$ of*

$$\chi^n(r, M) = (M^2 + |r|^2) c_T^{(\bar{n})}(M, r) \quad (4.8)$$

such that, with $w = q + r$ and

$$A_A = [\mu_n(q) \chi_{\text{reg}}^n(r)]_{j_A^{(n)}}, \quad B_A = [\mu_{\bar{n}}(r) \chi_{\text{reg}}^{\bar{n}}(q)]_{j_A^{(\bar{n})}}, \quad (4.9)$$

one has

$$(M_o^2 + |w|^2)A_z - (|w|^2 - M_o^2)B_z + 2M_o w_i B_i = P_q + P_r. \quad (4.10)$$

The degree of each polynomial is uniformly finite. The coefficients of P_q , polynomial in q , may depend on r ; those of P_r , polynomial in r , may depend on q .

Proof. Choose a positive subtraction point z_0 and retain the full finite covariant tensor family. Use one integer d larger than all its polynomial growth orders, perform the Cauchy subtraction on that family, and then contract with $nn, \bar{n}\bar{n}$ and the output frame. The subtracted Cauchy kernel is

$$\frac{(z - z_0)^d}{(z' - z_0)^d(z' - z)}.$$

Using (4.7), its value at $z = |r|^2$ is proportional to

$$\left[\mu_n(q) \chi^n(r, M) \left(\frac{|r|^2 - z_0}{-M^2 - z_0} \right)^d \right]_{j_A}. \quad (4.11)$$

Subtracting this Cauchy transform leaves zero jump. Its boundary Fourier transform is supported in both closed half-lines, hence at zero; the remainder is a polynomial in z .

Its dependence on the transverse parameter must also be controlled. A null rotation with parameter β , fixing n , changes

$$r \mapsto r + b\beta, \quad a \mapsto a + \beta \cdot (q + r) + \frac{1}{2}b|\beta|^2, \quad a_x \mapsto a_x + \beta \cdot q.$$

It fixes b, M_o, q, X, z and transports the output adapted frame. The remaining tensor contractions transform polynomially in β , by a matrix independent of z . This matrix commutes with the common-order Cauchy subtraction. Covariance of the split commutator therefore makes the coefficients of its subtraction polynomial polynomial in r at fixed q . Evaluation at $z = |r|^2$ preserves this property. The opposite calculation gives a polynomial in q at fixed r . The degrees are uniform because the distribution order in a rest frame and the degrees of the finite tensor representation are finite. The opposite calculation may hold its own boost coordinate a fixed; the evaluated contracted matrix has net boost weight zero, so it can then be written at the same b as the first calculation.

For completeness the wave regularization can be made explicitly. Set $R_2 = |r|^2$, $D = M^2 + R_2$. The change of spatial frame gives

$$c_T^{(\bar{n})} = l/3 + d_A d_B s_{AB}, \quad d_z = \frac{R_2 - M^2}{D}, \quad d_i = -\frac{2Mr_i}{D}. \quad (4.12)$$

Write $s_{ij}^{\text{TF}} = s_{ij} + \delta_{ij}s_z/2$. Expansion of (4.12) gives

$$\chi^n = \frac{D}{3}l + \left(R_2 - 5M^2 + \frac{6M^4}{D}\right)s_z + \left(-4Mr_i + \frac{8M^3r_i}{D}\right)s_{zi} + \frac{4M^2r_ir_j}{D}s_{ij}^{\text{TF}}. \quad (4.13)$$

In particular the radial coefficient follows from $s_{11} + s_{22} = -s_z$: $((R_2 - M^2)^2 - 2M^2R_2)/D = R_2 - 5M^2 + 6M^4/D$. The vector coefficient is $2Dd_zd_i$.

Choose a smooth mass cutoff ζ equal to zero below 1 and one above 2, and subtract $\zeta(M)$ times

$$\frac{M^2 + R_2}{3}l + (M^2 - 5R_2)s_z + (4Mr_i - 8r_iR_2/M)s_{zi} + 4r_ir_js_{ij}^{\text{TF}}.$$

The remaining large-mass terms are $O(M^{-2})$ or better on compact r sets, and hence belong to $L^2(M \, dM)$. At small mass, $|d| = 1$ in (4.12), so the norm is bounded by a constant times $M^2 + |r|^2$. Dominated convergence now proves continuity also at $r = 0$; the same formulas give polynomial growth in r .

Replacing (4.11) by this regularization changes only an L^2 -valued polynomial in r . Indeed

$$(R_2 - z_0)^d - (-M^2 - z_0)^d$$

is divisible by D , canceling the sole denominator in (4.13). Both regulated expressions are L^2 waves; polynomial interpolation then shows that every coefficient of their polynomial difference is an L^2 wave. Choose the same mass cutoff and subtraction prescription in the other horizon.

Finally the output frame rotation is

$$j_z^{(n)} = \frac{|w|^2 - M_o^2}{M_o^2 + |w|^2} j_z^{(\bar{n})} - \frac{2M_o w_i}{M_o^2 + |w|^2} j_i^{(\bar{n})}.$$

The common gap value from Lemma 4.2 and the common jump normalization therefore give (4.10) off $q = 0$ and $r = 0$.

There is no extra distribution supported on those axes. The left side is continuous in q, r , with values on compact mass tests. Take polynomial interpolation projections $\mathcal{P}_q, \mathcal{P}_r$ of degrees larger than the two degree bounds, using nodes away from zero. On the off-axis patch, $(1 - \mathcal{P}_q)(1 - \mathcal{P}_r)$ annihilates the left side. By continuity it does so everywhere. Decomposing with these two projections extends its separated-polynomial form to all q, r . $\square$

Fix the transverse Fourier convention

$$\widehat{g}(r) = (2\pi)^{-2} \int e^{-ir \cdot z'} g(z') \, d^2 z', \quad g(z') = \int e^{ir \cdot z'} \widehat{g}(r) \, d^2 r.$$

If both position tests vanish near the origin, every term $P_q + P_r$ in (4.10) is annihilated. For a real radial $g \in C_c^\infty(\mathbb{R}^2 \setminus \{0\})$, put

$$\Phi(g) = \int \widehat{g}(r) \chi^n(r, M) \, d^2 r, \quad \mu(g) = \int g(z') \mu(dz'). \quad (4.14)$$

The subtraction makes no difference in this formula.

Indeed, integrating a polynomial in r against $\widehat{g}(r)$ gives a finite linear combination of derivatives of g at zero, all of which vanish. With one such test in each transverse variable, this removes P_q and P_r separately, even though their remaining coefficients can depend on the other variable. The regularized formula supplies the meaning of this operation on the original wave. To perform the

horizon exchange explicitly, let R be the spatial half-turn about e_2 . It sends n to $\bar{n}$, $\bar{n}$ to n , and acts on transverse coordinates by $S = \text{diag}(-1, 1)$. The adapted tetrads obey

$$Re_z^{(n)}(p) = e_z^{(\bar{n})}(Rp), \quad Re_i^{(n)}(p) = S_{ij}e_j^{(\bar{n})}(Rp).$$

Thus the longitudinal label has no extra sign. In the covariantly identified output frames, with the same regularization on both horizons,

$$A_z(q, r) = B_z(Sr, Sq), \quad A_i(q, r) = S_{ij}B_j(Sr, Sq).$$

Write $\langle F \rangle_g = \int \hat{g}(q)\hat{g}(r)F(q, r) d^2q d^2r$. Radiality makes the test invariant under $(q, r) \mapsto (Sr, Sq)$, including the weight $|q + r|^2$. Consequently

$$\langle A_z \rangle_g = \langle B_z \rangle_g =: C_g, \quad \langle |q + r|^2 A_z \rangle_g = \langle |q + r|^2 B_z \rangle_g,$$

and (4.10) reduces to

$$2M_o^2 C_g + 2M_o \sum_i \langle (q_i + r_i) B_i \rangle_g = 0.$$

Our Fourier convention gives $q_i \hat{g}(q) = -i \widehat{\partial_i g}(q)$. Each of the two summands differentiates one test, with the same sign. On returning their vector contraction to the n frame, the transverse matrix S occurs twice and cancels. Division by $2M_o$ gives

$$M_o [\mu(g) \Phi(g)]_{jz} = i \sum_i [\mu(\partial_i g) \Phi(g) + \mu(g) \Phi(\partial_i g)]_{ji}. \quad (4.15)$$

Reversing the Fourier orientation would change the common phase, which will play no role; the relative coefficients are fixed. In particular, (4.13) shows that the radial and vector kernels surviving away from the position origin are $6M^4/(|r|^2 + M^2)$ and $8M^3 r_i/(|r|^2 + M^2)$. The two-dimensional massive Green transform gives respectively $M^4 K_0(M|z'|)$ and a fixed imaginary multiple of $M^4 \hat{z}'_i K_1(M|z'|)$. The scalar and transverse trace-free terms do not contribute to a radial test, nor to its first angular harmonic derivative, as appropriate.

4.3 Exact Hankel leakage

We now isolate the only obstruction to improving the order-two comparison on l to order zero. A test with zero first radial moment has no leakage beyond its Hankel support. It follows that every possible exterior tail depends on that single moment; dilation covariance will determine its exact shape.

Let $\mathcal{H}_\nu$ be the unitary Hankel transform

$$(\mathcal{H}_\nu F)(M) = \int_0^\infty J_\nu(Mv) F(v) v dv.$$

It diagonalizes h_ν as multiplication by v^2 . Put $\lambda = \sqrt{\kappa}$. On its positive spectral part, dilatation covariance permits a homogeneous realization $L^2(\mathbb{R}_+, \lambda d\lambda; \mathcal{K})$ with constant multiplicity space $\mathcal{K}$. Its zero spectral subspace is kept separately. Write $\mathcal{U}$ for this spectral realization, and f_c for the input radial wave with label c .

Lemma 4.4 (Exact tail). *For $c = l, j_z, s_z$ there is $d_c \in \mathcal{K}$ such that, if $F \in C_c^\infty(0, R)$, then*

$$\mathcal{U}(c\mathcal{H}_0 F)(\lambda) = \left(\int_0^\infty v F(v) dv \right) \lambda^{-2} d_c \quad \text{for almost every } \lambda > R. \quad (4.16)$$

The label map is linear, and

$$d_{j_z} = -d_l, \quad d_{s_z} = -d_l/3. \quad (4.17)$$

Proof. The Bessel recurrence $J_2(x) = 2J_1(x)/x - J_0(x)$ and $\int_0^v r J_0(Mr) dr = v J_1(Mv)/M$ give

$$(\mathcal{H}_2 \mathcal{H}_0 F)(v) = \frac{2}{v^2} \int_0^v r F(r) dr - F(v). \quad (4.18)$$

The identity first follows on smooth compact tests by the unitary Hankel inversion and then in $L^2(v dv)$. If F has zero first radial moment and support below R , the right side also has support below R . The order-two comparison in (4.3) therefore puts $c\mathcal{H}_0 F$ in the $\lambda \leq R$ subspace.

Consequently the restriction of the transfer to any interval strictly above R factors through the single functional $F \mapsto \int v F(v) dv$. Choose one annular test with this moment equal to one. Overlapping support bounds identify the resulting vector-valued coefficient at every $\lambda > 0$. In these coordinates internal dilatations act as $f(M) \mapsto e^s f(e^s M)$ on the input and $\psi(\lambda) \mapsto e^{-s} \psi(e^{-s} \lambda)$ on the output. Since the moment of the corresponding Hankel-side test scales by e^s , its coefficient must be $\lambda^{-2} d_c$. A reference test and its square-integrable tail ensure $d_c \in \mathcal{K}$. This proves an exact tail, not only its asymptotic order.

The order-zero comparisons on c_V and c_T say that $c_V \mathcal{H}_0 F$ and $c_T \mathcal{H}_0 F$ have no spectrum above the support radius of F . Linearity of the tail gives $d_l + d_{j_z} = 0$ and $d_l/3 + d_{s_z} = 0$. $\square$

4.4 The logarithmic pairing

The tail relations alone allow a nonzero vector d_l . We will insert a radial test covering many scales into (4.15). Its two scalar-channel tails produce a term proportional to $\log L \|d_l\|^2$, whereas the first angular harmonics give smaller terms. The estimates below keep this leading coefficient and the errors separate.

Choose $0 \leq \eta_L \leq 1$, smooth and supported in $(0, \log L)$, equal to one outside fixed-width neighborhoods of the endpoints, with uniformly bounded derivatives. Set

$$g_L(z') = |z'|^2 \eta_L(\log |z'|). \quad (4.19)$$

We retain the nonzero normalization constant c_* throughout the estimates; it is determined by the radial Fourier kernel.

Lemma 4.5 (Plateau estimates). *There are scalar functions f_L, u_L such that*

$$\Phi(g_L) = s_z f_L(M), \quad \Phi(\partial_i g_L) = M s_{zi} u_L(M), \quad (4.20)$$

and, writing $\mathcal{D} = M \partial_M$,

$$\|f_L - c_* \mathbf{1}_{(1/L, 1)}\|_{L^2(dM/M)} = O(1), \quad c_* \neq 0, \quad (4.21)$$

$$\|\mathcal{D}^a f_L\|_{L^2(dM/M)} + \|\mathcal{D}^a u_L\|_{L^2(dM/M)} = O(1) \quad (a = 1, 2), \quad (4.22)$$

$$\|\kappa \Phi(\partial_i g_L)\| = O(1), \quad \|f_L/M\|_{L^2(M dM)} = O(\sqrt{\log L}). \quad (4.23)$$

For $c = s_z, j_z$, in the positive λ representation,

$$\lambda \mathcal{U}(c f_L) = c_* \mathbf{1}_{(1, L)}(\lambda) \lambda^{-1} d_c + E_{c, L}, \quad \|E_{c, L}\|_{L^2(\lambda d\lambda; \mathcal{K})} = O(1). \quad (4.24)$$

Proof. The kernels computed after (4.15), with $t = M|z'|$, give explicitly in our transverse Fourier convention

$$f_L(M) = 6 \int_0^\infty t^3 K_0(t) \eta_L(\log t - \log M) dt, \quad (4.25)$$

$$u_L(M) = -4i \int_0^\infty t^2 K_1(t) (2\eta_L + \eta'_L)(\log t - \log M) dt. \quad (4.26)$$

Changing the Fourier orientation changes the fixed phase in the second formula only. The first has

$$c_* = 6 \int_0^\infty t^3 K_0(t) dt = 24$$

with the convention fixed above. In particular it cannot vanish. On the logarithmic variable these are convolutions with integrable kernels having finite first moment. At $t = 0$ their densities decay exponentially in $|\log t|$, and at infinity they decay faster; the relevant logarithmic derivatives have the same integrability. Convolution of the interval plateau differs from its kernel integral times that plateau by a uniformly bounded L^2 error: the difference is supported near the two edges up to integrable tails. Positive derivatives act on these two fixed-width edges. This proves (4.21)–(4.22). The functions themselves and their derivatives have integrable tails at zero and infinity for each fixed L .

Direct differentiation gives

$$h_1(Mu) = -M^{-1}(\mathcal{D}^2 + 2\mathcal{D})u, \quad \|h_1(Mu)\|^2 = \int_0^\infty |\mathcal{D}^2 u + 2\mathcal{D}u|^2 \frac{dM}{M}.$$

The order-one comparison proves the first estimate in (4.23); the second is (4.21).

To prove (4.24) without assigning a divergent form norm to an individual nonzero-moment test, fix a smooth annular F with $\int vF(v) dv = 1$ and put $G = \mathcal{H}_0 F$. Replace f_L by

$$c_* \mathcal{H}_0(F - L^{-2}F(\cdot/L))(M) = c_*(G(M) - G(LM)).$$

Since $G(0) = 1$ and G decreases rapidly at infinity, this difference has the same plateau and the same uniform edge estimates as f_L . Its error e_L satisfies

$$\|\sqrt{h_2}e_L\|^2 = \int_0^\infty (|\mathcal{D}e_L|^2 + 4|e_L|^2) \frac{dM}{M} = O(1).$$

The comparison (4.3) gives an $O(1)$ error after multiplication by λ . Above a fixed radius, the G term has the exact tail of Lemma 4.4. The norm of $G(LM)$ in $L^2(M dM)$ is $L^{-1} \|G\|$; hence its λ -weighted norm below CL is $O(1)$ for any fixed C . Above both Hankel support radii the two exact tails cancel, because their moments are both one. The remaining fixed lower interval and fixed-ratio upper interval contribute $O(1)$ to the weighted norm. This proves (4.24). The $\lambda = 0$ spectral subspace contributes nothing to that estimate.

Using the difference of the two dilated tests is essential here: their individual λ^{-2} tails need not have finite λ -weighted norm, but the exact cancellation above the larger support radius makes their difference admissible. No estimate has been assigned to either divergent weighted tail separately. $\square$

Proposition 4.6 (Order-zero scalar comparison). *The quadratic moment κ of the physical bounded-profile generators satisfies*

$$\mathbf{1}_{(s,\infty)}(\kappa)\mathbf{1}_{[0,s]}(h_0)|_l = 0 \quad (s \geq 0). \quad (4.27)$$

Proof. On the positive λ spectrum define

$$A_L(\lambda) = \lambda^{-2}\mu(g_L), \quad B_{i,L}(\lambda) = \lambda^{-1}\mu(\partial_i g_L).$$

The moments in (4.2) imply $0 \leq A_L \leq \mathbf{1}$ and $\|B_{i,L}\| \leq C$ uniformly: $|\partial_i g_L(z')| \leq C|z'|$, and positive-measure Cauchy–Schwarz bounds its integral by $C\sqrt{\kappa}$. Scale covariance identifies the law of z'/λ in each fiber with one fixed positive measure on $\mathcal{K}$. Its mean is zero and its trace-second moment is $\mathbf{1}$. Hence, on every fixed multiplicity vector,

$$A_L(\lambda) \longrightarrow \mathbf{1}, \quad B_{i,L}(\lambda) \longrightarrow 0 \quad \text{if } \lambda \longrightarrow \infty, \quad L/\lambda \longrightarrow \infty. \quad (4.28)$$

For the first limit use the integrable weight $|z'/\lambda|^2$; for the second use $|z'/\lambda|$ and the zero first moment. The limiting derivative integrand is $2z'_i/\lambda$. Bounded convergence for the positive measure proves strong convergence; no uniform rate is required.

These are statements about the fixed multiplicity vectors that enter the tail formula. Positivity and the second moment control the truncated integrands; projection-valuedness of μ is unnecessary. On $\ker \kappa$ the second moment is zero, so μ is concentrated at $z' = 0$ there. The profiles g_L and $\partial_i g_L$ vanish near that point, and hence this spectral subspace makes no contribution to the pairing.

Pair (4.15) with $f_L(M_o)/M_o$ and use (4.20). This is allowed although the identity was first tested on compact mass intervals. Cut off this test smoothly at zero and infinity. The approximants converge both in its $L^2(M_o dM_o)$ norm and after multiplication by M_o . For fixed L , the profile operators are bounded, and (4.23) controls the derivative wave, so every pairing has a limit.

The left side becomes

$$\begin{aligned} \langle j_z f_L, \mu(g_L) s_z f_L \rangle &= \langle \lambda \mathcal{U}(j_z f_L), A_L \lambda \mathcal{U}(s_z f_L) \rangle \\ &= |c_*|^2 \int_1^L \langle d_{j_z}, A_L(\lambda) d_{s_z} \rangle \frac{d\lambda}{\lambda} + O(\sqrt{\log L}). \end{aligned}$$

The exact normalization $|c_*|^2$ is common to the bra and ket. Equation (4.28) and boundedness imply, by averaging on the logarithmic interval, that this is

$$|c_*|^2 \log L \langle d_{j_z}, d_{s_z} \rangle + o(\log L). \quad (4.29)$$

For example, discard two logarithmic edge intervals of fixed width, apply the strong bulk convergence on the remaining interval, and then let the discarded width grow. This proves the asserted Cesaro limit directly.

For the first term on the right of (4.15), (4.28), (4.24) and uniform boundedness give

$$\|\mu(\partial_i g_L) s_z f_L\| = \|B_{i,L} \lambda \mathcal{U}(s_z f_L)\| = o(\sqrt{\log L}).$$

The output test $j_i f_L/M_o$ has norm $O(\sqrt{\log L})$. Their pairing is therefore $o(\log L)$. For the last term, $0 \leq \mu(g_L) \leq \kappa$ and commutativity give

$$\|\mu(g_L) \Phi(\partial_i g_L)\| \leq \|\kappa \Phi(\partial_i g_L)\| = O(1).$$

Its pairing is $O(\sqrt{\log L})$. Comparison with (4.29) forces

$$0 = \langle d_{j_z}, d_{s_z} \rangle = \frac{1}{3} \|d_l\|^2,$$

by (4.17). Thus $d_l = 0$. If either of the spin labels used above is zero, the same conclusion already follows from (4.17).

Lemma 4.4 now says that $l\mathcal{H}_0 F$ has no λ spectrum above R whenever F is supported below R . Smooth compact tests in $(0, R)$ are dense in $L^2((0, R), v dv)$. Hankel unitarity and closedness of spectral subspaces give (4.27) with $s = R^2$; at $s = 0$ the regular h_0 copy has no zero eigenvector. The result concerns the moment κ . It does not identify the full parabolic operator h with that moment. $\square$

5 Single insertions, quadratic charges, and time cones

Throughout this Section, l is the dimension-two vacuum wave constructed in Section 2. The letter L will initially denote only an insertion in a sesquilinear Wightman form. Neither an operator $L(f)$ nor a product containing two copies of L is being asserted. This distinction is essential when conserved currents are used below.

The section proceeds through three distinct constructions. We first extend the extreme vacuum values of l to local forms with one insertion. Their conserved currents then define charge forms, whose quadratic part is identified with the positive moment K_n on a specified domain. Finally the vanishing residual dilation charge identifies the actual cone modular groups and produces the actual time-Möbius action used for scalar extraction.

5.1 From scalar localization to one local insertion

The order-zero spectral comparison has a localization interpretation. There are two transports: from the scalar comparison to the moment κ , and from that moment to a physical bounded cut. We state both, together with the reverse containment for actual parabolic cuts that will be needed later.

Lemma 5.1 (Localization transport). *Let $X_0(W)$ be the scalar wedge real space in the l channel, ι its embedding into the physical Hilbert space, and $X(W)$ the physical wedge real space. For $p(z) = c|z - d|^2 + c_0$, $c > 0$, put*

$$K_{0,p} = b\{c(|Y - d|^2 + h_0) + c_0\}, \quad K_p = b\{c(|Y - d|^2 + \kappa) + c_0\}.$$

Then

$$\iota e^{iK_{0,p}} X_0(W) \subset e^{iK_p} X(W). \quad (5.1)$$

If g is bounded and smooth and $g \leq p$, the left side also lies in $X(W_g) = e^{iH_g} X(W)$. Conversely, the actual parabolic real space satisfies

$$X(W_p) \subset e^{iK_p} X(W). \quad (5.2)$$

These statements hold after translations of the null plane and reflection.

Proof. The scalar wedge conjugation is the real scalar conjugation; it is identified already on the dense waves generated by $M^2 l$. To prove (5.1), cancel the common external factors and consider $e^{-icb\kappa} \iota e^{icbh_0}$. Let E_i be the actual κ step projections and P_j the reference h_0 step projections. Clip both spectral variables at one common finite bound, then approximate the clipped κ from below and the clipped h_0 from above. If their step values are a_i, c_j , spectral inclusion gives $E_i \iota P_j = 0$ whenever $a_i > c_j$. Each surviving ordered term is therefore $E_i \iota P_j e^{icb(c_j - a_i)}$, with nonnegative translation parameter; no joint spectral resolution is used. The coefficients intertwine the boost and wedge conjugation, so they map the reference real wedge space into the actual one. The positive

translations preserve that actual space. At the common clipping endpoint both step values coincide, which preserves the same sign. Strong convergence follows from the separate spectral calculi as the mesh tends to zero and the clipping bound tends to infinity. Closedness of $X(W)$ proves the inclusion. This uses the ordered products of two separate spectral resolutions, not a joint resolution of κ and h_0 .

For the second transport, positivity, normalization, and the zero first moment of μ give

$$H_g = b \int g(Y + z) \mu(dz) \leq b\{c(|Y - d|^2 + \kappa) + c_0\} = K_p.$$

This inequality is in the common commuting calculus. Its difference is b times a nonnegative boost-independent operator commuting with wedge conjugation. Thus $e^{i(K_p - H_g)}$ is a direct integral of inward null translations and preserves $X(W)$. Apply it to (5.1) to obtain membership in $X(W_g)$.

Finally the actual parabolic generator is $H_p = b\{c(|Y - d|^2 + h) + c_0\}$. Since h and κ commute and $\kappa \leq h$, the same argument applies to $H_p - K_p = bc(h - \kappa) \geq 0$ and proves (5.2). The operators vanish on $b = 0$. Covariance and reflected agreement give the remaining placements. No equality of h with κ is used. $\square$

Here is the resulting extreme-ordering statement. First let $A_i(y_i)$ be homogeneous original physical fields. The two expressions with $L(x)$ at the extreme left or extreme right are defined by pairing the wave l with a physical polynomial vector. They agree whenever x is spacelike to all y_i and, for some future null n ,

$$n \cdot x < \min_i n \cdot y_i \quad \text{or} \quad n \cdot x > \max_i n \cdot y_i. \quad (5.3)$$

For the first alternative, place a null plane between x and the y_i and use the future shadow of a tip just before x as a dominating parabola. The past shadows of the y_i lie strictly below it by spacelikeness. Their differences from it tend quadratically to $-\infty$ at transverse infinity, so a bounded smooth truncation from below still separates the supports. These strict inequalities persist in small neighborhoods. Locality of bounded operations in that cut, followed by Tomita adjunction of the whole physical product, proves the equality. The opposite alternative uses the reflected construction. The scalar insertion is even, so no additional graded sign occurs. For a permitted compact joint kernel, the same statement follows by the tensor-test approximation and closed Tomita-graph limit in the discussion of (2.2); it does not require an affiliated realization for the joint-smeared polynomial itself.

The remaining task is to place this insertion between arbitrary physical factors while retaining their ordered spectra. Causal solutions at adjacent interchanges will have the required wave source, but their compatibility leaves a homogeneous-wave defect. The next lemma recovers such a defect from its support and spectator analyticity. We then apply it to assemble the insertion with all its ordering cones.

Lemma 5.2 (Recovery of a homogeneous wave defect). *Let $H(x, y_1, \dots, y_m)$, $m \geq 1$, be a tempered, jointly translation-invariant distribution with $\square_x H = 0$ and*

$$\text{supp}_x H \subset C_y \cup K_y, \quad C_y = \bigcup_i J(y_i), \quad K_y = \bigcap_{n \text{ future null}} \{x : \min_i n \cdot y_i \leq n \cdot x \leq \max_i n \cdot y_i\}. \quad (5.4)$$

For a compact transfer-momentum test $\varphi(k)$, let $H_\varphi(y)$ be its pairing with the partial Fourier transform in x , using e^{-ikx} . Suppose these pairings are holomorphic in the ordered spectator tube and, for $\text{supp } \varphi \subset \{|k| \leq K\}$, $0 < K \leq 1$, obey

$$|H_\varphi(y)| \leq C p_N(\varphi) (1 + |y| + \delta(y)^{-1})^N \exp(CK |\text{Im } y|). \quad (5.5)$$

Here p_N is one fixed finite Schwartz seminorm and $\delta(y)$ is the minimum distance of the successive imaginary gaps from the cone boundary ($\delta = 1$ for $m = 1$). Then H is a finite sum

$$H(x, y) = \sum_{i, P} C_{iP}(y) P(\partial_x) \Delta_0(x - y_i),$$

where the coefficients have tempered boundaries with the ordinary ordered spectator spectra. They are invariant under common spectator translation. The polynomials P may be chosen independently modulo k^2 . We normalize the massless commutator solution by $\Delta_0(0, \mathbf{r}) = 0$ and $\partial_0 \Delta_0(0, \mathbf{r}) = \delta(\mathbf{r})$.

Proof. Normalizing $n = (1, \omega)$ and taking $\omega = \pm e_a$ shows that K_y is compact, uniformly on compact spectator sets. We first remove its part outside C_y , then recover the coefficients of the remaining wave solutions with bounds that preserve the unshifted spectator spectra.

We give the support argument explicitly. Choose an open spectator patch around distinct equal-time spatial extremal vertices, so that each y_i is a strict maximum and a strict minimum for open sets of future null directions. If $x \in K_y \cap J^+(y_i)$, the maximizing directions give $0 \leq n \cdot (x - y_i) \leq 0$. A causal vector orthogonal to an open set of null directions is zero. The minimizing directions treat $J^-(y_i)$. Hence

$$K_y \cap C_y = \{y_1, \dots, y_m\} \quad (5.6)$$

on a smaller patch. Fix a test neighborhood U spacelike to all the vertices, and choose small vertex neighborhoods N_i whose closures remain spacelike to U . There is a compactly supported smooth $\chi(x)$ equal to one near K_y whose derivatives meet C_y only in $N = \bigcup_i N_i$. Positive separation margins make one such cutoff work uniformly after shrinking the spectator patch. A compact spectator cutoff causes no change to $\square_x$. Thus $v = \chi H$ has compact x -support and

$$\square_x v = [\square_x, \chi] H =: f, \quad \text{supp}_x f \subset N.$$

For compactly supported distributions the exact identity $v = E_{\text{ret}} * \square_x v$ follows by moving $\square$ through convolution. Its right side vanishes on U , since $J^+(N) \cap U = \emptyset$. Near K_y one has $v = H$; elsewhere the original support statement already suffices. This removes $K_y \setminus C_y$ from (5.4), without any decomposition of distributions on touching closed sets.

In a common thin time slab the cone components are disjoint. A spatial partition separates them with derivatives off their supports. Each component still solves the wave equation there. In $r = x - y_i$ coordinates its displacement and normal-derivative data on $r^0 = 0$ are supported at $\mathbf{r} = 0$. Distributional solutions of the wave equation have distributional Cauchy data on this non-characteristic slice. The point-support structure theorem therefore writes both data as finite sums of spatial delta derivatives with spectator-distribution coefficients. On a smaller relatively compact patch their orders are uniformly bounded. With $\Delta_0(0, \mathbf{r}) = 0$ and $\partial_0 \Delta_0(0, \mathbf{r}) = \delta(\mathbf{r})$, their evolutions are derivatives of Δ_0 . Cauchy uniqueness first in the slab and then globally in x gives

$$H(x, y) = \sum_{i, P} C_{iP}(y) P(\partial_x) \Delta_0(x - y_i). \quad (5.7)$$

Choose the finitely many polynomials P independently modulo k^2 . Enlarging the finite family if necessary, use homogeneous bases $P_{q\alpha}$ of the quotient by k^2 in each degree $q = 0, \dots, N$.

We next justify continuation of the coefficients, including its growth bound. This is needed to recover the unshifted spectator spectra. Smear the Fourier transform of $P(\partial_x) \Delta_0(x - y_i)$ against compact k -tests. Its column is, up to constant factors,

$$\langle P(k) \hat{\Delta}_0(k) e^{iky_i}, \varphi(k) \rangle.$$

All augmented minors expressing that the corresponding tests of H lie in the span of these columns vanish on the initial patch. They are holomorphic in the ordered spectator tube; boundary uniqueness makes them vanish throughout that tube. The columns have full rank at every interior point, with a quantitative bound as follows. Choose finitely many fixed positive null vectors $n_r = (1, \omega_r)$ such that, for every $q \leq N$, the evaluation matrix

$$(E_q)_{r\alpha} = P_{q\alpha}(n_r)$$

is injective. Such a choice exists: a homogeneous polynomial vanishing on every positive null ray is zero modulo k^2 , and finite dimensionality permits a finite separating set of evaluations. Taking the union of these sets for $q \leq N$ gives one collection of rays. Fix a left inverse for each E_q .

On the r th ray the polynomial-exponential column sum is

$$\sum_{i=1}^m \sum_{q=0}^N a_{iqr} s^q e^{i s n_r \cdot y_i}, \quad a_{iqr} = \sum_{\alpha} C_{i,q\alpha}(y) P_{q\alpha}(n_r).$$

Let $d = m(N+1)$, $\varepsilon = c/(1+|y|)$, and use the d samples $s = t\varepsilon$, $t = 1, \dots, d$. The matrix from the coefficients a_{iqr} to these samples is

$$(V_r)_{t,(i,q)} = (t\varepsilon)^q z_{ir}^t, \quad z_{ir} = e^{i\varepsilon n_r \cdot y_i}.$$

Its determinant is the confluent Vandermonde determinant

$$\det V_r = c_{m,N} \varepsilon^{mN(N+1)/2} \prod_i z_{ir}^{(N+1)(N+2)/2} \prod_{i < j} (z_{jr} - z_{ir})^{(N+1)^2}, \quad c_{m,N} = \left(\prod_{q=0}^N q! \right)^m. \quad (5.8)$$

For example this follows by replacing powers of t by falling factorials and using the derivative columns of the ordinary Vandermonde matrix. Thus the ray inversion distinguishes the degree q and center i before the fixed left inverses of E_q recover each coefficient $C_{i,q\alpha}$. This describes a left inverse for the full rectangular multi-ray matrix; independence on each ray alone would not suffice.

The scale of the samples is also significant. Keeping them at fixed nonzero momenta would preserve the exponential type of the shifted spectator cones. Taking $s = O((1+|y|)^{-1})$ instead keeps that exponential bounded. The determinant estimate ensures that this shrinking choice costs only polynomial powers, which are compatible with tempered spectral boundary values.

For the tube distance $\delta(y)$ in the statement, the moduli of the z_{ir} are bounded above and below, and for $i \neq j$

$$|z_{ir} - z_{jr}| \geq ||z_{ir}| - |z_{jr}|| \geq c'\varepsilon\delta(y) \quad (i \neq j),$$

because the imaginary gaps are strictly ordered and the directions are fixed. The determinant formula and cofactor bounds therefore give the full left inverse polynomial norm in ε^{-1} and $1+\delta(y)^{-1}$. To turn these point samples of the known columns into legitimate compact k -tests, replace each sample $t\varepsilon n_r$ by a smooth bump of width w , normalized to have integral one on the positive null shell. The centers are a distance comparable to ε from its vertex. For $w \leq c_1\varepsilon$, its shell Jacobian and the prescribed test derivatives therefore give

$$p_N(\varphi) \leq C\varepsilon^{-a} w^{-b}$$

for fixed finite a, b . Derivatives of the column functions $P(k)e^{ik \cdot y}$ cost polynomial powers of $1+|y|$ and ε^{-1} , while their exponential is bounded on these supports. If A_0 is the point-sample matrix and $L_0 A_0 = 1$, choose

$$w = c_2 \varepsilon (1 + |y| + \delta(y)^{-1})^{-J}$$

with fixed J sufficiently large and c_2 sufficiently small. The resulting test-column matrix A then satisfies $\|L_0(A - A_0)\| < 1/2$. Its left inverse $(L_0 A)^{-1} L_0$ has the same polynomial bound, and the displayed seminorm cost remains polynomial.

This proves full rank using genuine test columns. On a neighborhood of any fixed interior point, keep a full-rank finite set of tests fixed; its inverse and the vanishing augmented minors define holomorphic coefficients. Uniqueness identifies them on overlaps. The parameter-dependent test choices above serve only to estimate these already defined holomorphic functions. Their supports satisfy $|k| = O(\varepsilon)$ and their seminorms have polynomial cost. Applying (5.5) with $K = O(\varepsilon)$ therefore costs only polynomial powers: the remaining exponential is bounded by $\exp(O(\varepsilon |\operatorname{Im} y|)) = O(1)$. The left inverse consequently gives

$$|C_{iP}(y)| \leq C(1 + |y|)^N (1 + \delta(y)^{-1})^N \quad (5.9)$$

for some C, N on the tube. The coefficients have ordinary tempered ordered spectral boundaries. Testing all compact k -sets, including a neighborhood of zero, proves (5.7) globally. Translation covariance also shows that the coefficients are invariant under common spectator translation. $\square$

The construction is first carried out for a fixed list of homogeneous original entries. We will then extend it by finite multilinearity in their scaling components. Descent to polynomial vectors will be checked for the whole summed bra or ket, so cancellations between different homogeneous terms remain allowed.

Proposition 5.3 (One-insertion forms). *There are unique tempered, Hermitian, Poincaré- and scale-covariant Wightman forms with exactly one insertion L and arbitrarily many physical insertions, whose extreme vacuum values are those of l and which satisfy $\square L = \Theta$. They have the ordinary ordered spectra, are local under every allowed adjacent interchange, and depend only on the physical polynomial vectors on the two sides of L .*

Proof. Causal source solutions. The empty physical product has only the specified zero vacuum expectation, since l has no zero-momentum component. Suppose $m \geq 1$ and fix an order of homogeneous original entries $A_i \in \mathcal{F}_{\Delta_i}$, $i = 1, \dots, m$. Use the Fourier convention $e^{-ikx - i \sum_i p_i y_i}$, and put $q_h = \sum_{i \leq h} p_i$, with $q_0 = 0$. For an insertion just after A_j , the required spectral conditions are

$$q_h \in \bar{V}_+ \quad (1 \leq h \leq j), \quad k + q_h \in \bar{V}_+ \quad (j \leq h \leq m-1), \quad k + q_m = 0. \quad (5.10)$$

Redundant zero partial sums at the endpoints may be included. Write E_R, E_L for the known extreme values. For each i , take the difference of the physical distributions with Θ immediately before and immediately after A_i . Its support, in $r = x - y_i$, is causal. Split it into forward and backward parts by a finite-order Taylor subtraction in r alone, and apply respectively the retarded and advanced wave fundamental solutions. This gives a tempered B_i with the same causal support and the prescribed wave source. The convolutions are proper on the respective cones: two future causal vectors, or two past causal vectors, are bounded on the inverse image of a bounded set under addition. The same cone geometry gives the polynomial bounds needed for tempered convolution.

This construction preserves precisely the common spectral cones

$$q_h \in \bar{V}_+ \quad (h < i), \quad k + q_h \in \bar{V}_+ \quad (h \geq i). \quad (5.11)$$

Indeed in the coordinates $(r, y_1, \dots, y_m)$ the spectator momenta are p_j for $j \neq i$ and $k + p_i$ for $j = i$. All the conditions in (5.11) involve only those spectator momenta. Multiplication, subtraction, and

convolution in r do not alter them. This is why solving the wave equation separately in the external momentum before doing causal splitting would be insufficient.

Removing the homogeneous defect. Set

$$R_j = E_R + \sum_{i>j} B_i, \quad L_j = E_L - \sum_{i\leq j} B_i.$$

Both have the wave source appropriate to the j th position. The first has all the initial cones in (5.10); the second has all its final cones. Their difference $H_* = R_j - L_j$ is independent of j and satisfies $\square_x H_* = 0$. After a compact Fourier test in k , its spectator partial sums lie in $\bar{V}_+$ up to bounded shifts of size $O(\sup |k|)$. It consequently has an analytic continuation to the ordinary ordered spectator tube, with polynomial bounds multiplied by an exponential of type $O(\sup |k|)$. Equation (5.3) and the causal support of the B_i give the support condition $\text{supp}_x H_* \subset C_y \cup K_y$ of (5.4).

Its shifted spectator cones also give (5.5): for a transfer test supported in $|k| \leq K$, every partial cone is shifted by at most a fixed multiple of K . Fourier–Laplace damping contributes $\exp(CK|\text{Im } y|)$; the finite-order tempered bound on the joint distribution controls the remaining terms by one fixed seminorm $p_N(\varphi)$ and polynomial powers of $|y|$ and the inverse tube distance. This estimate is uniform in the test, including when its support shrinks toward $k = 0$. Lemma 5.2 therefore gives the expansion (5.7) for H_* .

Let H_i denote the summand centered at y_i . In Fourier variables it shifts p_i by k , and no other spectator momentum. Consequently it satisfies exactly the common cones (5.11). The required distributions are

$$W_j = R_j - \sum_{i>j} H_i = L_j + \sum_{i\leq j} H_i. \quad (5.12)$$

The first expression supplies the initial cones and the second the final cones; $W_m = E_R$, $W_0 = E_L$, and $W_j - W_{j-1} = -B_j + H_j$ is supported in $J(y_j)$.

Locality and descent to polynomial vectors. For an adjacent interchange of physical fields in the suffix, smear all suffix variables with any compact test supported where that pair is spacelike. Keep this test fixed. The two orderings have the same partial spectral cones in the prefix and x : $q_h \in \bar{V}_+$ for $h \leq j$ and $q_j + k \in \bar{V}_+$. Smearing the suffix does not change those cones. Their difference is therefore the boundary of a holomorphic function in the ordered tube of the $j + 1$ variables $y_1, \dots, y_j, x$, with the last imaginary difference measured from the real boundary. On the nonempty open set where x is spacelike to every prefix point, the already proved adjacent L interchanges move it to the extreme left. At that extreme, physical locality makes the fixed suffix swap zero, regardless of the prefix. Boundary uniqueness in this partial tube makes the difference zero for all prefix and x variables. Since the fixed suffix test was arbitrary on the spacelike-pair patch, this proves the required distributional locality. For a swap in the prefix, hold its test fixed instead and move L to the extreme right. The common final cones, equivalently the negative reversed partial sums by total momentum conservation, give the opposite ordered tube in x and the suffix variables. The same argument applies.

For a fixed homogeneous list, locality moves L to an extreme position on a spacelike open set. Ordered tube uniqueness therefore shows that the specified extreme values determine the form uniquely. In particular, the result is multilinear in the homogeneous labels and is covariant, independently of the auxiliary causal splittings. For arbitrary original entries write

$$A_i = \sum_{\Delta \in S_i} A_{i,\Delta}, \quad |S_i| < \infty,$$

and take the finite sum of the constructed forms over $S_1 \times \cdots \times S_m$, using the same joint test kernel in every term. The common ordered spectral cones, locality, wave equation, and extreme pairings survive this finite sum. Its dilation covariance is the termwise action on the labels, not homogeneity of one total degree. The distribution orders and constants may depend on this fixed finite list.

The vector-dependence assertion uses exactly the preceding choice of which variables to hold fixed, now for the whole finite summed form. If a fixed ket polynomial $A\Omega$ is zero, even by cancellation between component terms or word lengths, vary only the opposite prefix and x . Once the entire ket has been smeared and held fixed, the finite sum has the same prefix-and- x tube, regardless of the word lengths on that fixed side. On the open set where the L interchanges through that prefix are spacelike, the extreme-left value pairs l with $C^*A\Omega = 0$. The partial-tube argument just given makes the insertion zero everywhere. It never requires a null polynomial vector to remain zero under independent translations of its individual factors. If the bra polynomial is zero, hold it fixed and use the extreme right and the opposite tube. For a permitted joint kernel, distributional continuity first supplies the summed form and its common partial-tube boundary for that fixed whole kernel. The zero extreme pairing is then used for the whole kernel, not for its approximants. Hermiticity and the stated covariance follow from uniqueness of the whole form. This proves a statement about forms with one L , without a positivity assertion for repeated insertions.

In particular, the construction may now be differentiated in the L slot and paired with arbitrary physical polynomial vectors. These are precisely the operations needed to form the conserved currents below; none requires composing two L insertions. $\square$

5.2 Conserved charge forms and their vacuum flux

Use Proposition 5.3 to define the conserved one-insertion forms

$$t_{\mu\nu} = T_{\mu\nu} + \frac{1}{3}(\partial_\mu\partial_\nu - \eta_{\mu\nu}\square)L, \quad J_\mu = V_\mu - \partial_\mu L. \quad (5.13)$$

Thus t is symmetric and traceless and $\partial^\mu J_\mu = 0$. For a conformal Killing vector X , $X^\nu t_{\mu\nu}$ is conserved. Its commutator flux through a compact Cauchy slab around a compact physical polynomial A , paired with $C\Omega$, defines a charge form $q_X(C, A)$. The cutoff equals a time step on the relevant causal support. Locality and conservation permit compact deformations of this slab. To move the charge from ket to bra requires an additional statement: a complete vacuum flux through a physical product must vanish even when X is quadratic.

The compact commutator flux and this complete vacuum flux have different roles. Conservation and locality make the former independent of a compact deformation. The latter is a limit over increasing scales and requires an infrared estimate. We establish that estimate uniformly in the mass band so that the bands can subsequently be removed.

Lemma 5.4 (Low-energy estimate). *Let Z be an even Hermitian compact polynomial vacuum vector, smooth under translations. For an invariant-mass spectral band $I \in (0, \infty)$ and $\Lambda > 0$,*

$$\|1_{P^0 < \Lambda} E_I Z\| \leq C_Z \Lambda \langle Z, P^0 E_I Z \rangle^{1/2}, \quad (5.14)$$

where the constant is independent of I and Λ .

Proof. The mass projection commutes with the wedge modular data and hence preserves all wedge real spaces containing Z . Wedge separation therefore makes the self-commutator function of $E_I Z$ vanish at bounded times and sufficiently large spatial separation, with the spatial bound independent of I . Its time derivative at zero has compact spatial support and is bounded in absolute value by

$2\langle Z, P^0 E_I Z \rangle$. If μ_I is the positive translation spectral measure of $E_I Z$, the spatial Fourier transform of this derivative, after division by its fixed imaginary factor, is the sum of the positive marginal measures of $p^0 d\mu_I(p)$ at $\mathbf{p}$ and $-\mathbf{p}$. Fourier transformation of the compactly supported bounded function bounds each of these positive measures by $C_Z \langle Z, P^0 E_I Z \rangle d^3 \mathbf{p}$. On the positive-energy cone $p^0 \geq |\mathbf{p}|$. Hence

$$\int_{p^0 < \Lambda} d\mu_I \leq C_Z \langle Z, P^0 E_I Z \rangle \int_{|\mathbf{p}| < \Lambda} \frac{d^3 \mathbf{p}}{|\mathbf{p}|} \leq C'_Z \Lambda^2 \langle Z, P^0 E_I Z \rangle.$$

The integrable singularity at $\mathbf{p} = 0$ causes no problem; the positive mass band has no atom there. Taking square roots proves the estimate. $\square$

A fully scaled smooth vacuum flux for a quadratic X is of the form $R U_D (\log R) F$, where F is a fixed purely timelike, translation-smooth wave. For a compact mass band of F , the paired part of Z lies in $I_R = [c/R, C/R]$. Equation (5.14), with $\Lambda = N/R$, gives a bound proportional to $N \langle Z, P^0 E_{I_R} Z \rangle^{1/2}$, which tends to zero. To remove the compact-energy and compact-mass restrictions, write $F_N = 1_{N < P^0 < 2N} F$ for dyadic $N > 0$. The same argument without a mass restriction bounds the pairing on this dyad by $C_Z N \|F_N\| \langle Z, P^0 Z \rangle^{1/2}$, uniformly in R . The stress-tensor vacuum amplitude is $O((p^0)^2)$ and the four-momentum measure of a dyadic cone segment is $O(N^4)$. Thus $\|F_N\| = O(N^4)$ at zero. At infinity the smooth flux test gives arbitrarily rapid decrease. Hence $\sum_N N \|F_N\| < \infty$. On a fixed dyad, compact timelike mass approximation has arbitrarily small norm, and its complement is controlled by the same uniform bound. Dominated convergence now proves the full flux assertion. Lower-degree X have a smaller scaling factor. Polarization into even Hermitian tests proves that every complete vacuum flux under consideration vanishes.

The left and right charge prescriptions consequently agree and q_X is Hermitian. For an affine Killing field the improvement changes the current by the superpotential

$$\frac{1}{3} \partial^\lambda (X_\lambda \partial_\mu L - X_\mu \partial_\lambda L + (\partial_\lambda X_\mu) L). \quad (5.15)$$

Its compact commutator flux is zero. Thus the translation and Lorentz charge forms are the physical ones. Integration by parts for dilatations gives

$$D' = D + Q, \quad Q = q_J. \quad (5.16)$$

At this point D' and Q are charge forms, not asserted self-adjoint operators. Their dependence on two separately translated polynomial tests is tempered and has each test's ordinary vacuum translation spectrum. For the bra spectrum one lets the compact flux follow the ket, translating the polynomial vector field X as well; Hermiticity supplies the ket spectrum. In particular strictly energy-damped collective translations are legitimate arguments of these forms.

5.3 Identification of the quadratic charge and its domains

The comparison with the physical current will first be made on opposite wedge polynomials, smoothed under translations and rotations. Compact smooth field tests have both kinds of derivatives; the angular derivatives will control the distant caps. We keep the large longitudinal cutoff and the growing transverse cutoff separate: the first is removed at fixed transverse radius, and only then is that radius allowed to grow.

Write

$$k_n(x) = 2(n \cdot x)x - x^2 n = |y|^2 n + 2uy + 2u^2 \bar{n}.$$

We shall identify its charge with the already constructed positive self-adjoint moment $K_n = b(|Y|^2 + \kappa)$. First take opposite compact Hermitian wedge polynomials, as in the bounded-profile Ward identity of Section 3. Locality identifies the commutator flux through $u < 0$ with the extreme vacuum commutator of $(t \cdot k_n)\Omega$ against their product. Translation smoothing makes these matrix elements smooth and supported in the union of the corresponding causal supports. Conservation gives a plane term $|y|^2 t_{vv}$ and the following cap terms.

On the lower cap, the nonzero part with positive step function is to the future of the right wedge polynomial. It has bounded u , $|y| \asymp R$, and $v \gtrsim R^2$. Product cutoffs can therefore be chosen with $\partial_u \chi_R = 0$ there. On a dyad $v \asymp S$, $S \gtrsim R^2$, the transverse cap amplitudes obey

$$C_N R S (R^2 b \sqrt{ab} + Rab + a \sqrt{ab}) (1 + bS)^{-N}. \quad (5.17)$$

The factors follow respectively from the three terms of k_n , the stress-tensor component bounds, the transverse cutoff derivative, and the cap volume. Integration in v gives the last factor. With sufficiently many inverse energy weights these amplitudes have uniformly bounded Hilbert norms. More explicitly, p^0 is comparable to $a + b$, and the allowed transverse momenta have area at most Cab . In the d^4p realization the squared norms of the three terms of (5.17), weighted by $(1 + p^0)^{-m}$ with $m > 5/2$, are bounded respectively by

$$C(R^6 S^{-3}, R^4 S^{-2}, R^2 S^{-1}). \quad (5.18)$$

For example the first squared integrand, after transverse integration, is bounded by $CR^6 S^2 a^2 b^4 (1 + a + b)^{-2m} (1 + bS)^{-2N}$; its b integral is $O(S^{-5})$ and its a integral is finite. The other two have powers $a^3 b^3$ and $a^4 b^2$. Since $S \gtrsim R^2$, all three bounds are uniform. There is also a positive power gain when paired with a test vector smooth under rotations. Fix a sufficiently small $\varepsilon > 0$. The region $a < S^{-\varepsilon}$ gains at least $S^{-3\varepsilon/2}$ in norm by the same integral; $b > S^{-1+\varepsilon}$ is negligible by increasing N . In the remaining region $b/p^0 \leq CS^{-1+2\varepsilon}$, so the momentum direction lies in a spherical cap of power-small area. Rotation Sobolev estimates bound the spectral projection of the fixed test vector onto this cap by a positive power of its area. This follows by averaging the squared projected norms over rotations and then applying Sobolev evaluation on the rotation group; the same estimate holds after each of the finitely many required rotation derivatives and energy weights. The angular cap has area $O(S^{-1+2\varepsilon})$, so its projection norm is $O(S^{-1/2+\varepsilon})$. For example any $0 < \varepsilon < 1/2$ gives $\gamma = \min(3\varepsilon/2, 1/2 - \varepsilon) > 0$. The paired dyads are bounded by $CS^{-\gamma}$, and summing $S \gtrsim R^2$ gives $O(R^{-2\gamma})$.

A regulator at large v can first be removed at fixed R . Its derivative has size S^{-1} , its volume is $O_R(S)$, and the orbital coefficients are then bounded independently of S . The rapid bS decay gives a vanishing energy-weighted norm. On the plane, locality restricts the $|v| \asymp S$ tails to $|y| \lesssim \sqrt{S}$. Their amplitudes are bounded by

$$C_N S^3 b^2 (1 + bS)^{-N}. \quad (5.19)$$

Here transverse integration bounds the squared weighted norm by $CS^6 \int ab^5 (1 + a + b)^{-2m} (1 + bS)^{-2N} da db$, which is uniform. Restriction to $a < S^{-\varepsilon}$ gives $O(S^{-\varepsilon})$ in norm. The preceding angular argument therefore again gives summable paired tails, uniformly when the growing quadratic spatial profile is truncated by a bounded-factor smooth cutoff. The term $\partial_v^2 L$ integrates to zero: the derivative of the separating step is in a spacelike gap. Moving both derivatives onto a distant tail cutoff gives the scalar-wave amplitude $C_N S (1 + bS)^{-N}$. Its squared weighted norm is bounded by $CS^2 \int ab (1 + a + b)^{-2m} (1 + bS)^{-2N} da db$, and has the same low- a and angular gains. The surviving plane integral is therefore the growing second moment of the bounded-profile Ward identity. It equals the form of K_n .

This identity must be extended on a genuine domain. We record the domain argument rather than treating an unbounded charge form as an operator.

Its two inputs are geometric Tomita localization and smoothness under the boost group. They first yield power domains for vectors with strictly one-sided supports. Positive-energy analyticity then transports these estimates to damped translates, where the charge identity can be compared on one common dense domain.

Lemma 5.5 (An affine Hardy estimate). *Let $B = -i\partial_\tau$ on $L^2(\mathbb{R}, d\tau; \mathcal{K})$, where $\mathcal{K}$ is any Hilbert space. Suppose $f \in \bigcap_m H^m(\mathbb{R}; \mathcal{K})$ and $g(\tau) = e^{-ice^\tau} f(\tau) \in \text{Dom } e^{-\pi B}$, with $c > 0$. For every $r > 0$ there is a finite integer m such that*

$$\|e^{r\tau} f\| \leq C_{r,c}(\|f\|_{H^m} + \|e^{-\pi B} g\|). \quad (5.20)$$

The constants are uniform when c varies in a compact subinterval of $(0, \infty)$.

Proof. Choose $\chi \in C_c^\infty(\mathbb{R})$ equal to one on $[0, 1]$, put $\chi_j(\tau) = \chi(\tau - j)$, and write $L = e^j$, $j \geq 0$. At Fourier frequencies $\zeta < -\delta L$, split the convolution $\widehat{\chi_j} * \widehat{g}$ at source frequency $\nu = -\delta L/2$. The Tomita-domain assumption gives

$$\|\mathbf{1}_{\nu < -\delta L/2} \widehat{g}\| \leq e^{-\pi\delta L/2} \|e^{-\pi B} g\|.$$

For the remaining source frequencies, $|\zeta - \nu| > \delta L/2$. The Schwartz L^1 tail of $\widehat{\chi_j}$ is $O_N(L^{-N})$, uniformly in j , since translation changes only its phase. Young's inequality therefore gives

$$\|\mathbf{1}_{\zeta < -\delta L} \widehat{\chi_j g}\| \leq C_N L^{-N} (\|g\| + \|e^{-\pi B} g\|).$$

On $\zeta \geq -\delta L$, take δ smaller than half the minimum of $ce^{\tau-j}$ on $\text{supp } \chi_j$. The phase $-ce^\tau - \zeta\tau$ then has derivative of absolute value at least $C(L + |\zeta|)$. Repeated integration by parts in the Hilbert-valued integral gives

$$\|\widehat{\chi_j g}(\zeta)\| \leq C_N (L + |\zeta|)^{-N} \sum_{k \leq N} \|f^{(k)}\|_2.$$

Its squared integral on this half-line is $O(L^{-2N+1})$ times the displayed Sobolev bound squared. Since $\|\chi_j f\| = \|\chi_j g\|$, these two estimates give arbitrary inverse powers of e^j for the localized norm. Choose $N > r + 1$ and sum over j ; on $\tau < 0$ the weight is at most one. This proves (5.20) and the stated uniformity. $\square$

Lemma 5.6 (Domains for the null moment). *A smooth compact physical polynomial vacuum vector whose supports lie strictly on one side of a null plane is in $\text{Dom } K_n^r$ for every $r > 0$, locally uniformly in such placements. Every strictly energy-damped translate of a compact physical polynomial vector also has this property. Applying K_n^r to the latter vectors preserves strong tube holomorphy and the indicated one-sided boundary continuity.*

Proof. On the lower side choose $c > 0, d \in \mathbb{R}$ so that all supports lie in W_{cp_0+d} . Tomita affiliation, after undoing $U_H(c)U_b(d)$, gives the domain condition for $e^{-\pi B}$. For a compact joint kernel, the same condition is the closed Tomita-graph limit of factorized tests from Section 2. Collective boost derivatives act on the smooth joint kernel, so their norms are controlled by the old joint-distribution seminorms. No affiliated joint-smear operator is needed for this vector statement. Remove the fixed constant translation $U_b(d)$ first; it commutes with H and preserves boost smoothness of these field

vectors. On the positive part of the affine representation use $H = e^\tau$, $B = -i\partial_\tau$. The resulting vector f satisfies

$$e^{-ice^\tau} f \in \text{Dom } e^{-\pi B},$$

where the original vector f has Sobolev derivatives of every order in τ . Lemma 5.5 gives $e^{r\tau} f \in L^2$ for every $r > 0$. The $H = 0$ part is harmless. Since $K_n \leq H$ strongly and the two commute, this proves the assertion for the lower placement. Reflection and equality of the reflected moments prove the upper placement. The choices have positive geometric margins, giving locally uniform bounds.

For $U(x + i\eta)A\Omega$, $\eta \in V_+$, consider the bounded Hilbert-valued holomorphic function

$$1_{(L,\infty)}(K_n)U(x + s\eta)A\Omega, \quad \text{Im } s > 0.$$

A real interval of s places the entire translated support on one side of the plane. On that interval its norm is $O(L^{-N})$ for every N , locally uniformly; elsewhere it is bounded by the original vector norm. The subharmonic logarithmic norm estimate with harmonic measure of that interval yields any prescribed inverse power at every fixed interior point, by starting with a sufficiently large N . Spectral-tail integration gives all the power domains, locally uniformly. Closedness of K_n^r and uniform convergence of spectral truncations give strong holomorphy and the stated boundaries. $\square$

Let $\mathcal{D}$ be the linear span of all vectors obtained from these strictly energy-damped compact polynomial translates by finitely many real translations, dilatations, and compact smooth dilation convolutions

$$\int \rho(s)U_D(s)\xi \, ds, \quad \rho \in C_c^\infty(\mathbb{R}).$$

The locally uniform bounds of Lemma 5.6 and closedness allow each such integral in the graph of every required K_n^r , also after translating the origin; iteration gives the same conclusion for every finite sequence of these operations. The space is dense and invariant under real translations and dilatations by construction. Its defining smooth kernels also place it in $\text{Dom } D$; the compact dilation convolutions form a graph core for D , so $\mathcal{D}$ is such a core. The preceding charge identity extends to $\mathcal{D}$ by tube uniqueness. Thus

$$q_{k_n}(C, A) = \langle C\Omega, K_n A\Omega \rangle \tag{5.21}$$

on this common domain, for every origin and future null normal. The linear relations among conformal vector fields give vector relations among the K_n there: test them against the dense domain and use (5.21). In particular

$$K_n(s) := U_a(s)K_n U_a(-s) = K_n - 2s(D' - B) + 2s^2 a \quad \text{on } \mathcal{D}. \tag{5.22}$$

This equation realizes D' and $Q = D' - D$ as symmetric vector operators on $\mathcal{D}$, by a difference of the indicated operators. Only each $K_n(s)$ is being used as self-adjoint; no essential self-adjointness of Q is required.

5.4 The canonical action on the one-field space

The charge identity supplies more information than the spectral comparison alone. Its translated quadratic dependence first kills the nonnegative comparison defect in the scalar channels. The spectral support statement then upgrades equality of compression forms to a reducing-subspace identity. Only after that step do we treat the remaining helicities.

Theorem 5.7. *The one-field space Π reduces each K_n . On Π , K_n equals the null special-conformal generator of the auxiliary positive conformal representation carried by l, J, t . Moreover $\langle Q\phi, \psi \rangle = 0$ for $\phi \in \mathcal{D}$, $\psi \in \Pi$.*

Proof. We first treat the scalar labels l, j_z, s_z . The comparison (4.27), together with $c_V = l + j_z$ and $c_T = l/3 + s_z$, gives order-zero comparison on each of them. For smooth compact waves with $b, M > 0$ compare the compression of K_n with $K_{0,n} = b(|Y|^2 + h_0)$. Their nonnegative expectation difference, evaluated on $U_a(-s)\psi$, has zero second derivative in s . Indeed the second-difference identity from (5.22) transfers to these waves by symmetry against $\mathcal{D}$; the domains follow from the spectral comparison. Direct differentiation gives the same second difference for $K_{0,n}$. A nonnegative affine function on the whole real line is constant.

Testing with separated compact multipliers in b, k now shows that the positive radial form $h_0 - \kappa_{\text{compr}}$ is invariant under multiplication by e^{itM^2} . Its distribution kernel is annihilated by $M^2 - M'^2$. On $M, M' > 0$ this forces a diagonal kernel; positivity makes it a positive measure $\nu(dM)$. It is bounded by the h_0 form. For any compact interval $I \Subset (0, \infty)$, choose logarithmic cutoffs $f_L = 1$ on I , vanishing outside successively larger compact intervals, with

$$\int_0^\infty M |f'_L(M)|^2 dM \rightarrow 0.$$

For example piecewise linear ramps in $\log M$ have energy $O(L^{-1})$, and smoothing preserves that bound. Consequently $\nu(I) \leq \int |f_L|^2 d\nu \rightarrow 0$. The compression forms agree.

This is also equality of operators on a reducing copy. The spectral comparison gives

$$\|1_{(s,\infty)}(\kappa)\psi\| \leq \|1_{(s,\infty)}(h_0)\psi\|.$$

Equality of the first moments makes the squared tail norms equal almost everywhere. The support inclusion underlying this inequality and equality in the projection norm inequality give equality of the projected vectors. Spectral continuity gives it at all continuity points, hence equality of all spectral measures and reduction. Comparing the odd translation differences with the canonical scalar formula shows that $Q\mathcal{D}$ is orthogonal to these labels. Varying the null direction spans all polarizations of Π , proving the last assertion of the Theorem.

For clarity, the projection argument above does not require the two spectral measures to commute. Write ι for the scalar-copy embedding, $E_s = 1_{(s,\infty)}(\kappa)$ and $P_s = 1_{(s,\infty)}(h_0)$. Spectral comparison gives $E_s \iota = E_s \iota P_s$. Equality of the first moments is equality of the integrals of $\|E_s \iota \psi\|^2$ and $\|P_s \psi\|^2$; their pointwise difference is nonnegative, so it vanishes almost everywhere. Equality in the contraction estimate for the orthogonal projection E_s therefore gives $E_s \iota \psi = \iota P_s \psi$. One first uses a countable dense form core and then spectral continuity to obtain the operator identity for every threshold. This proves reduction rather than merely equality of expectations.

We recall explicitly the reference generator on the remaining polarizations. On transverse helicity of absolute weight d it is

$$K_{c,n} = b(|Y|^2 + h_d). \quad (5.23)$$

For a rank- r primary of dimension $r + 2$, $r = 1, 2$, contract $r - d$ slots with n and use $\partial_v O_i - \partial_i O_v$ in each of the other slots, discarding joint transverse traces. The wave is a constant multiple of $b^r M^d$ times its helicity label. Under the null conformal map of Section 2, the transverse Jacobian is $\gamma(e_i + 2cy_i n)$. The curl cancels the $y_i n$ terms; derivatives of those terms contribute only the discarded traces. The transformed wave thus has the effective scalar weight $2 + d$, giving (5.23) by the scalar computation. This uses only the positive primary two-point kernels and also acts on their descendant combinations V, T . Its origin translation formula contains D , without Q .

On smooth compact timelike Π waves the actual domains follow from (2.26). Put $X_n = K_n - K_{c,n}$ there. Symmetry-testing (5.22) on $\mathcal{D}$, using the orthogonality just proved, shows that X_n intertwines translations and is linear in n . The same is true on the energy-damped compact V, T waves in $\mathcal{D}$: symmetry now gives $Q = 0$ as a vector on these waves, and their reference domains follow from the primary and descendant tube action. The translation formula gives polynomial graph bounds. Integrating translates against Schwartz functions therefore commutes X_n through compact timelike momentum multipliers. Such multiples of the damped V, T waves span the smooth compact timelike tests of Π .

It follows that X_n is a measurable fiber coefficient map. Linearity in n supplies a covector index λ . The coefficient tensors

$$X_\lambda V_\mu(p) = X_\lambda J_\mu(p), \quad X_\lambda T_{\mu\nu}(p) = X_\lambda t_{\mu\nu}(p)$$

are locally square integrable up to the nonzero light cone from inside: this follows first with the damping and smooth test multipliers, and then locally without them by choosing those multipliers nonvanishing. Their dimensions are respectively $d' = 2, 3$, and they are covariant tensors. A rest-frame boost component of weight ω has norm proportional to $M^{d'-2-\omega}$ when boosted to a fixed nonzero energy and $M \downarrow 0$. The transverse mass measure is proportional to $M dM$. Local integrability therefore excludes every weight $\omega \geq d' - 1$.

For the first tensor, contracting the μ slot with a null vector leaves the λ covector parallel to its lowered null vector: its transverse and same-null contractions have the forbidden weights. At rest conservation makes the μ slot purely spatial. Reversing its spatial direction changes the contracted tensor by a sign, but would require it to be parallel to a different null covector. It is zero. For the second tensor, contract both μ, ν slots with the same null vector. The forbidden weights again leave only that parallel covector. Both slots are spatial at rest, so reversing direction leaves the contraction unchanged and again forces it to be zero. Varying the direction and using symmetry recovers every spatial tensor coefficient. Hence both tensors vanish and $K_n = K_{c,n}$ on compact timelike tests.

For $d \geq 1$, conjugation by $M^{1/2}$ turns the radial operator into $-\partial_M^2 + (d^2 - 1/4)M^{-2}$, which is limit-point at both endpoints. Compact smooth radial tests, combined with smooth transverse cores and compact b cutoffs, are thus graph cores for (5.23). The $d = 0$ part was already reducing with its specified Hankel extension. The closed graph of the actual self-adjoint K_n contains the graph of the reference self-adjoint operator on each copy. Resolvent uniqueness gives reduction of their sum Π and equality there, as claimed. $\square$

5.5 The residual dilation charge vanishes

The remaining charge Q is not assigned an independent self-adjoint realization. Instead, future and past cone comparisons provide two families of genuine self-adjoint generators whose first moments contain the same Q . Opposite exponential bounds on their limiting measures will force both families to concentrate on the physical dilation spectrum.

Fix $n, \bar{n}$ and write $K = K_n$. Truncate the future time cone by $u < R$ and the past cone by $u > -R$. Their actual real spaces are contained in comparison real spaces constructed with K in place of the full parabolic generator. This is (5.2) for the translated parabolic cut; reflection treats the past cone. Let T_R^+ be the comparison future modular generator and let $-T_R^-$ be the comparison past modular generator. The affine product formulas and (5.22) give self-adjoint operators satisfying

$$T_R^\pm = D + Q \mp \frac{K}{2R} \quad \text{on } \mathcal{D}. \quad (5.24)$$

For example the future expression is $B(R) - K(R)/(2R)$; the past modular expression is $-B(-R) - K(-R)/(2R)$. Their spectral measures have uniformly bounded second moments on $\mathcal{D}$ as $R \rightarrow \infty$. The same is true on a smooth single-field vector supported compactly in either time cone: the relevant planes avoid its support, so Lemma 5.6 and the affine product domains apply. Testing against $\mathcal{D}$ shows that

$$T_R^\pm \xi \pm K\xi/(2R)$$

is a Hilbert vector independent of R and of the sign.

Proposition 5.8. *The charge form Q vanishes on all physical polynomial vectors. In particular $Q = 0$ on $\mathcal{D}$.*

Proof. Average the spectral measures of $T_R^\pm$ over $S < \log R < 2S$, with normalized Lebesgue measure, and take common weak limit points as $S \rightarrow \infty$. One may first test continuous functions on the compactified real line. The uniform second moments on dense vectors remove mass at infinity. The limits are normalized positive operator-valued measures, not assumed projection-valued. Dilatations shift $\log R$ by a fixed amount, whose boundary contribution in the averages is $O(S^{-1})$. Thus the limits commute with D . They have joint positive measures $E^\pm(dd, ds)$ with the spectral measure of D . Uniform second moments also give uniform integrability of first moments, including polarized cross moments on the vectors just described.

Only these first moments will be compared with $D + Q$. The logarithmic averaging is used to obtain commutation with the known self-adjoint operator D , while the uniform second moments prevent loss of first moments in the weak limit. Neither operation turns the limiting positive operator-valued measures into spectral measures.

Let O be a Hermitian scaling component of real dimension α and $\xi = O(f)\Omega$, with f real, smooth and compact in the future cone. The scaling orbit continues for $0 < \text{Im } z < \pi$ because $e^z x$ has forward timelike imaginary part there. Compact interior-cone support and integration by parts give Hilbert boundary values at both ends; real scale covariance gives uniform norm bounds along the strip. Consequently

$$e^{-\pi D} \xi = e^{i\pi\alpha} O(f^-)\Omega, \quad f^-(x) = f(-x). \quad (5.25)$$

We need this strip identity also for complex linear combinations of real dilation translates. Write $\xi_s = U_D(s)\xi$ and let ξ_s^- be its reflected-test wave. Hermitian conjugation of the two-point function, with both test coordinates reflected, gives

$$\langle \xi_s^-, \xi_t^- \rangle = \langle \xi_t, \xi_s \rangle.$$

For $p(D) = \sum_j c_j U_D(s_j)$ put $p^\#(D) = \sum_j \bar{c}_j U_D(s_j)$; the real parameters s_j are unchanged. With $\omega_\alpha = e^{i\pi\alpha}$ and $\eta = e^{-\pi D} \xi = \omega_\alpha \xi^-$, the reflected Gram identity yields

$$\|p(D)e^{-\pi D} \xi\| = \|p^\#(D)\xi\|, \quad \|p^\#(D)\eta\| = \|p(D)\xi\|. \quad (5.26)$$

For all sufficiently large truncations, the comparison Tomita partner of $p(D)\xi$ is $p^\#(D)\xi$, and that of $p(D)\eta$ on the past side is $\omega_\alpha^{-2} p^\#(D)\eta$. Their norms are therefore exactly the two norms in (5.26). The finite set of dilated supports remains compactly in the respective cone. In particular, we do not require a complex filter to preserve its real space.

Real-space containment and these Tomita norm bounds, followed by the spectral-measure limits in the future and past cones, give

$$\int |p(d)|^2 e^{-2\pi s} d\langle E^+ \rangle_\xi \leq \|p(D)e^{-\pi D} \xi\|^2, \quad (5.27)$$

$$\int |p(d)|^2 e^{-2\pi d} e^{2\pi s} d\langle E^- \rangle_\xi \leq \|p(D)\xi\|^2. \quad (5.28)$$

Truncation followed by monotone convergence justifies the unbounded exponential tests in the limiting measures. The first moments in s , with every such filter, agree for the two signs by (5.24) and the uniform moment bounds. Exponential polynomials determine finite measures: polarization recovers every exponential e^{itd} , and Fourier uniqueness then recovers the measure. Thus these equalities and inequalities hold conditionally on the common d marginal μ_ξ .

To spell out the first-moment assertion, the finite filters $p(D)$ preserve the class of smooth cone-supported single-field vectors; they also commute with $e^{-\pi D}$. Thus both signs admit every same filter. Their limiting first-moment measures in d are finite by the uniform second-moment bounds. Taking $p(d) = 1 + ze^{itd}$ and polarizing in $z \in \mathbb{C}$ shows that these two measures have identical Fourier transforms, hence agree. This is equality of conditional means, rather than just equality of their integrals over d .

Disintegrate the two scalar joint measures as $\mu_\xi(dd)\nu_d^\pm(ds)$. Their conditional first moments $m_\pm(d)$ agree almost everywhere, while (5.27)–(5.28) say

$$\int e^{-2\pi s} \nu_d^+(ds) \leq e^{-2\pi d}, \quad \int e^{2\pi s} \nu_d^-(ds) \leq e^{2\pi d}.$$

Jensen gives $m_+(d) \geq d \geq m_-(d)$. Equality of the means forces both to equal d ; strict convexity then forces both conditional laws to be δ_d . This argument requires only positivity and normalization of the measures, not multiplicativity of the operator-valued limits. Positivity also gives Cauchy–Schwarz for each measurable set, so cross measures against ξ are supported on $s = d$. Their first moments, paired with $\phi \in \mathcal{D}$, now give the following identity. Indeed, if a measurable set is disjoint from $s = d$, its positive operator has zero quadratic value on ξ , and Cauchy–Schwarz makes every mixed value against ξ zero as well. The mixed measure is therefore concentrated on the diagonal; its d marginal is the mixed spectral measure of D . Its first moment is consequently that of D , while the uniformly integrable first-moment limits from (5.24) give the form of $D + Q$. Subtracting the two gives

$$\langle Q\phi, \xi \rangle = 0. \tag{5.29}$$

The strip and Jensen argument just completed used a Hermitian homogeneous field of dimension α with a real test. Splitting a complex test into its real and imaginary parts extends (5.29) to all tests. A complex homogeneous field is the sum of its Hermitian real and imaginary parts,

$$O_\Delta = \frac{O_\Delta + O_\Delta^\dagger}{2} + i \frac{O_\Delta - O_\Delta^\dagger}{2i},$$

which have the same grade because adjunction preserves real scaling dimensions. Linearity therefore extends (5.29) to every homogeneous original field. For $O = \sum_{\Delta \in S_O} O_\Delta$, its finite scaling support then extends the conclusion to O with the same smearing test. This step sums conclusions; it does not assign the nonhomogeneous field one strip phase or apply Jensen’s inequality to cross-grade terms.

It remains to pass from single fields to products. Define the insertion $\delta_Q O(y)$ by the commutator with $J(y + r)$ integrated against a compact flux cutoff in r . Conservation and locality shrink the flux as needed. This insertion is relatively local to physical fields and is tempered with the ordered spectra: in this cutoff operation the adjacent momenta are first combined, so the spectator spectral cones are unchanged. Its extreme vacuum values against every physical polynomial vanish by (5.29) and the charge identity. To justify that passage, damp both collective translations first, remove the ket damping on the smooth future-cone single-field test, and remove the bra damping distributionally. Translation covariance removes the future-cone support restriction. Move the insertion to the extreme on a spacelike open set; ordered tube uniqueness makes the zero equality

global. Finally the commutator flux obeys the derivation rule over every finite list of product slots. This rule is the telescoping of the adjacent single-insertion commutators just constructed, not a new operator axiom for J or L . For a joint test kernel the telescoping identity is first an identity of joint distributions and is then tested with that same kernel. Each summand is zero, proving the Proposition. $\square$

5.6 Modularity of time cones and the actual time action

We now return from comparison real spaces to algebras of actual bounded operations. The equality $Q = 0$ determines the limiting comparison generator, but identifying the physical cone modular group also requires a KMS identity and separation of the vacuum. Both will be obtained for the original cone algebra before using half-sided modular inclusion for line diamonds.

Theorem 5.9. *The vacuum is cyclic and separating for the actual future and past cone algebras. Their modular generators, with the normalization of Section 2, are respectively D and $-D$.*

Proof. By Proposition 5.8, $T_R^\pm \psi \rightarrow D\psi$ on $\mathcal{D}$. This is a core for D : it is dense and invariant under its unitary group, and smoothing by compactly supported smooth functions of the group parameter supplies a graph core. The self-adjoint core convergence criterion therefore gives $T_R^\pm \rightarrow D$ in strong resolvent sense, and their real unitary groups converge strongly.

Take two bounded operations in a fixed compact subregion of the future cone. For all sufficiently large R , their vacuum and adjoint-vacuum vectors belong to the comparison standard space and its complexification with the required Tomita domains. Writing the two operations as A, B , the standard-space two-vector identity gives the scalar holomorphic continuation of $\langle B^* \Omega, e^{itT_R^+} A \Omega \rangle$ to $0 \leq \text{Im } z \leq 2\pi$. Its values are defined by

$$F_R(t + iy) = \langle e^{-yT_R^+/2} B^* \Omega, e^{itT_R^+} e^{-yT_R^+/2} A \Omega \rangle, \quad 0 \leq y \leq 2\pi,$$

with the opposite boundary

$$F_R(t + 2\pi i) = \langle e^{itT_R^+} A^* \Omega, B \Omega \rangle.$$

One obtains the width 2π by assigning up to π of imaginary evolution to each vector and using $e^{-\pi T_R^+} = J_R S_R$; the comparison conjugation disappears from the displayed boundary formula. The split-vector pairing uses only $\text{Dom } e^{-\pi T_R^+}$; no vector $e^{-2\pi T_R^+} A \Omega$ is required. The bounds depend only on the four vector norms. Normal-family convergence on the strip, together with the strong convergence on the real edge, gives the vacuum KMS identity for D on these actual operations. The identical argument in the past gives the identity for $-D$. Bounded strong-star approximation extends the identity from compactly localized operations to the cone algebras.

Scaling already acts as an automorphism of each actual cone algebra. The vacuum is cyclic by Reeh–Schlieder. For clarity, faithfulness is part of the conclusion: the KMS identity implies that a bounded operation annihilating the vacuum has adjoint annihilating it. Apply this to BA for arbitrary B if $A\Omega = 0$. Then $A^*B^*\Omega = 0$; cyclicity gives $A^* = 0$. Thus the vacuum is separating. Uniqueness of the modular automorphism group of this faithful vector state identifies the generators as claimed. $\square$

Theorem 5.10 (Actual time-Möbius action). *For every future unit timelike vector e , the actual Hilbert space carries a positive-energy representation of the universal covering group of the time-Möbius group. Locally it maps physical diamond algebras with tips on the line $\mathbb{R}e$ to the corresponding diamond algebras. Its affine subgroup is the physical time translation and dilation subgroup. The canonical lift of the connected stabilizer of each finite interval is its actual diamond modular group.*

Proof. Choose two line parameters $a < b$. Let A, B_t be the dilation generators centered at ae, be , and let C be the modular generator of the physical diamond with those tips. Inclusion of the diamond into the future cone at ae and the past cone at be is half-sided modular by Theorem 5.9 and isotony. The common vacuum is cyclic and separating for both inclusions; no cyclicity of a relative commutant is needed. The half-sided inclusion theorem gives the affine relations corresponding to

$$A = (z - a)\partial_z, \quad B_t = (z - b)\partial_z, \quad C = \frac{(z - a)(b - z)}{b - a}\partial_z. \quad (5.30)$$

More specifically, $A - C \geq 0$ scales with weight -1 under A , with the negative inward inclusion convention $N = U_{A-C}(-1)MU_{A-C}(1)$; and $-B_t - C \geq 0$ scales with weight $+1$ under $-B_t$. The pair A, B_t already has the physical affine relation with $P_e \geq 0$.

These are relations of unitary groups, not just commutators on a tentative common domain. Ordered products of the three classical one-parameter groups are coordinates near the identity. The three affine relations give exact pair-switching identities, so reordering products proves a local unitary homomorphism. It integrates uniquely to the simply connected cover. This is the three-subgroup integration criterion of [22, Lemma 1.1]; see also [34, Lemmas 2.3–2.4]. In its circle formulation the intervals are (a, b) , (b, ∞) , and (∞, a) , with stabilizer fields $C, B_t, -A$. Their three pair relations are precisely the two cone–diamond relations above and the physical affine relation between the two cone groups. Its translation subgroup is the physical one, so the representation has positive energy.

Affine covariance transports C to the modular generator of every line diamond. A sufficiently small Möbius transformation relative to a fixed finite interval factors into an affine map of its tips and an element of its connected stabilizer, with the lift near the identity. Both factors have the asserted action on its actual algebra. This proves the local geometric action, without replacing the physical net by a larger net.

The construction is made for each chosen time direction using its actual cone and diamond algebras. Its local geometric statement is the one needed below: the connected interval stabilizer acts by that diamond’s modular group. Compatibility sufficient to integrate all spacetime conformal transformations on the full net has not been used in this argument. $\square$

We finish by recording the representation data needed for extraction. Separate the trivial representation. The nontrivial irreducible positive-energy representations of the simply connected cover are the lowest-weight modules with $h > 0$ [29, Theorem 1.5.3 and the following direct-integral discussion]. The general representation may contain a direct integral of these modules, rather than a discrete sum. To obtain the decomposition, disintegrate the central unitary of a full rotation. Within each central character, compact energy lies in one integer translate; positivity and the raising and lowering norm relations decompose it into orthogonal lowest-weight ladders. The real ladder coefficients give measurable multiplicity spaces $\mathcal{K}_h$.

Each nontrivial irreducible module restricts to the unique strictly positive irreducible affine representation [36, Corollary 2.33 and equation (6)]. Multiplication by $E^{1/2}$ converts its $L^2(dE)$ realization to the common affine realization

$$\mathcal{H}_h = L^2(\mathbb{R}_+, dE/E), \quad E = P_e, \quad (U_D(s)\psi)(E) = \psi(e^s E), \quad \psi_{h,z}(E) = E^h e^{izE}. \quad (5.31)$$

The last wave, for $\text{Im } z > 0$, transforms as a primary of weight h : its inner product kernel is

$$\int_0^\infty E^{2h-1} e^{iE(z-\bar{w})} dE = \frac{\Gamma(2h)}{[-i(z-\bar{w})]^{2h}},$$

with the branch fixed in the upper half-plane. Kernel covariance on the dense span of these waves fixes the primary realization and its lift to the cover. Here h denotes a lowest-weight parameter, distinct from the internal parabolic operator used in the null-cut comparison. On the nontrivial summand, put

$$\mathcal{K} = \int_{h>0}^\oplus \mathcal{K}_h d\nu(h), \quad \mathcal{H}_{\text{nontriv}} = L^2(\mathbb{R}_+, dE/E) \otimes \mathcal{K}, \quad hv_h = hv_h.$$

Thus a measurable weight-band projection is $\mathbf{1} \otimes \mathbf{1}_I(h)$. No discreteness of this weight operator follows from, or is required by, the field-label spectrum.

Lemma 5.11. *On any bounded weight band contained in $h > 0$, the real standard space of a line diamond is generated by the primary waves (5.31) smeared with real smooth compact interval tests and multiplicity vectors real for the appropriate phased conjugation. Weight-band projections commute with the diamond Tomita operator.*

Proof. Let J be the actual future-cone modular conjugation. The positive-translation relation gives $JP_eJ = P_e$. The half-sided cone–diamond relation gives the same formula for its positive special-conformal partner, while $JDJ = -D$. These three generating groups show that J implements time reflection on the integrated Möbius representation.

In each lowest-weight ladder choose the canonical reflection conjugation, measurably in h ; its existence and multiplicity form are given in [29, Theorem 1.6.3 and its universal-cover extension]. The product of this conjugation with J commutes with the full representation. Inequivalent lowest weights cannot be mixed by that commutant, so J preserves every measurable weight band. In the common affine realization it is pointwise energy conjugation together with an energy-independent multiplicity conjugation j , preserving the spectral projections of h . This argument distinguishes weights h and $1-h$ even when their Casimir values coincide.

For a real compact half-line test f , the strip continuation gives

$$e^{-\pi D}(E^h \hat{f}(E)v) = e^{i\pi h} E^h \hat{f}(-E)v.$$

Hence its Tomita fixed-point condition is $v = e^{-i\pi h} jv$, the phased real structure in the statement. These tests are complex cyclic by Fourier uniqueness. Their real closure is invariant under dilatations and contained in the half-line real space. Gaussian smoothing in D gives graph density for Tomita, hence equality of the two real spaces. On compact positive weight bands these operations have uniform bounds, and exhaustion treats a bounded band approaching zero. The half-sided unitary sends the half-line space onto the finite-interval space and transports the tests by primary covariance. All these operations commute with the weight-band projections. Each such projection therefore commutes with the diamond conjugation, strongly commutes with its modular operator, and preserves the domain and action of its Tomita operator. $\square$

6 Extraction of a localizable scalar wave and accessible products

Throughout this Section, l remains the scalar *vacuum wave* constructed from the trace. In particular, division by M^2 on that vacuum wave has not defined an operator on other vectors. We first extract

from the physical virial field a scalar wave S with actual local approximants. Only subsequently do we construct products of such accessible waves. The notation L in a formal insertion still refers to the single-insertion forms of Proposition 5.3; a physical field L will be reconstructed in Section 8.

The construction has two distinct localization steps. We first use the canonical two-point spaces to select the part of the virial vacuum wave with actual time weight two. The actual time-Möbius unitaries then express that part through operations in physical diamonds. After proving independence of the chosen time direction, we quantify these operations by bounded approximants and construct their analytic products. These products have real boundaries on spacelike patches; the remaining real-axis reconstruction is a separate task.

6.1 Projection into canonical diamond localization

Write $X_{\text{phys}}(O)$ for the closed real vacuum space of the physical algebra in O , and $X_c(O)$ for the additive canonical real space on the one-field representation Π . The latter is determined by the positive scalar, conserved-vector and traceless-tensor two-point kernels of Section 2.

Lemma 6.1. *For every bounded diamond O ,*

$$P_{\Pi}X_{\text{phys}}(O) \subset X_c(O). \quad (6.1)$$

This assertion uses conformal covariance only of the auxiliary two-point representations.

Proof. The actual parabolic containment (5.2) of Lemma 5.1, followed by the reduction in Theorem 5.7, gives the desired projected membership for every parabolic diamond N containing $\bar{O}$: actual localization is contained in the comparison localization, and the comparison generator acts canonically on Π . The conjugation on Π is the primary reality operation by wedge Bisognano–Wichmann; the same statement holds in the reverse orientation. We explain why these containments determine the bounded diamond.

The point of the geometry is to recover a bounded diamond from the larger parabolic regions already controlled by the null-charge comparison. We carry out this recovery in the canonical real spaces, where the complementary diamond can be described explicitly.

Use the Einstein cylinder, with coordinates $(\tau, x) \in \mathbb{R} \times S^3$ and round distance d . By a Poincaré transformation and a dilation, the physical patch and the chosen diamond can be written

$$|\tau| + d(x, N_0) < \pi, \quad O = \{ |\tau| + d(x, N_0) < \alpha \}, \quad 0 < \alpha < \pi.$$

On the canonical representations,

$$X_c(O)' = X_c(O^{\perp}), \quad O^{\perp} = \{ d(x, N_0) > \alpha + |\tau| \}, \quad (6.2)$$

where prime denotes symplectic complement. Indeed, a connected conformal transformation carries the wedge and its opposite to this diamond pair. Transporting wedge modular data proves the equality. There is no assumption here about a physical algebra’s duality.

For completeness, these transports are well-defined across the required charts. The two-point kernels are rational functions with the finite Jacobian and weight factors of their integral bosonic primaries. On strict complex point waves in sufficiently small boxes the rational covariance identity preserves every inner product. Their span is dense, so it defines local unitaries; the same identity gives local composition and hence integration to the connected cover. On nonsingular real patches, the differentials preserve the future cone, so smaller strict imaginary approach cones have distributional boundary values after transport. On overlaps, both boundary prescriptions are continuations of the

same rational complex kernel. In strict spacelike separation the two prescriptions agree. Additivity therefore transports the real smear spaces, even when the opposite diamond meets several charts.

Here is an explicit covering of the complement needed in (6.2). Let a target have $\tau \geq 0$ and $\theta = d(x, N_0) > \alpha + \tau$. Choose $\alpha < \beta < \theta - \tau$ and take the past tip of N to be $(-\beta, N_0)$. Its future tip is $(\pi - r, q)$, where $d(q, N_0) = r < (\pi - \alpha)/2$; it lies on the future null boundary of the physical patch. This parabolic diamond contains $\overline{O}$ with strict margins. The target is outside the causal future of its past tip because $\theta > \beta + \tau$. To put it outside the causal past of the future tip, it suffices to choose q so that

$$d(x, q) + r > \pi - \tau.$$

At fixed r , the maximum left side is $\min(\theta + 2r, 2\pi - \theta)$. As r increases to $(\pi - \alpha)/2$, this tends to $\min(\theta + \pi - \alpha, 2\pi - \theta)$, which exceeds $\pi - \tau$, except when $(\tau, x) = (0, -N_0)$. Time reversal treats $\tau < 0$. Consequently the spacelike shadows of these N cover $O^\perp$ except possibly its central antipode.

If a vector symplectically annihilates all these shadows, its pairing with canonical real point waves vanishes on their union. The resulting distribution cannot be supported only at the omitted point: locally its singular covectors are causal, from the positive-energy tube and its conjugate, whereas a nonzero point-supported distribution has all nonzero covector directions in its wavefront set. It therefore annihilates $X_c(O^\perp)$. Taking symplectic complements in (6.2) proves (6.1). $\square$

6.2 Timelike restrictions of the original fields

The time-weight argument uses point waves restricted to a timelike line. Later, bounded approximation will also require uniform moments of the original source field as a transverse smearing shrinks to that line. Both facts follow from the same ordered-spectrum estimate.

Lemma 6.2. *Every fixed finite original vacuum distribution has a restriction to parallel timelike lines, continuous in joint Schwartz temporal tests. The restrictions of its fixed transverse derivatives exist as well. Transverse mollification converges in these distributions, and in the one-field vacuum and adjoint-vacuum vectors, with error $O(\delta)$ at width δ . Each bound uses finitely many temporal test seminorms, uniformly on compact families of line positions. In particular every fixed source-field polynomial moment is uniformly bounded under this mollification. For a Hermitian field, a real line test has its vacuum vector in $X_{\text{phys}}(O)$ whenever the compact line support is contained in the double cone O .*

Proof. Choose the line direction as the time axis and write the individual Fourier variables as $k = (\omega, \mathbf{k})$. On the ordered spectral support Γ_n , positive energy of the partial sums gives $\sum_i |\mathbf{k}_i| \leq C_n \sum_i |\omega_i|$. Choose a smooth cutoff $\chi = 1$ near Γ_n , supported where $|\mathbf{k}| \leq C'_n(1 + |\omega|)$, with polynomially bounded derivatives. For a joint Schwartz temporal test h , $\chi(k)\widehat{h}(\omega)$ is therefore Schwartz in all variables, with each Schwartz seminorm bounded by finitely many temporal ones. Compact families of spatial translations and fixed transverse derivatives introduce only polynomial factors and obey the same bound.

For the tempered Fourier distribution $\widehat{W}_n$, fix a finite Schwartz seminorm p_m controlling its action. Transverse mollifier multipliers are uniformly bounded in the required weighted derivative seminorms, and

$$|\partial_{\mathbf{k}}^\gamma(\widehat{\rho}(\delta\mathbf{k}) - 1)| \leq C_\gamma \delta(1 + |\mathbf{k}|), \quad 0 < \delta \leq 1.$$

After multiplication by $\chi\widehat{h}$, this gives an $O(\delta)$ bound in p_m , with finitely many additional temporal seminorms. It constructs the restriction and proves the stated distributional convergence. For the squared one-field vacuum error, the doubled correlation has two mollifier differences; its seminorm is $O(\delta^2)$, giving vector error $O(\delta)$. The adjoint and all fixed alternating source-field words are covered

by the same original vacuum distributions. Thus every fixed moment required below is uniformly bounded. This argument tests tempered distributions; it assumes no ordinary Fourier density for an n -point function.

For a real line test of a Hermitian field, sufficiently thin compact transverse mollifications are supported in O . Their vacuum vectors belong to its closed real space by affiliation. Their Hilbert limit has the same membership. Translation and Lorentz covariance give the assertion for all the lines under consideration. $\square$

6.3 Actual and auxiliary time weights

Fix a future timelike unit e and put $E = P_e > 0$. Use the common affine realization $L^2(\mathbb{R}_+, dE/E; \mathcal{K})$, where translations multiply by e^{itE} and dilations rescale E . In a lowest-weight module of weight $h > 0$, the point wave is $E^h e^{izE} v$. Here h is a scalar time lowest weight, unrelated to the internal parabolic operator h of Proposition 2.4 or the radial comparison operators h_ν . Theorem 5.10 and Lemma 5.11 constructed the *actual* time-Möbius action and its interval real spaces. The auxiliary canonical representation has the same affine convention, with its own primary weights s . Write H for the actual lowest-weight operator on the affine multiplicity space. We will compare interval real spaces as well as affine actions: the affine actions alone do not distinguish lowest weights.

Lemma 6.3 (Interval Paley regularity and weight selection). *Let $X_s(I)$ be the closed real interval space of a primary of weight $s > 0$. Dividing a wave in $X_s(I)$ by E^s gives an entire Fourier transform; evaluation and each derivative on a compact complex-frequency set are continuous in its real-space Hilbert norm. The bounds are uniform on compact positive weight bands. In particular:*

1. *Suppose a bounded affine intertwiner j_s from the actual representation to a weight- s primary representation satisfies $j_s X_{\text{act}}(I) \subset X_s(I)$ for bounded intervals I . In the common affine realization write $j_s = \mathbf{1} \otimes T_s$. Then*

$$T_s \mathbf{1}_{\mathbb{R} \setminus (s + \mathbb{Z}_{\geq 0})}(H) = 0.$$

2. *A physically real localized vacuum point wave of scale dimension $d > 0$ has actual weights only among $h = d - j > 0$, $j \in \mathbb{Z}_{\geq 0}$.*

Proof. For real $f \in C_c^\infty(I)$ the primary norm, up to a positive constant, is

$$\int_0^\infty E^{2s-1} |\widehat{f}(E)|^2 dE.$$

Reality determines negative frequencies. To evaluate $\partial_z^k \widehat{f}(z)$, choose a compactly supported smooth test which agrees on I with $(it)^k e^{izt}$. Modify it outside I to annihilate its first m Fourier jets at zero, with $m > s - 1$. Such a modification exists: finitely many disjoint small bumps outside I , with distinct centers, have a moment matrix arbitrarily close to a nonsingular Vandermonde matrix. Cauchy–Schwarz with dual weight $|E|^{1-2s}$ now bounds the pairing. Its high-frequency integral converges by smoothness, and its low-frequency integral converges by the vanished jets. The construction and all bounds are uniform on compact sets of z and compact bands of s . They extend to the real Hilbert closure. The extended evaluations satisfy the Cauchy–Riemann equations and give the asserted entire function.

The regularity at zero frequency now detects the allowed weights. In the common affine realization, changing the primary weight changes only a power of E . Entire continuation permits precisely the nonnegative integral powers of that variable.

An intertwiner commuting with E and its rescalings is an energy-independent bounded multiplicity map T_s . For a real multiplicity section v in a compact H -band, projection of the actual localized wave has quotient

$$\hat{f}(E)T_s E^{H-s}v. \quad (6.3)$$

The assumed real-space inclusion and the preceding continuity make this holomorphic at $E = 0$. Choose $\hat{f}(0) \neq 0$. On a compact band $T_s \exp((H-s)z)v$ is an entire function of $z = \log E$. Single-valued continuation in E implies

$$T_s(e^{2\pi i(H-s)} - \mathbf{1})v = 0.$$

Complexifying real sections and applying the spectral theorem shows that T_s is supported on $h-s \in \mathbb{Z}$. Only finitely many such values occur on the band. Their finite Laurent expansion in (6.3) has no pole, so its negative powers vanish. Exhausting the positive weight axis proves the first assertion.

For the second, write a dimension- d point wave as $E^d u$ in the actual affine representation. Project onto compact H -bands, which commute with interval Tomita data. In each fiber division by its primary power gives $\hat{f}(E)E^{d-h}u_h$. The same argument forces $d-h \in \mathbb{Z}_{\geq 0}$. There is no translation-invariant component of a positive-dimension point wave: its zero-momentum vacuum amplitude would have to transform by a nonunit-modulus dilation character. For original fields, the required timelike restrictions and real localization are supplied by Lemma 6.2. The same assertion for a subsequently constructed wave uses its stated Hilbert-valued restriction and physical real localization. $\square$

The norm-continuous evaluations above concern the closed real space. For a complex localized operator vector, use its Hermitian real and imaginary parts; the bound then involves the sum of the vacuum and adjoint-vacuum norms. No Hilbert-norm evaluation bound on an arbitrary completion of $X_s(I) + iX_s(I)$ is being used.

Here the intertwiners are specific canonical projections. Project by P_Π , then onto an auxiliary time-primary summand of weight s . Lemma 6.1 and the canonical interval Tomita data give the required real-space inclusion for this map j_s . The same construction applies when the canonical comparison space is enlarged below. Taking adjoints of the selection identity gives

$$\mathbf{1}_{\mathbb{R} \setminus (s + \mathbb{Z}_{\geq 0})}(H)T_s^* = 0.$$

Write $l(0) = E^2 u_l$. Its auxiliary temporal primary has weight two, so its actual affine label is $u_l = T_2^* v_l$ for its canonical label v_l ; it has actual weights in $2 + \mathbb{Z}_{\geq 0}$. Likewise $J_e(0) = E^3 u_{J_e}$ has $u_{J_e} = T_3^* v_{J_e}$ and actual weights in $3 + \mathbb{Z}_{\geq 0}$. These are adjoint support statements, not commutation of actual and auxiliary weight projections. The temporal virial wave has E^3 -label $iu_l + u_{J_e}$. Its physical real localization and dimension three bound its actual weights above by three. It therefore has only weights two and three. The weight-two label is iu_{S_e} , where

$$u_{S_e} = P_{h=2}^{\text{act}} u_l, \quad S_e(0) = E^2 u_{S_e}. \quad (6.4)$$

This is a vacuum-wave definition.

The derivative of a primary of weight two has an inhomogeneous transformation term. Differentiating its primary transformation, and adding the weight-three part of V_e , gives for a small real non-affine time-Möbius map g

$$U_e(g)V_e(ze)\Omega = (g'(z))^3 V_e(g(z)e)\Omega + 2g'(z)g''(z)S_e(g(z)e). \quad (6.5)$$

Choose an interval on which $g'g'' \neq 0$. After changing variables and multiplying by smooth real coefficients, this equation expresses any smearing of S_e there as a difference of a physical virial smearing and an actual modular transform of one. Consequently, for a real test supported in a line interval whose diamond lies in O , $S_e(f) \in X_{\text{phys}}(O)$. Complex linear extension gives its adjoint-wave reality relation; this is the meaning of a Hermitian wave here. It is even and physically real localized on all translated lines. Translation covariance defines it as a spacetime Hilbert-valued distribution. For example $\|e^{-\epsilon P_e} S_e(0)\| = O(\epsilon^{-2})$, and $|\mathbf{P}| \leq P_e$ supplies all required spacetime test bounds. The family is Lorentz covariant in e , a rotational scalar about e , and homogeneous of dimension two.

6.4 Independence of the time direction

Proposition 6.4. *The waves S_e agree for all future timelike unit vectors e . Their common value S is a purely timelike Lorentz scalar of dimension two, with physical real localization. It has the canonical null special-conformal action. It can be included in the auxiliary representation and in Lemma 6.1.*

Proof. There are three issues to settle: the timelike spectral densities, the possible massless component, and the action of the actual null generators on the resulting scalar. The projection identity (6.4) will link the first two issues: comparison on one energy slice fixes both the normalization and the dependence on the time direction.

We first prove a two-point claim: the timelike entries of the joint spectral measure of the family S_e and l are constant multiples of the ordinary measure d^4p on the future timelike cone. No density is assumed in this claim. Write $\mu_{ee'}$ for a joint spectral entry, including l as an additional label. The corresponding commutator distribution is

$$C_{ee'}(x) = \int e^{ip \cdot x} d\mu_{ee'}(p) - \int e^{-ip \cdot x} d\mu_{e'e}(p).$$

Every invariant-mass projection commutes with wedge Tomita data. Wedge separation and reality therefore give causal support for this commutator after projection to any compact mass interval. For l , the real wedge data are the canonical scalar data already identified in Section 2; physical point-field membership is not needed.

Homogeneity disintegrates the timelike measure as $M^3 dM d\nu(p/M)$, with a polynomially bounded measure on the unit hyperboloid. Testing arbitrary mass intervals shows that the shell commutator has causal support for almost every mass; rescaling then gives the same statement for the reference shell. This is a disintegration of distributions, not an assumed field on a fixed mass shell. The shell commutator solves the Klein–Gordon equation. Its displacement and normal-derivative Cauchy data at zero time are supported at the spatial origin, and hence are finite sums of delta derivatives. Fourier solution of this Cauchy problem expresses its shell measure as the ordinary commutator measure times a polynomial in p/M , modulo $p^2 = M^2$. In particular this proves absolute continuity rather than presupposing it.

For a self-pair the shell polynomial is even under $p \mapsto -p$. Take its representative of least degree modulo the shell relation. If its highest degree is k , that homogeneous term is nonzero on an open set of null directions; otherwise divisibility by p^2 would reduce the degree. At fixed nonzero energy and $M \downarrow 0$ it contributes order M^{-k} to the density with respect to d^4p . The transverse mass integration is $M dM$. Positivity and local finiteness therefore require $k < 2$. Evenness leaves only a constant. Positivity of the matrix measure bounds each cross density by the geometric mean of the self densities. A polynomial on the unit hyperboloid with such a global bound is constant by the same highest homogeneous term argument. All timelike density entries are therefore constant, proving the two-point claim. Covariance makes the self constant of S_e independent of e .

There is no zero-momentum part. We must also exclude a massless part; positivity alone would allow one. Let $\rho = \sqrt{-W_\mu W^\mu}$ be the massless Pauli–Lubanski length and project the entire Lorentz family onto a compact band $0 < a \leq \rho \leq b < \infty$. This projection commutes with wedge modular data. Its commutators are causal massless solutions. The preceding point-supported Cauchy-data argument makes them derivatives of the massless commutator, hence supported on the light cone in four dimensions.

On the complex cyclic span of this projected family, let X_+ and X_- be the real spaces generated in the future and past cones. Translation-spectrum uniqueness makes both spaces complex cyclic: a vector orthogonal to smearings on an open cone is orthogonal to all translates by tube uniqueness. Huygens support gives $X_- \subset X'_+$, so X_+ is standard. Lorentz transformations preserve it and hence commute with its modular group. Every future translation maps it into itself. The positive-generator modular translation relation therefore scales all P_μ by one common nontrivial exponential under that modular group. Because the Lorentz generators commute with the group, ρ scales by the same factor. No nonzero spectrum contained in $[a, b]$ is invariant under all these rescalings. The projected space is zero. Exhausting positive bands excludes nonzero Pauli–Lubanski length.

The remaining massless representations have trivial little-group translations and decompose into helicities. A rotational scalar about e has only helicity zero, since a spatial rotation fixing both e and a massless momentum acts trivially on its wave. Use the scalar massless invariant measure. The possible internal part of dilations is a momentum-independent unitary group W on scalar multiplicities. Lorentz covariance and dimension two give the amplitude

$$S_e(p) = (p \cdot e)W(\log(p \cdot e))\beta. \quad (6.6)$$

For two directions the commutator argument makes the cross density a homogeneous quadratic shell polynomial. A quadratic scalar invariant under rotations fixing e, e' is, on the null shell, a linear combination of $(p \cdot e)^2$, $(p \cdot e)(p \cdot e')$ and $(p \cdot e')^2$. It follows that

$$\varphi(u) = \langle \beta, W(\log u)\beta \rangle, \quad u = \frac{p \cdot e'}{p \cdot e},$$

is a linear combination of $u^{-1}, 1, u$ on the attainable interval. As the relative rapidity grows these intervals exhaust $(0, \infty)$; the coefficients agree on overlaps. Since $|\varphi(u)| \leq \|\beta\|^2$, both nonconstant coefficients vanish. Thus the massless cross density is $c(p \cdot e)(p \cdot e')$, with $c = \|\beta\|^2 \geq 0$.

The only massless possibility left is the scalar density just displayed. Its growth with the relative rapidity of e and e' is incompatible with the bounded timelike cross densities. The next energy-slice calculation makes this comparison with its exact normalization.

Now apply the defining projection in (6.4) to $F = S_{e'}$. A dimension-two wave with physical real localization has actual weights at most two; u_l has actual weights $2, 3, \dots$. In the multiplicity fiber at $P_e = 1$,

$$\langle u_F, u_{S_e} \rangle = \langle u_F, u_l \rangle. \quad (6.7)$$

Let $B(e', e)$ and $D(e')$ denote the constant timelike density entries for $(S_{e'}, S_e)$ and $(S_{e'}, l)$, respectively. Write $b = B(e, e)$, which is independent of e by covariance, and let ρ_l be the self density of l . With timelike measure d^4p and massless measure $\theta(p^0)\delta(p^2)d^4p$, (6.7) reads explicitly

$$\frac{4\pi}{3}B(e', e) + 2\pi c(e \cdot e') = \frac{4\pi}{3}D(e'). \quad (6.8)$$

Indeed the timelike slice $p \cdot e = 1$ is the unit spatial ball in the rest frame of e , of volume $4\pi/3$. Lorentz invariance of d^4p makes this normalization independent of the slicing direction. On the

massless slice the angular integral is $\frac{c}{2} \int_{S^2} (e^0 - \mathbf{e}' \cdot \mathbf{n}) d\Omega = 2\pi c(e \cdot e')$. Positivity gives $|B(e', e)| \leq b$ and $|D(e')| \leq \sqrt{b\rho_{ll}}$ uniformly. Arbitrarily large relative rapidity therefore forces $c = 0$. Setting $e = e'$ in (6.8) now gives $D(e') = b$, and hence $B(e', e) = b$ for all pairs. Thus $\|S_e(f) - S_{e'}(f)\|^2 = 0$ for every test f . It also shows that the constant two-point density of S with $l - S$ is zero. This is orthogonality of the scalar wave channels, including all their spacetime translates; it does not assume that the actual time-weight projection commutes with spatial translations.

Finally, physical real localization supplies the scalar order-zero parabolic comparison for S . We can also apply the proof of Lemma 5.6 to its smooth compact vacuum smearings in the same strictly one-sided placements, and then to all their strictly energy-damped translates. That proof requires real localization, giving the Tomita bound after parabolic transport, and boost smoothness of every order. Both are available here: localization was obtained from (6.5), and the scalar covariance just proved identifies every iterated boost derivative with another compact smooth S -smearing. Its Tomita and Hardy estimates therefore give the asserted K_n domains without treating S as an operator. Symmetry against the same dense core transfers the even second difference of (5.22), $K_n(s) + K_n(-s) - 2K_n = 4s^2a$, to these vectors as a form identity. The D' term cancels, so no D' operator domain for S is assumed. The second-difference and zero-capacity argument of Theorem 5.7 therefore applies: the nonnegative comparison defect is translation-diagonal in M^2 , and is bounded by $\int_0^\infty M|f'(M)|^2 dM$. Logarithmic cutoffs equal to one on a prescribed compact mass interval make this bound tend to zero. Spectral comparison then gives reduction and the exact canonical null action. Its wedge reality and two-point kernel are scalar, so the proof of Lemma 6.1 includes this enlarged canonical space as well. $\square$

Corollary 6.5. *Put $\ell = l - S$. Its insertion at imaginary infinity annihilates every compact physical vacuum polynomial:*

$$\lim_{R \rightarrow \infty} R^4 \langle \ell(iRe), A\Omega \rangle = 0. \quad (6.9)$$

Proof. The multiplicity label u_ℓ has actual weights $2 + j$, $j \geq 1$. First take a Hermitian polynomial localized in a diamond on the line. Its vacuum vector ψ is in the corresponding real space. The scalar quotient

$$F(E) = E^{-2} \langle u_\ell, \psi(E) \rangle$$

is regular at zero, with continuous evaluation, by Lemmas 6.1 and 6.3. For compact actual weight-band projections of ψ , its overlap with u_ℓ has only positive integral powers E^{h-2} , so $F(0) = 0$. These projections preserve the interval real space and converge in Hilbert norm; continuity of evaluation proves the same equality without the band cutoff. With the common dE/E normalization,

$$\langle \ell(iRe), \psi \rangle = \int_0^\infty E^3 e^{-RE} F(E) dE.$$

The contribution near zero is $o(R^{-4})$, and the complementary contribution is exponentially small by Cauchy–Schwarz. Every compact polynomial is a complex linear combination of Hermitian ones, and its support fits in a line diamond. For a compact joint polynomial kernel, the same real-space statement is the closed Tomita-graph limit of factorized tests, using the old joint-distribution continuity from Section 2. This proves the claim. $\square$

At this point S is a direction-independent scalar vacuum wave with physical real localization, and $l - S$ has the stated decay against physical polynomial vectors. To multiply S with other entries we need quantitative bounded operations, including control of both their vacuum and adjoint-vacuum errors.

6.5 Bounded approximants for accessible waves

Call a vacuum wave *accessible* when it has the approximation property below, including the corresponding adjoint wave. Physical fields and S have this property. This definition does not assert that an accessible wave already acts on a common field domain. In the following norm estimates, $a(f)$ denotes a vector: for an original field it is its smeared vacuum vector, and for S it is its Hilbert-valued smearing. The notation $a^\dagger(\bar{f})$ denotes the corresponding adjoint-vacuum vector.

Proposition 6.6. *Let a be a physical field or S , and let f have support in a small real source patch. Given a double-cone neighborhood of its support, there are bounded operations $A_R(f)$ in that neighborhood, for dyadic $R \rightarrow \infty$, such that for one fixed exponent q*

$$\|A_R(f)\| \leq CR^q, \quad (6.10)$$

$$\|A_R(f)\Omega - a(f)\| + \|A_R(f)^*\Omega - a^\dagger(\bar{f})\| \leq C_N R^{-N} \quad (N = 1, 2, \dots). \quad (6.11)$$

For each prescribed N , constants are uniform on balls of finitely many test seminorms. When patches and localization margins shrink to size ε in a fixed compact set, these constants cost at most fixed inverse powers of ε . The approximants need not depend linearly on f .

Proof. For an ordinary field smearing with closed affiliated realization $A = v|A|$, take its polar cutoff $A_R = v|A|\mathbf{1}_{[0,R]}(|A|)$. It has norm at most R . The tails on the vacuum and adjoint vacuum satisfy, for every positive integer m ,

$$\begin{aligned} \|(A - A_R)\Omega\| &\leq R^{-m} \| |A|^{m+1} \Omega \|, \\ \|(A^* - A_R^*)\Omega\| &\leq R^{-m} \| |A^*|^{m+1} \Omega \|. \end{aligned}$$

Here the absolute-value moments follow from the original realization hypothesis together with the generated-domain action check in Section 2. Both A and A^* preserve the dense invariant vacuum-generated domain $\mathcal{D}_{\text{old}}$ and agree there with the physical field and its polynomial adjoint. Induction therefore gives

$$\mathcal{D}_{\text{old}} \subset \text{Dom}(A^*A)^m \cap \text{Dom}(AA^*)^m \quad (m = 1, 2, \dots),$$

and these powers on that domain are the corresponding alternating polynomial words. Their vacuum norms are finite and are bounded by finitely many seminorms of the source tests, by the joint Wightman distributions. Since $|A|^{2m} = (A^*A)^m$ and $|A^*|^{2m} = (AA^*)^m$, spectral calculus also gives every lower integer absolute-value moment used in the tail bounds. This uses the adjoint-extension clause of Assumption 2.1 and does not require essential self-adjointness.

The invariant polynomial domain supplies these moments, uniformly on finite test-seminorm balls; subdivision supplies the requested local neighborhoods. Use a Hermitian basis or adjoint-paired cutoffs to preserve adjunction and grading.

For a nonhomogeneous original source, any comparison of shrinking patches by dilation is made on its finitely many homogeneous components and then summed. For this fixed field and requested accuracy, a maximum of the finitely many seminorm orders and powers is sufficient. This does not assign the source a single scaling dimension or require a bound uniform in the set of dimensions, and it does not change the fixed polynomial norm exponent in (6.10).

For S the order of construction matters. We truncate the ordinary source field before applying the actual modular unitary, and then add the resulting bounded operations. We will estimate the sum by its absolute quadrature weights, so no domain or moment estimate for an unbounded sum of modularly transformed fields enters.

For S , use (6.5) on parallel lines. After undoing g in the test, a line smearing of S is a fixed linear combination of a line virial smearing and a modularly transformed line virial smearing, with smooth real coefficients. By Lemma 6.2, transverse approximation at width $\delta_R \leq e^{-R}$ has vacuum error $O(\delta_R)$, and every fixed source-field moment is bounded uniformly as $\delta_R \downarrow 0$. Its adjoint data obey the same estimates.

Polar truncate each such source field at R and apply the actual Möbius unitary. It fixes the vacuum and transports the bounded operation into the destination diamond, so both estimates in (6.11) survive. A transverse quadrature of mesh at most e^{-R} approximates a full spacetime smearing. Although its number of terms grows, the sum of absolute quadrature weights is bounded. Hence the sum of the operator norms is $O(R)$, and the sum of the vacuum errors remains rapidly small. No moment bound for a sum of noncommuting transformed fields is being assumed.

All the operations just described use finitely many derivatives for each requested accuracy. Rescaling tests and using diamonds with margins proportional to ε therefore introduces only inverse powers of ε . The fixed non-affine map can be chosen nonsingular on the entire compact parameter patch, with $g'g''$ bounded away from zero. Take $\delta_R \leq c\varepsilon e^{-R}$ and sufficiently fine quadrature to retain these margins. Subdivision completes the construction for an arbitrary compact source support. $\square$

The norm exponent in (6.10) is fixed before an error order N is requested. Increasing N changes the finite collection of moments and test seminorms used to estimate the error, but does not change that exponent. This distinction allows rapid vacuum errors to dominate the finitely many polynomially growing factors in a product.

6.6 Holomorphic products, ties, and unsmearing

For p accessible entries, use increasing imaginary translations:

$$\text{Im } z_1 \in V_+, \quad \text{Im}(z_{j+1} - z_j) \in V_+ \quad (1 \leq j < p). \quad (6.12)$$

The first inequality is needed for vector products. With tests fixed in sufficiently small patches, form actual bounded vector products

$$F_R(z) = U(z_1)A_{1R}U(z_2 - z_1)A_{2R} \cdots U(z_p - z_{p-1})A_{pR}\Omega. \quad (6.13)$$

They are holomorphic in (6.12), with norm at most a fixed power of R . Their scalar vacuum expectations have the larger tube determined only by the successive imaginary gaps: vacuum invariance removes the first factor $U(z_1)$. Equivalently, add a common future imaginary translation to put all the arguments in the vector tube, then pair with Ω . The scalar answer is independent of this common translation. We use W for its limit and retain Ψ for the vector limit.

Proposition 6.7. *The vectors (6.13) have a unique, locally uniform holomorphic limit. It is multilinear in the accessible waves and has Hilbert-valued distributional boundaries at mutually strictly spacelike real placements. These boundaries are graded symmetric. The limits extend to common holomorphic neighborhoods of imaginary ties: successive blocks may have equal imaginary translation when real placements within each block are pairwise spacelike and all inequalities between blocks remain strict. The limits unsmear to holomorphic point vectors, denoted Ψ , depending only on the total complex insertion arguments. For physical entries alone they agree with the usual Wightman vectors. For any two finite accessible words, with each word in its strict vector tube, their scalar functions satisfy*

$$\begin{aligned} & \langle \Psi[A_1(z_1) \cdots A_p(z_p)], \Psi[B_1(w_1) \cdots B_q(w_q)] \rangle \\ &= W(A_p^\dagger(\bar{z}_p), \dots, A_1^\dagger(\bar{z}_1), B_1(w_1), \dots, B_q(w_q)). \end{aligned} \quad (6.14)$$

An empty word denotes Ω . Source tests on the bra word are conjugated and reversed, together with their field labels. The identity holds for distinct words and their finite linear combinations at every finite total length. It asserts analytic Hilbert splittings and their spacelike real boundaries, without yet asserting a distributional boundary at every real configuration.

Proof. We first prove convergence where locality lets us estimate the whole product by vacuum errors. Holomorphic propagation then carries that convergence into the ordered tube. Finally, bounded-product continuity at a spacelike edge gives the common neighborhoods needed for ties; translation compatibility removes the source smearing.

At real translations making all approximation patches mutually spacelike, telescope two bounded products, whether at $R, 2R$ or from different approximation choices. Strong graded locality moves each difference to the vacuum. Its vacuum error is rapidly small by (6.11), while all other factors have polynomial norm growth by (6.10). The product difference is therefore $O_N(R^{-N})$ for every N , uniformly on smaller placement patches and appropriate test-seminorm balls. This proves independence, graded symmetry and multilinearity at these real placements.

We record the propagation estimate, since vacuum convergence alone would not justify the products elsewhere. For a holomorphic vector difference G_R bounded by CR^q , restrict to a complex line $z = x + \zeta\eta$ with η in the ordered cone. A real interval I can be chosen inside a good spacelike-placement patch. The subharmonic logarithm of its norm and harmonic measure give

$$\log \|G_R(x + \zeta\eta)\| \leq \omega(\zeta, I)(\log C_N - N \log R) + (1 - \omega(\zeta, I))(\log C + q \log R). \quad (6.15)$$

On compact subsets reached this way ω is bounded below by a positive number. Interior three-circle chains propagate the same conclusion to every fixed compact subset of the connected tube. As N is arbitrary, convergence is rapid there. Above a smaller good real patch it is rapid uniformly up to the boundary, because ω has a uniform positive lower bound for bounded approach directions. Summing the dyadic differences proves existence.

For ties, first keep the common translation of each block and its spectators strictly inside their allowed complex domains. The relative translations within a block have wedges of opposite imaginary-gap cones for its order and its reversal. At their common real edge the bounded products agree, with the graded sign, by strong locality. Each bounded product has a norm-continuous extension to that edge: the real translation groups and the positive-energy contraction semigroups are strongly continuous, and only finitely many bounded factors occur. On fixed smaller wedges and their common edge its norm is $O(R^q)$. The continuous-edge form of the local edge-of-the-wedge theorem therefore applies directly; its continuation neighborhood depends only on the wedges and the edge patch, and its interior bounds are controlled by the supremum bounds on those wedges and their edge; see [8, Theorems 1' and 2].

To make uniformity explicit, apply the theorem after scalarization to the Hilbert-valued family $(R^{-q-1}F_R)_R$ in the square-summable direct sum, with dyadic R . Its common edge is norm continuous: finitely many components are continuous and the remaining squared norms are bounded uniformly by a convergent tail of $\sum_R R^{-2}$. Scalarization gives a bounded holomorphic Hilbert-valued continuation on one common neighborhood, and each unweighted component there has a bound $O(R^{q+1})$. The same supremum-bound argument is uniform on the prescribed test-seminorm balls; no estimate from boundary distribution seminorms alone is being asserted. Interior propagation from the strict tube now gives rapid convergence on compact subsets of these neighborhoods. Adjacent permutations within a block are treated on overlapping smaller neighborhoods, where uniqueness glues the continuations.

By uniqueness from the spacelike real patch, multilinearity persists throughout these domains. The seminorm bounds and the test-function kernel theorem give joint distributions in the source

variables, holomorphic in the translation variables. Translate a source test within its patch. At real spacelike placements this is the same as translating its insertion argument; the equality involves only the vacuum wave and hence survives all approximation choices. Analytic uniqueness extends this equality. If x denotes source variables, the kernel in coordinates $(x, \operatorname{Re} z + x, \operatorname{Im} z)$ is therefore independent of its first variables in the distributional sense. It satisfies the Cauchy–Riemann equations in the remaining variables. Thus it is a holomorphic Hilbert-valued function of $x + z$, rather than merely a distribution in x . This proves unsmearing. The same identity glues source patches and their translations.

For the split identity first keep all source tests smeared, and use the same bounded approximants on the two sides, taking the adjoints of the bra approximants in the scalar word. This is an admissible choice because both vacuum and adjoint-vacuum errors were controlled. Bounded adjunction gives (6.14) before taking limits. The displayed scalar word lies in its ordered scalar tube: its reversed bra gaps and ket gaps are future directed, and its central gap has imaginary part $\operatorname{Im} w_1 + \operatorname{Im} z_1 \in V_+$. Local uniform convergence of the two vector factors and of the scalar word therefore proves the identity there. The preceding unsmearing then gives the point formula. Conversely, at any cut between two successive entries of an ordered scalar tuple, choose a common imaginary shift strictly between their imaginary parts. Subtracting that shift puts the reversed-adjoint prefix and the suffix in the two strict vector tubes of (6.14). Thus the formula supplies the analytic Hilbert splitting of every ordered scalar tuple at every cut. Finite linear combinations retain all cross terms. At mutually spacelike real placements, the same identity follows from the Hilbert-norm limits already proved on those patches; no unrestricted real-axis limit is needed.

For physical fields, polar truncations converge on the polynomial domain as well as on the vacuum. At mutually spacelike real placements their product limit is consequently the ordinary smeared product. For a joint source kernel, finite tensor-test approximation and the joint distribution estimates extend this equality; the kernel itself need not have finite tensor rank. Tube uniqueness identifies their point vectors everywhere. Finally the estimates above give distributional boundary convergence locally uniformly on the good real patches. In any closed strict subcone of approach, point bounds near such an edge are polynomial in inverse distance: test the real variables on balls at that scale and use the holomorphic mean-value estimate on smaller complex balls. $\square$

6.7 A boundary pullback rule

The preceding products will be composed with analytic coordinate changes before their real limits are taken. The following formulation specifies the required boundary operation.

Lemma 6.8. *Let a Hilbert-valued holomorphic function have a local distributional boundary u and polynomial distance bounds in closed strict approach subcones. Let Φ be a real analytic local diffeomorphism whose complexification takes a smaller wedge strictly into such approach directions. Then its composition with Φ has boundary $\Phi_{\mathbb{R}}^* u$ on that smaller wedge.*

Proof. Shrink the real patch and the approach cone so that the image of a small imaginary translate of size δ is a graph at distance comparable to δ inside one fixed strict convex subcone. Interpolate that graph to a straight imaginary translate there. Suppose the holomorphic function is $O(\delta^{-m})$ on this interpolation. Extend a compact smooth test almost analytically to order N : its $\bar{\partial}$ is $O(\delta^N)$, uniformly with the finitely many derivatives required on the patch. Stokes’ formula for the holomorphic volume form compares the two integrals. Their error is bounded by a constant times δ^{N+1-m} ; take $N > m + 1$.

There is a minor test-function issue in this comparison. The desired coefficient on the moving graph is a fixed smooth real test, whereas an almost-analytic extension evaluated on that graph has

nonzero imaginary Taylor terms. Include the holomorphic Jacobian, and choose the real test on the straight surface by inverting these Taylor terms through order N . More explicitly, the graph evaluation operator on test jets is $\mathbf{1} + \delta B_1 + \cdots + \delta^N B_N$, where each B_j is a differential operator with smooth coefficients. It has a unique inverse modulo δ^{N+1} , obtained successively in each power of δ . This is the negative-imaginary-displacement Taylor correction. Its corrected real tests converge in every smooth seminorm to the ordinary real change-of-variables test; the remaining graph error is $O(\delta^{N+1-m})$. The boundary limit on the straight surface is therefore precisely the distributional real pullback. Pairing with arbitrary Hilbert vectors proves the weak statement, and the uniform estimates prove the asserted Hilbert-valued distributional convergence. $\square$

7 Short products and the return to the accessible scalar

The scalar vacuum wave l has not yet been realized as a physical field. In this Section, L is an insertion symbol whose one-vacuum wave is l ; it is not an operator on the polynomial field domain. We construct only products of total length at most four, including derivatives, and prove exactly the factorization statements needed below. The symbols

$$J = V - \partial L, \quad t = T - \frac{1}{3}(\eta \square - \partial \partial) L, \quad \ell = L - S \quad (7.1)$$

initially have this same meaning. At the end we prove $l = S$ as vacuum waves. Construction of mixed products at every finite length and of the physical operator L is deferred to Section 8.

We use the ordered-vector-tube convention of Section 6: the vacuum is on the right, and

$$\Im z_1 \in V_+, \quad \Im(z_{j+1} - z_j) \in V_+. \quad (7.2)$$

The strict first inequality is necessary for vector products. Scalar vacuum functions depend only on differences and require only the remaining inequalities. A derivative always acts on its own insertion. All entries used in this Section are even. Tensor indices and source smearings in the following constructions are retained, even when suppressed in the notation.

The length bound is chosen to support a specific Hilbert-space argument. Actual bounded products and finite interpolation will first construct the short vectors and their split identities. Their four-point scalar functions then determine inner products of pairs, which give a conformal representation on the pair space. Its radial positivity and the normalized current fluxes will force the residual wave $l - S$ to vanish.

7.1 Finite interpolation on spacelike patches

Lemma 7.1 (Common wedge for small null flows). *Fix any finite family of sufficiently small, mutually spacelike real source patches. There are a point o strictly to their future, an open set of future null directions n , and, for each finite set of sufficiently small null special-conformal parameters in that set, an opposite open cone O_- with the following property. Bounded operations on the source patches, or on their inverse coordinate images, after conjugation by the actual centered null-charge unitaries lie in wedges spacelike to O_- . The vectors $C\Omega$, with $C \in \mathcal{M}(O_-)$, form a total set. These choices persist under sufficiently small changes of the source patches and Lorentz transformations.*

Proof. Choose a future null direction m separated from the directions n and use m as the conjugate wedge axis. Work strictly below the null cut plane centered at o . In the resulting coordinates put $p(y) = |y|^2$. Each source patch fits, with a strict margin, inside a cut $W_{\alpha p+c}$ with $\alpha > 0$

sufficiently small and c sufficiently negative. The same containment holds for the finitely many inverse coordinate images when the flow parameters are small enough.

Here is a two-sign bound uniform in quadratic truncation. Choose smooth bounded profiles $0 \leq q_R \leq p$, increasing to p , and write $U_R(t) = e^{itH_{q_R}}$. For $0 < t < \alpha$, cut inclusion and the constant-shift identity give

$$\mathcal{M}(W_{\alpha p+c}) \subset \mathcal{M}(W_{tq_R+c}) = \text{Ad}(U_R(t))\mathcal{M}(W_c).$$

Thus conjugation by $U_R(-t)$ sends the source algebra into the fixed wedge algebra $\mathcal{M}(W_c)$. For positive t , the source algebra is already contained in $\mathcal{M}(W_c)$, and

$$\text{Ad}(U_R(t))\mathcal{M}(W_c) = \mathcal{M}(W_{tq_R+c}) \subset \mathcal{M}(W_c).$$

Both signs therefore have the same containing wedge for $|t| \leq t_0 < \alpha$, independently of R . This argument uses $q_R \leq p$, not a bound on the diverging norms $\|q_R\|_\infty$. The positive moment construction of Section 3 gives $U_R(t) \rightarrow e^{itK_n}$ strongly. Weak closedness of the fixed wedge algebra now preserves the containment for each bounded conjugated operation in this limit. Translations give the centered version used here.

For finitely many choices, take $m \cdot x$ sufficiently negative, then add a sufficiently large positive multiple of m . This leaves $m \cdot x$ unchanged and makes all the finitely many $n \cdot x$ sufficiently positive. A small open double cone about such an x lies in the required opposite region. All inequalities are strict, hence stable under the small changes asserted in the statement. Local vacuum cyclicity in this double cone proves totality. No cyclicity of a relative commutant is used. $\square$

Let g be the standard null special-conformal coordinate map centered at o , with small parameter vector $b_g = sn$, and let U_g denote the actual unitary generated by the corresponding centered K_n . On the one-field space these actions agree canonically by Theorem 5.7. This does not assert covariance on the full physical Hilbert space. Reparametrize an input at $g^{-1}x$ and multiply its test and indices by the inverse primary factors at x . The transformed one-vacuum data from accessible inputs S, V, T are then

$$S(x), \quad V(e_j; x) + \frac{4b_g \cdot e_j}{1 + 2b_g \cdot (x - o)} L(x), \quad t(x) + \mathcal{P}_g L(x). \quad (7.3)$$

Here e_j is a fixed target polarization, and $\mathcal{P}_g$ is a tensor-valued differential operator of order at most two with analytic coefficients on the patch. It is obtained by transforming $(\eta\Box - \partial\partial)L/3$ as derivatives of a scalar primary of weight two. In particular $\mathcal{P}_0 = (\eta\Box - \partial\partial)/3$. The coefficient in the vector formula is the derivative of the scalar primary factor $(1 + 2b_g \cdot (x - o))^2$; this fixes its normalization. All these identities also hold after differentiation and source smearing.

Lemma 7.2 (Finite null interpolation). *For $q \leq 4$ and generic independent polarizations $e_1, \dots, e_q$, finite evaluations at arbitrarily small future null b distinguish the 2^q functions*

$$b \mapsto \prod_{j \in I} \frac{4b \cdot e_j}{1 + 2b \cdot (x_j - o)}, \quad I \subset \{1, \dots, q\}, \quad (7.4)$$

uniformly for x_j in sufficiently small compact source patches. The inverse matrix is analytic in the positions, also in a complex neighborhood where its determinant stays nonzero.

Proof. Complete the polarizations to a generic basis and put $z_j = b \cdot e_j$. A polynomial in the square-free monomials z_I that vanishes on an open part of the null cone vanishes on the full complex

quadric and is divisible by its nondegenerate quadratic equation $Q(z)$. This last assertion follows, for example, by solving the quadric for one variable on a nonsingular chart and applying polynomial division; the remainder vanishes on an open set and is zero. In a generic basis Q has a nonzero z_j^2 coefficient for every j . If QR were nonzero and multi-affine, then $\deg_{z_j}(QR) = 2 + \deg_{z_j} R \geq 2$, a contradiction. Thus the square-free monomials are independent on every such cone patch. There is consequently a finite nonsingular evaluation matrix. Varying ray lengths separates its homogeneous degrees.

Scale all its sample vectors by $\varepsilon > 0$ and divide column I by $(4\varepsilon)^{|I|}$. As $\varepsilon \downarrow 0$, the matrix in (7.4) converges uniformly on compact source patches to the nonsingular monomial matrix. Its determinant remains nonzero for sufficiently small ε . Matrix inversion gives the last assertion. Thus the interpolation involves multiplication of distributions by smooth functions, never pointwise inversion of a distribution. $\square$

The limit $\varepsilon \downarrow 0$ in the preceding proof is used only to find a nonsingular evaluation matrix. In the construction below we fix one sufficiently small positive scale for the compact placement or continuation geometry under consideration. We then shrink the position neighborhoods so that its determinant stays bounded away from zero, and only afterward take the bounded- approximation and boundary limits. The inverse constants may depend on this fixed scale; a uniform inverse as the scale tends to zero is not needed.

Generic polarizations also suffice for exact tensor recovery. The decomposable tensors coming from admissible generic polarization tuples span the full tensor product: a linear functional vanishing on that open set is a multilinear polynomial vanishing identically. Choose a finite spanning subset once. Every requested tensor component is then a fixed linear combination of those evaluations, without inverting an evaluation matrix at a nongeneric tuple.

Proposition 7.3 (Hilbert products through length four). *For entries from t, S, V, L , and hence J, ℓ and their derivatives, there are Hilbert-valued product distributions on mutually spacelike real patches through total length four. They are multilinear, permutation-symmetric, and covariant under translations, Lorentz transformations and dilations. They satisfy every leg-local linear identity of their vacuum waves. With C in a common opposite cone, they obey*

$$\langle C\Omega, \Psi[O_1 \cdots O_p] \rangle = \langle O_1^\dagger \Omega, C^* \Psi[O_2 \cdots O_p] \rangle. \quad (7.5)$$

Smearings on the left side of the inner product are conjugated. The scalar functions factor at every split of total length at most four. Explicitly, for $p + q \leq 4$ and both words in their vector tubes,

$$\begin{aligned} & \langle \Psi[A_1(z_1) \cdots A_p(z_p)], \Psi[B_1(w_1) \cdots B_q(w_q)] \rangle \\ &= W(A_p^\dagger(\bar{z}_p), \dots, A_1^\dagger(\bar{z}_1), B_1(w_1), \dots, B_q(w_q)). \end{aligned} \quad (7.6)$$

Here an empty word is Ω , and bra source tests are conjugated and reversed. The reversed bra gaps, the ket gaps, and the central gap $\text{Im } w_1 + \text{Im } z_1$ are all future directed. Thus the scalar word is in its gap-only tube. In particular the case $p = q = 2$ gives the pair-pair Gram identities. All these products extend holomorphically to (7.2) and the spacelike tie neighborhoods of Proposition 6.7; the split identities persist wherever both vector factors are defined.

Proof. Proceed by induction on length, and at a fixed length by increasing number of t slots. Length one is the given one-vacuum construction. For accessible inputs use the localized bounded approximants of Proposition 6.6. Conjugate every factor of an approximating real product by the same U_g . Lemma 7.1 allows the first factor to commute past C . Vacuum-vector and adjoint-vacuum

convergence for this factor, and convergence of the shorter tail, therefore yield the right side of (7.5). By the shorter induction assertions this is the multilinear expansion of (7.3), initially as a functional of $C\Omega$.

Use T in positions intended eventually to contain t . Terms with fewer t slots have representing vectors already and can be subtracted: replacing a transformed T by $\mathcal{P}_g L$ lowers this induction index. Interpolation is then needed only for the remaining V/L choices. At the remaining V positions the unknown functionals are indexed by the subsets replaced by L , with the coefficients (7.4). Lemma 7.2 expresses each unknown as a finite smooth linear combination of already existing Hilbert-valued distributions. They therefore have representing vectors. Generic polarization choices suffice: multilinearity, followed by a finite choice of generic bases, recovers all components. This proves (7.5) at the new length.

The same argument with any other sufficiently small g proves the expansion formula, because it proves equality against the total family $C\Omega$ in a common opposite cone. Subdivision of patches or a change in all auxiliary choices gives the same vectors recursively by (7.5). Comparing a construction with a small translated, Lorentz-transformed or rescaled construction proves the similarity laws. Interpolating the symmetry of transformed accessible products proves spacelike permutation symmetry, again first subtracting terms with fewer t slots. Inserting a zero vacuum wave into (7.5), after permuting it to the first position, gives zero. Hence, in particular,

$$\square L = \partial \cdot V = \Theta, \quad \partial \cdot J = 0, \quad \partial^\mu t_{\mu\nu} = 0, \quad t^\mu{}_\mu = 0 \quad (7.7)$$

hold in these products. This argument concerns vector distributions, not operator-domain equalities.

For bounded approximants the vacuum expectation of a product splits at any position by Hilbert-space adjunction. The same is true for its common unitary conjugate. Expand the full scalar product and both factors by (7.3); real transformations respect adjunction. Subtract the terms already split by induction in the number of t slots. The identical finite interpolation now proves (7.6) for every remaining entry list, with the same adjoint and coordinate convention as (6.14). Thus short-length associativity and positivity come from actual bounded products.

It remains to continue the vectors, rather than merely their matrix coefficients. Fix a compact continuation path from a spacelike boundary patch to an interior tube point or one of the allowed ties. Choose the parameter scale in Lemma 7.2 small enough that every inverse image $g^{-1}z_j$ stays in the accessible product domain along that path. Near a tie use its open holomorphic neighborhood. The analytic inverse matrices remain nonsingular on a small neighborhood of the compact path. Applying U_g to the accessible analytic vectors and performing the same successive subtractions and inversions constructs the desired holomorphic vectors. The boundary calculus of Lemma 6.8 identifies their distributional boundary with the real construction: the coordinate maps take a smaller approach wedge strictly into the original wedge.

There is no choice of branch. On overlapping connected compact continuation neighborhoods choose parameters sufficiently small for both constructions; their boundaries agree on a smaller original patch, and distributional edge-of-the-wedge uniqueness makes the holomorphic vectors agree. Exhaustion gives the full ordered tube. At a tie the accessible germs and previously constructed terms agree across the relevant permutations, so the interpolated germs agree as well. All leg identities and similarity laws continue by the same uniqueness. Finally, the two sides of each split identity have equal boundary distributions when bra and ket are separately smeared on the original spacelike patches and approached in strict subcones. Uniqueness in their common split tube proves the asserted continuation of the identity. Only a local distributional boundary and an open convex approach subcone are needed here; no global boundary estimate for arbitrary L products has been assumed. $\square$

Lemma 7.4 (Comparison with the single-insertion forms). *The analytic pairs LV, VL, JV, VJ, tV, Vt agree, when paired with an ordinary physical polynomial bra, with the corresponding insertions of the tempered single- L forms of Proposition 5.3.*

Proof. First take a finite sum of factorized compact physical bra words and translate it into a common opposite cone. Its affiliated bounded truncations can be used as C in (7.5). With L first, the adjoints of these truncations acting on an ordinary smeared $V\Omega$ converge on the polynomial domain. Locality of the single-insertion forms therefore gives equality on an open real spacelike patch. For a compact joint bra kernel, approximate by factor tests in the permitted smooth topology, preserving the opposite-cone margin. Continuity of the vacuum and adjoint-vacuum graph vectors and of the single-insertion distributions extends this equality to that kernel. Give bra and pair strictly separated ordered imaginary translations. The known single-insertion spectral cones give analyticity in the two pair arguments and the common translation of the bra, with its internal spectator tests fixed. Analytic uniqueness extends the equality, and one may remove the separate bra damping while keeping the pair damped. The other pairs follow from their linear differential definitions. $\square$

7.2 The Ward rows and the alternating exception

The vector construction supplies finite transformation identities for products formed from the accessible entries. To obtain the canonical Ward identity for each interpolated entry separately, we must recover every row of the triangular system mixing V and L . The possible four-slot alternating term is the one case not removed by the square-free interpolation argument; its elimination uses the purely timelike spectrum of the last scalar wave.

Proposition 7.5 (Conformal covariance of the short scalar functions). *All scalar vacuum functions W of total length at most four obey the infinitesimal conformal Ward identities, with L, S, J, t primary of weights 2, 2, 3, 4 and spins 0, 0, 1, 2, respectively. The laws for V, T are their derivative laws from (7.1).*

Proof. Fix the t and S positions. At the remaining $m_0 \leq 4$ positions start with polarized vectors $V(e_j)$, and let W_I be the function obtained by replacing the positions in I by L . The canonical special-conformal Ward operator $\widehat{K}_a$ acts diagonally by the primary differential actions and has off-diagonal part

$$(\widehat{K}_a W)_I|_{\text{off}} = 4 \sum_{j \notin I} (a \cdot e_j) W_{I \cup \{j\}}. \quad (7.8)$$

These operators commute: they are the canonical commuting special transformations of primaries and their derivatives. They may be centered at o , since the difference from another center is an affine Ward operator, already zero on every row.

Unitarity and the expansion in Proposition 7.3 give

$$(e^{s\widehat{K}_n} W)_{\emptyset} = W_{\emptyset} \quad (7.9)$$

for small s and the open family of null n . Initially the tensor positions contain T . Induct in their number, substitute $T = t + \mathcal{P}_0 L$, and subtract terms with fewer t positions, whose covariance is already established. This gives (7.9) with the indicated t positions.

Write $X_I(a) = (\widehat{K}_a W)_I$. Differentiating (7.9) once and using the linear span of the open null family gives $X_{\emptyset}(a) = 0$ for every a . Suppose that $X_I = 0$ when $|I| < k$. Commuting two Ward

operators on a row H of size $k - 1$ and differentiating (7.9) $k + 1$ times give, respectively,

$$\sum_{j \notin H} z_j(a) X_{H \cup \{j\}}(b) = \sum_{j \notin H} z_j(b) X_{H \cup \{j\}}(a), \quad (7.10)$$

$$\sum_{|I|=k} z_I(n) X_I(n) = 0, \quad (7.11)$$

where $z_j(a) = a \cdot e_j$ and $z_I = \prod_{j \in I} z_j$. Indeed all diagonal differentiations of already zero rows vanish; the first surviving paths through the triangular off-diagonal array are exactly the $k!$ orders of adding the same k indices. The common nonzero factors 4 and $k!$ have been canceled.

Complete a generic set of polarizations to a basis and write $X_I(a) = \sum_{h=1}^4 z_h(a) x_{Ih}$. Equation (7.10) says that, for each H , the matrix with rows outside H equal to $x_{H \cup \{j\}, h}$ and its other rows zero is symmetric. If $k \geq 2$, choose $H = I \setminus \{j\}$. For $h \in I$ choose $j \neq h$; its zero row kills x_{Ih} . A completed-basis row $h > m_0$ likewise kills that entry. For $h \notin I$, symmetry identifies all the surviving entries with a coefficient $c_{I \cup \{h\}}$, depending only on the union. Thus (7.11) becomes

$$(k+1) \sum_{|B|=k+1} c_B z_B(n) = 0. \quad (7.12)$$

The square-free independence proved in Lemma 7.2 kills every coefficient, including the case where there is no such union.

For $k = 1$, (7.10) instead gives a symmetric matrix. Its quadratic form vanishes on the null cone, so it equals a scalar multiple of the metric quadratic. If $m_0 < 4$, its zero completed-basis rows force that multiple to vanish. If $m_0 = 4$, let $G_{ij} = e_i \cdot e_j$. The coefficient matrix is βG^{-1} , which says

$$X_{\{i\}}(e_j) = \beta \delta_{ij}. \quad (7.13)$$

The function β is multilinear in all four polarizations: row i is independent of e_i , and evaluation on e_i restores its linear factor. It vanishes when two polarizations coincide, by (7.13) and continuity from generic choices. It is therefore alternating. The sole possible exception is

$$X_{\{i\}}(a) = \lambda \epsilon(e_1, \dots, e_{i-1}, a, e_{i+1}, \dots, e_4). \quad (7.14)$$

Here there are exactly four V/L positions and no other entries. Divergence in position i of $X_\emptyset = 0$ also gives

$$\square_i X_{\{i\}} = 0. \quad (7.15)$$

For clarity, the canonical array preserves the relation $\partial \cdot V = \square L$: its difference is the divergence of the primary dimension-three vector J , whose divergence transforms homogeneously. This is the intertwining identity used in deriving (7.15).

We eliminate (7.14) using its last position, $i = 4$. Smear the first three positions on a real mutually spacelike patch, keeping the last argument in the future tube. Split the scalar function into the triple bra and the one-wave ket. Its boundary in the first three positions is obtained from negative ordered bra dampings. Move the center of the generator sufficiently far to the future of these compact supports; affine covariance leaves X unchanged. The last-position canonical differential action on L is exactly the canonical generator applied to $l(z_4)$. The ordinary physical triple VVV bra belongs to the domains of these centered actual K_n by Lemma 5.6. Moreover Π reduces these generators and their actions on Π are canonical by Theorem 5.7. Projecting the bra to Π thus allows the last-position generator to be moved to the fixed bra. Linear combinations of the available null parameters span arbitrary a .

Every other term of the smeared $X_{\{4\}}(a)$ pairs the unchanged last wave with a fixed constructed triple bra, possibly differentiated or with replacements. Consequently the whole expression is $\langle \eta, l(z_4) \rangle$ for a fixed Hilbert vector η , with no coordinate multiplier left on the last wave. Its translation spectrum is purely timelike, since that of l is. Equation (7.15) says that this spectrum is supported on $p^2 = 0$; these disjoint supports force it to vanish. More explicitly, after any fixed future damping it is a finite spectral pairing, and multiplication by p^2 kills it only if that pairing is zero off the null shell. There is no shell or zero-momentum component of l . Boundary uniqueness in the first three arguments and generic polarization then give $\lambda = 0$.

The induction on k is now complete, followed by the induction in the number of t positions. This proves every row of the canonical Ward system. $\square$

7.3 Euclidean continuation and the residual at infinity

Wick coordinates are $(\tau, \mathbf{r}) \mapsto i\tau e + \mathbf{r}$. Sort points by increasing τ and use the tie neighborhoods when heights coincide. A distinct Euclidean configuration has only spacelike equal-height ties, so the resulting scalar functions are real analytic on the full distinct-point configuration space. Use complex-linear tensor pullback with basis $C_0 = ie$ and real spatial bases C_j . The identities become

$$V = J + \partial L, \quad T = t + \frac{1}{3}(\delta\partial^2 - \partial\partial)L. \quad (7.16)$$

Here ∂^2 is the Euclidean Laplacian. Integrating the Ward identities along nonsingular local conformal flows gives their finite covariance, with reordering through tie neighborhoods as required.

For a scalar primary A of dimension two define

$$W(A_\infty \cdots) = \lim_{R \rightarrow \infty} R^4 W(A(Ru) \cdots), \quad |u| = 1. \quad (7.17)$$

The limit exists and has the primary transport law. To verify this, choose a Euclidean special-conformal transformation carrying infinity to a finite point disjoint from the spectators. Join it to the identity by a flow whose pole ray avoids the finite set of points, including the moving large point. At the final transformation every argument is distinct and finite, so ordinary local analyticity and the weight-two factor give the limit. This argument is uniform on small complex neighborhoods of strictly time-ordered spectator configurations and proves locally holomorphic convergence there. It also proves independence of the direction in (7.17).

Proposition 7.6 (Residual infinity decoupling). *For the residual scalar $\ell = L - S$,*

$$W(\ell_\infty O_1 \cdots O_j) = 0, \quad O_i \in \{S, V, T\}, \quad j \leq 3, \quad (7.18)$$

at all distinct Euclidean points, also after differentiating any spectator insertion.

Proof. We justify passage from the localized-vector result of Corollary 6.5 to these products; unrestricted timelike real boundaries for accessible products are unnecessary. For a real vector localized in a fixed line diamond, the projected Paley multiplier in that Proposition has zero value at energy zero. The same local Paley bound controls its first derivative there. Thus it is $O(E) \|\xi\|$ at low energy. Integrating with the weight-two energy powers against e^{-RE} in the measure dE/E , and using exponential damping beyond a fixed energy, gives

$$R^4 |\langle \ell(iRe), \xi \rangle| \leq CR^{-1} \|\xi\|, \quad R \geq 1. \quad (7.19)$$

For complex localized operator data the right side is bounded by the sum of the vacuum and adjoint-vacuum norms. Consequently the estimate passes to mutually spacelike accessible products in that diamond by the two bounded-approximation limits of Proposition 6.7.

Choose tests on small patches about equal-time mutually spacelike points and continue their translations a short distance into strictly increasing imaginary time. Before using the improved real-edge estimate, the corresponding smeared pairing multiplied by R^4 is $O(R^2)$ uniformly up to a smaller good real patch: the one-wave norm is $O(R^{-2})$ and the smeared product bounds are uniform there. Apply the subharmonic logarithmic estimate on a half-disk with this real edge. If its harmonic measure at an interior point is ω , the bound becomes $O(R^{-\omega+2(1-\omega)}) = O(R^{2-3\omega})$. There is an interior open set with $\omega > 2/3$, and hence convergence to zero on that set. The positive imaginary times can be fixed before further shrinking the source tests. On its strict Euclidean configurations (7.17) already converges locally holomorphically. Small real source offsets therefore commute with that limit. Unsmear uniqueness gives (7.18) on an open set, and real analyticity continues it to every distinct configuration. Derivatives follow from the same locally holomorphic identity. $\square$

7.4 The pair representation and normalized fluxes

The four-point split identities now supply a Hilbert-space conformal action on pairs. Its normalized current fluxes will control the possible singular terms in the radial argument below.

Proposition 7.7 (Conformal representation on the pair space). *Let $\mathcal{H}_{\leq 2}$ be the closed span of Ω and the one- and two-insertion tube vectors for L, S, J, t . There is a strongly continuous unitary representation of the universal cover of the connected conformal group on this space, with the canonical local primary transformations. Translations coincide with the actual positive-energy translations. Its timelike special-conformal generator K_e^{rep} is nonnegative.*

Proof. For each length restrict its arguments to a relatively compact open box in its tube. These vectors span a dense subspace: a vector orthogonal to them gives a holomorphic scalar function zero on an open set, and hence zero throughout the tube. For all sufficiently small real conformal transformations, map these vectors by their primary transformation laws. The resulting inner products are unchanged by the split identities of Proposition 7.3 and the integrated scalar Ward identities of Proposition 7.5. Total length four is exactly what is needed for pair-pair inner products. The image is dense by the same open-box argument; thus the map extends to a unitary. On smaller boxes the local composition identity is exact. Holomorphic point dependence gives strong continuity. Continuing these local unitaries along paths integrates the action to the universal cover.

Uniqueness extends the local laws to all tube points and, for sufficiently small transformations, to Euclidean configurations in the positive-time half-space. The point vectors there are smooth group vectors. On translations the construction agrees with the actual translation action, whose spectrum is future directed. The time-Möbius subgroup conjugates its positive time-translation generator to K_e^{rep} , proving the final assertion. No conformal action on the full physical Hilbert space is asserted here. $\square$

For a target $A(y)$ in the positive Euclidean time half-space let Σ be a small outward-oriented sphere about y , wholly within that half-space. Define Hilbert-valued fluxes from the analytic pair vectors by

$$F_J(A; y) = \int_{\Sigma} d\Sigma^a \Psi[J_a(x)A(y)], \quad F_X^t(A; y) = \int_{\Sigma} d\Sigma^a X^b(x) \Psi[t_{ab}(x)A(y)]. \quad (7.20)$$

At each point of the sphere use the Euclidean order, with the tie germs providing the transitions. There is no factor of the sphere area in (7.20). Conservation and tracelessness make the corresponding holomorphic flux forms closed off the target, so the contour may be deformed, or held fixed when differentiating the target nearby. If one instead rescales both currents' fluxes by a common conventional constant, the constant c below rescales with them.

Proposition 7.8 (Short-length Ward fluxes). *With one consistent nonzero Ward normalization c ,*

$$F_J(A) = 0, \quad A = V, L, J, \quad (7.21)$$

$$F_{\partial_b}^t(A) = c \partial_b A \Omega, \quad A = V, L, J. \quad (7.22)$$

For primary targets L, J and every conformal Killing field X ,

$$F_X^t(A) = c \delta_X A \Omega. \quad (7.23)$$

Here δ_X is the canonical differential action, including spin and weight. In particular, a special transformation centered at the target has zero flux, while centered dilation about L gives $2cL\Omega$. With outward orientation in (7.20), Wick time basis $C_0 = ie$, the convention $i[P_\nu, A] = \partial_\nu A$, and upper-minus-lower order $At - tA$ in the cap difference, one has $c = -1$.

Proof. First take $A = V$ and pair with an arbitrary ordinary physical polynomial bra. Lemma 7.4 identifies the analytic pairs with tempered single-insertion forms. Deform Σ to a cylinder whose relative Euclidean time caps are $\pm\delta$ and whose spatial disk has radius a . Give the whole relative time a small real Minkowski shift s , with $|s| < a$. The shifted contour remains in the holomorphic domains; where the imaginary relative time crosses zero on the side, strict spatial separation and tie gluing provide a common neighborhood.

Average smoothly and with integral one over a and s , keeping the shift support strictly inside $|s| < a$. The side contribution tends to zero as $\delta \downarrow 0$ because it lies on a compact nonsingular spacelike set and its time thickness tends to zero. The two caps now have smooth Cauchy-slab tests and converge to the opposite Lorentzian order boundaries at a common target damping. This use of the single-insertion boundary is legitimate even at a point in the common complex target translation: its total pair momentum is exponentially damped, while the smooth relative test decays rapidly in the relative momentum. The difference is the commutator charge with the time average and a spatial cutoff equal to one on its causal support. The J charge is zero by Proposition 5.8; the translation charge of t is the original translation charge by Section 5. They give (7.21)–(7.22) for V .

The Wick time component contributes i , and the tensor translation index contributes the pullback C_b^ν . With the order and charge convention in the statement, $i[V, P_b] = -\partial_b V$, hence $c = -1$. This calculation fixes one common normalization for all components and all subsequently transported Killing fields. Pairing with the dense polynomial bras proves equality of the Hilbert vectors.

A constant-current flux around a primary target extends to a holomorphic single-wave function of its Minkowski argument throughout the future tube: choose a relative sphere smaller than its common damping. Translation covariance makes it a translation wave; dilations give its flux dimension. Small Lorentz transformations act on the pair and its contour; homotopy of the transformed contour back to a small sphere gives the tensor transformation. Differentiate the identities for V in its target index and use $\partial \cdot V = \square L$ while keeping the enclosing contour fixed. One obtains the homogeneous wave equation for both

$$F_J(L), \quad F_{\partial_b}^t(L) - c \partial_b L \Omega. \quad (7.24)$$

These are, respectively, a scalar wave of dimension two and a vector wave of dimension three. Translation covariance and positive-energy holomorphy realize their values after any fixed future

damping as vacuum spectral amplitudes; further damping agrees by uniqueness. Their wave equation confines their spectrum to $p^2 = 0$. Scale covariance excludes a zero-momentum amplitude, and the light-cone boost-weight test of Section 2 excludes the nonzero shell: in these cases $d - 1 > \max \omega$ for the finite Lorentz multiplet. Only local finiteness after exponential damping is needed for that test. Thus both defects vanish. Subtraction of ∂L from V proves the identities also for J .

For $A = L, J$, conjugate the translation identity by any sufficiently small real conformal transformation in Proposition 7.7. Since t is a conserved traceless primary of dimension four, its flux transforms exactly to the flux for the pushed-forward vector field: three surface powers, one vector-field power, and the two orthogonal index factors cancel its weight. Deform the transformed sphere back locally. The target primary factors are invertible, and the differential action on the right transforms by the same Lie algebra adjoint action. Translations and all their such adjoints span the conformal Lie algebra (successive conformal brackets give rotations, dilations and special transformations). Linearity and complexification give every Euclidean conformal Killing field and prove (7.23). $\square$

The flux laws remain valid inside scalar Euclidean functions up to the available split length, with exterior spectators, including ℓ_∞ . First place those spectators earlier than the whole pair by a Euclidean time plane and use the split identity, representing infinity first by a large negative-time insertion. Then continue in the distinct spectator and target positions, choosing a smaller contour where necessary. One immediate consequence is

$$W(\ell_\infty J_a(x) L(y)) = 0. \quad (7.25)$$

Indeed rotations, translations and dilations give the only possible form $d(x - y)_a / |x - y|^4$. Its J flux about L is $d|S^3|$, and (7.21) makes $d = 0$. Primary covariance transports this vanishing to any placement of the infinity insertion.

We now have the pair representation and current flux identities with one fixed normalization. The remaining argument turns radial singularities of short functions into spectral statements on that pair space. Only the representation classifications at energies zero, one and two will be needed for this step, so the analysis does not require a discrete expansion of all radial states.

7.5 Radial Hilbert calculus and its low energies

The function to be tested is $W(\ell_\infty \ell(x) V_\mu(y) V_\nu(z))$. Residual infinity decoupling will make it harmonic in x . We will identify each singular radial power with an actual low-energy spectral projection on the pair space. The low-energy bounds and a Maxwell test leave only dipoles; a tangential reflection then removes those dipoles from a positive pair norm. The resulting norm bound eliminates the remaining low-energy overlaps, and the normalized current fluxes will finally test the residual scalar's own norm. The following calculus supplies these spectral and positivity statements without a complete expansion into radial eigenstates.

Write a Euclidean point as $y = (y_0, \mathbf{y})$. The conformal map

$$c_{\text{ball}}(y) = \frac{(1 - |y|^2, 2\mathbf{y})}{1 - 2y_0 + |y|^2} \quad (7.26)$$

takes the unit ball to the positive-time half-space. Define radial one- and two-insertion kets by primary pullback of the corresponding half-space vectors; obtain descendants by differentiation. Radial adjunction reflects y to $Iy = y/|y|^2$ with its full tensor and weight factors. Indeed direct substitution in (7.26) gives $c_{\text{ball}}(Iy) = (-c_{\text{ball}}(y)_0, c_{\text{ball}}(y)_{\text{spatial}})$, exactly Wick reflection. Scalar covariance therefore identifies radial splitting with Hilbert adjunction. To transport arbitrary

separated interior and exterior groups, each of size at most two, join the required proper Euclidean conformal transformations by paths avoiding poles on the finite point set. A pole condition has codimension four, so such paths can be chosen after a small perturbation. Points at infinity are reached by (7.17). In particular the radial ket $|\ell\rangle = |\ell(0)\rangle$ has bra ℓ_∞ .

On the pair representation put

$$H = \frac{1}{2}(P_e + K_e^{\text{rep}}). \quad (7.27)$$

This is the self-adjoint compact-time generator on the covering group. Its derived-generator identity holds on group-smooth vectors, a core for this generator. Both terms on the right are positive, so $H \geq 0$. The real Minkowski vector field of H , in Wick half-space coordinates, is $i((\tau^2 - |\mathbf{r}|^2 - 1)/2, \tau \mathbf{r})$. The derivative of (7.26) takes iy to this field. Thus positivity and analytic continuation of the local group law give

$$r^H |A_1(y_1) \cdots A_p(y_p)\rangle = r^{d_1 + \cdots + d_p} |A_1(ry_1) \cdots A_p(ry_p)\rangle, \quad 0 < r \leq 1, \quad p \leq 2. \quad (7.28)$$

One first continues through a short contraction, where both analytic laws apply, and then iterates through interior configurations. Radial rotations act unitarily with the usual $SO(4)$ tensor law.

The real-form involution on complex radial transformations is complex conjugation followed by conjugation with I . Write p_a^+, p_a^-, H, M_{ab} for the derived actions corresponding, respectively, to radial translation, $2y_a y - |y|^2 e_a$, dilation, and $y_a e_b - y_b e_a$. On smooth vectors,

$$(p_a^+)^{\dagger} = p_a^-, \quad [H, p_a^{\pm}] = \pm p_a^{\pm}, \quad [p_a^-, p_b^+] = 2\delta_{ab}H + 2M_{ab}. \quad (7.29)$$

These are active-point conventions. For example, $M_{ab}|A_d\rangle = \delta_{ad}|A_b\rangle - \delta_{bd}|A_a\rangle$ at the origin. Inversion takes translations to negative special transformations, giving the displayed matrix adjunction.

The positivity calculation below is the low-energy part of the four-dimensional conformal unitarity-bound method of Mack [31]; see also the level-one Gram-matrix derivation in [33, Section 2.5]. We give the required finite-array calculation on the present pair representation, including its smooth-density justification.

Lemma 7.9 (The required low radial energies). *The $H = 0$ eigenspace is a trivial representation. The $H = 1$ eigenspace has only scalar rotation types. At $H = 2$, among the types in a tensor product of two vectors, only scalars and antisymmetric two-forms can occur. Every smooth antisymmetric energy-two array obeys*

$$\sum_a p_a^+ |F_{ab}\rangle = 0, \quad \sum_a p_a^+ |(*F)_{ab}\rangle = 0. \quad (7.30)$$

No discreteness of the full radial spectrum is required.

Proof. We specify the smooth-density issue first. The element $Z = e^{2\pi i H}$ is central, since its adjoint action on the conformal Lie algebra is the identity by (7.29). Its fixed subrepresentation contains every integer H eigenspace. The projection onto that subrepresentation commutes with the group, preserves smooth vectors and has dense smooth image. There the spectrum of H is supported on the integers, and

$$E_H(\{h\})v = \frac{1}{2\pi} \int_0^{2\pi} e^{it(H-h)} v \, dt \quad (7.31)$$

preserves smoothness. Averaging also over the compact rotations gives dense smooth vectors in each rotation type and dense smooth finite tensor-equivariant arrays. Rotations descend to $SO(4)$, since their action on the dense radial kets is the ordinary tensor action. The computations below therefore extend by density to the stated eigenspaces.

At zero energy the expectations of both positive terms in (7.27) vanish. They annihilate a smooth zero-energy vector. The translation spectrum cone then makes every translation generator annihilate it, and the conjugate cone makes every special partner annihilate it. Their brackets annihilate it as well, proving triviality. If $Hv = v$, lowering has energy zero and $\langle w, p_a^- v \rangle = \langle p_a^+ w, v \rangle = 0$ for every zero-energy w ; hence $p_a^- v = 0$.

For any smooth lowest array of energy h , raising norms give

$$\left\| \sum_a p_a^+ v_a \right\|^2 = 2 \sum_{a,b} \langle v_a, (h\delta_{ab} + M_{ab}) v_b \rangle \geq 0. \quad (7.32)$$

A two-by-two plane-rotation block gives $|m| \leq h$ for each plane weight. An $SO(4)$ tensor type has two $SU(2)$ spins j_+, j_- , their sum an integer, with maximal plane weight $j_+ + j_-$. At $h = 1$ only scalars, vectors and the two chiral two-form types survive this test. The divergence arrays exclude the last two. In detail, the tensor index rotation rule gives

$$\sum_b M_{ab} |A_b\rangle = -3 |A_a\rangle, \quad (7.33)$$

$$\sum_c M_{ac} |F_{cb}\rangle = -3 |F_{ab}\rangle - |F_{ba}\rangle + \delta_{ab} \sum_c |F_{cc}\rangle. \quad (7.34)$$

The matrix eigenvalues in (7.32) are therefore -3 for the vector divergence, -4 for a symmetric traceless rank-two divergence, and -2 for an antisymmetric rank-two divergence. Their positivity bounds are $h \geq 3, 4, 2$, respectively. At $h = 1$ this leaves only scalars.

At $h = 2$ the types in two vectors are even under the central radial rotation $y \mapsto -y$. Their lowered arrays are odd but would have to belong to the scalar-only energy-one space, so they are lowest arrays. The -4 bound excludes the symmetric traceless type. The antisymmetric bound is saturated and makes the first norm in (7.30) zero. Apply the same argument to the Hodge dual, which intertwines rotations, to obtain its second identity. This proves exactly the needed low-energy statements without an expansion in a complete family of local operators. $\square$

Lemma 7.10 (The exterior Maxwell test). *An antisymmetric energy-two radial array has zero exterior overlap with the pair $\ell_\infty, V_\nu(z)$, with z outside the radial ball.*

Proof. Let $B_\nu(z)$ be the reflected ket of this exterior data; it is a smooth pair vector. Its exterior translation and similarity laws, including the full derivative reflection of V , are

$$\partial_{z^a} B_\nu = -p_a^- B_\nu, \quad (z \cdot \partial + 1) B_\nu = -H B_\nu. \quad (7.35)$$

Translation fixes the insertion at infinity; reflection changes its generator to the lowering generator with the indicated sign. For the second formula the exterior vector contributes weight -6 and the reflected interior pair dimension five, giving total degree $-1 - H$. These statements need only similarity covariance of V .

For a smooth Maxwell array, set $Q_{\nu ab}(z) = \langle B_\nu(z), F_{ab} \rangle$. Equations (7.30) and (7.35) make Q homogeneous of degree -3 and make it and its Hodge dual in a, b divergence-free. Rotation covariance and antisymmetry leave precisely

$$Q_{\nu ab} = A |z|^{-4} (\delta_{\nu a} z_b - \delta_{\nu b} z_a) + B |z|^{-4} \epsilon_{\nu abd} z_d. \quad (7.36)$$

For $n = z/|z|$, the divergence of the first tensor is $|z|^{-4} (\delta_{\nu b} - 4n_\nu n_b)$, while that of the second is zero. Hence $A = 0$. Hodge duality interchanges these tensors up to orientation sign, so dual divergence gives $B = 0$. Dense smooth arrays in the energy-two rotation types and continuity of the fixed exterior pairing prove the general assertion. $\square$

7.6 Harmonic removal of the residual

Consider the short Euclidean scalar function

$$G_{\mu\nu}(x; y, z) = W(\ell_\infty \ell(x) V_\mu(y) V_\nu(z)), \quad x, y, z \text{ distinct.} \quad (7.37)$$

By (7.18) and the leg identities,

$$\partial_x^2 G = 0, \quad \partial_{x^a} G_{\mu\nu} = -W(\ell_\infty J_a(x) V_\mu(y) V_\nu(z)). \quad (7.38)$$

For the second equation use $\partial \ell = V - J - \partial S$; the accessible terms vanish. For the first, $\partial^2 \ell = \Theta - \partial^2 S$, whose two terms likewise vanish in this correlation.

For fixed y, z , G is bounded as $|x| \rightarrow \infty$. To see the quantitative bound, transport the two scalars colliding at infinity by a proper spherical inversion, adjusted by a reflection, to a finite radial ball, placing the vector pair exterior. The separation r' of the inner scalar pair is comparable to $|x|^{-1}$. Its norm is at most $C(r')^{-4}$ by (7.28) and $H \geq 0$, starting from a fixed inner radius uniformly in angle. The scalar inversion factor of the moving point is $O(|x|^{-4})$, and cancels this bound. The fixed exterior pair has finite norm, proving boundedness.

We next determine the possible singularities at the two finite points y and z . Harmonicity gives two radial powers in each spherical harmonic. Pair-space positivity will identify a singular power with an actual low-energy spectral projection; the preceding representation and flux results can then test that projection.

Project the harmonic function onto a spherical harmonic of degree q about y . With $r = |x - y|$, its radial coefficient has the form

$$g_q(r) = A_q r^q + D_q r^{-q-2}. \quad (7.39)$$

Choose a small radial ball centered at y , let v_q be the harmonic projection of the inner pair ket $|\ell(y + r_0 n) V_\mu(y)\rangle$ at fixed r_0 , and let B be the fixed exterior ket. The contraction identity gives

$$r^5 g_q(r) = r_0^5 \langle B, (r/r_0)^H v_q \rangle. \quad (7.40)$$

The spectral pairing $\mu(E) = \langle B, E_H(E) v_q \rangle$ is a complex measure of finite total variation, at most $\|B\| \|v_q\|$. Thus the right side is a Laplace transform of a finite measure on $[0, \infty)$. The two powers on the left have exponents $q + 5$ and $3 - q$. If $q > 3$, boundedness of this transform at $r = 0$ kills D_q . Otherwise uniqueness of finite-measure Laplace transforms identifies D_q with the actual point spectral projection at energy $3 - q$. One can verify uniqueness by setting $u = e^{-h}$: all integer Laplace parameters give the polynomial moments on $[0, 1]$, which determine the finite measure. In particular, a continuous spectral component cannot imitate the singular power in (7.39).

The projected array transforms in $\mathcal{Y}_q \otimes \mathbf{4}$, where $\mathcal{Y}_q$ is the degree- q harmonic representation. At $q = 3$ the possible energy is zero, and at $q = 2$ it is one. Only scalars occur there by Lemma 7.9, whereas $\mathcal{Y}_q \otimes \mathbf{4}$ contains a scalar only for $q = 1$. Both coefficients therefore vanish. At $q = 1$ the energy is two; the possible types are the scalar and Maxwell two-forms. The latter have zero exterior overlap by Lemma 7.10. The unique scalar intertwiner leaves precisely a dipole parallel to the vector index. The remaining $q = 0$ monopole is killed by integrating (7.38) over a small sphere about y : $F_J(V) = 0$, whereas the radial derivative of $D_0 r^{-2}$ has flux $-2|S^3|D_0$. Repeat at z .

Convergent harmonic expansions on punctured-ball annuli now show that subtraction of these two dipoles removes both singularities. The remainder is entire and bounded at infinity, hence constant in x . We have proved

$$G_{\mu\nu}(x; y, z) = D_\nu(y, z) \frac{(x - y)_\mu}{|x - y|^4} + \tilde{D}_\mu(y, z) \frac{(x - z)_\nu}{|x - z|^4} + C_{\mu\nu}(y, z). \quad (7.41)$$

The constant is the limit at x infinity, so its similarity laws are immediate: it depends only on $y - z$ and is homogeneous of degree -6 .

The only possible finite singularities are now the two displayed dipoles. A tangential polarization kills them simultaneously after placing the two vector points at inverse radial positions. This choice turns the bounded remainder into a bound on a genuine pair norm, which excludes its low radial energies.

Take radial center $x = 0$, choose $|u| = 1$, put $y = u/r$, $z = ru$, and contract both vector indices with a fixed $w \perp u$. Both dipoles in (7.41) vanish, and its last term is $O(r^6)$. For clarity, the full radial reflection law for $V = J + \partial L$ is

$$V_\mu(x)^\dagger = |x|^{-6} R_{\mu\nu}(x) V_\nu(Ix) - 4x_\mu |x|^{-6} L(Ix), \quad R_{\mu\nu} = \delta_{\mu\nu} - 2\hat{x}_\mu \hat{x}_\nu. \quad (7.42)$$

The second term is the derivative of $|x|^{-4}$ in the scalar reflection formula. Tangential contraction removes it and leaves $Rw = w$. Radial splitting therefore gives

$$\| |\ell(0) V_w(ru) \rangle \|^2 = r^{-6} G_{ww}(0; u/r, ru) = O(1). \quad (7.43)$$

This bounds an actual Hilbert norm. If ψ is the corresponding ket at a fixed radius r_0 and $s = r/r_0$, contraction gives

$$\| |\ell(0) V_w(ru) \rangle \|^2 = s^{-10} \int_{[0, \infty)} s^{2h} d \langle \psi, E_H(h) \psi \rangle. \quad (7.44)$$

A nonzero spectral mass on $[0, 5 - \varepsilon]$ would make this at least a positive constant times $s^{-2\varepsilon}$, contrary to (7.43). Taking all $\varepsilon > 0$ excludes the whole spectrum below five, in particular overlap with $|J_\rho\rangle$ at energy three.

Use (7.25) and its conformal transports to remove the ∂L part of V in that overlap. The remaining three-point function has the similarity form

$$\langle J_\rho | \ell(0) J_\mu(z) \rangle = |z|^{-2} (a \delta_{\rho\mu} + b \hat{z}_\rho \hat{z}_\mu). \quad (7.45)$$

Tangential testing forces $a = 0$. Its divergence is $|z|^{-3} (b - 2a) \hat{z}_\rho$, so conservation forces $b = 0$. Primary conformal transport yields

$$W(\ell_\infty J J) = 0. \quad (7.46)$$

Expand the already zero $W(\ell_\infty V V)$ using $V = J + \partial L$. Equations (7.25) and (7.46) leave the mixed second derivative of $W(\ell_\infty L(y) L(z))$ equal to zero. Translation covariance makes this a function of $y - z$ with zero Hessian, and its degree -2 excludes a nonzero affine function. Consequently

$$W(\ell_\infty L L) = 0. \quad (7.47)$$

The residual already decouples from the JL , JJ and LL three-point functions. To test its own scalar norm, we need one more identity involving the stress tensor. We first remove the allowed Jt tensor structure with a centered special flux, and then use a centered dilation flux on the remaining Lt function.

It remains to use the stress flux. Write $R = |z - y|$ and $n = (z - y)/R$. Rotation and scale covariance, symmetry in ρ, σ , and the absence of a nonzero parity-odd symmetric tensor give the general form

$$W(\ell_\infty J_\mu(y) t_{\rho\sigma}(z)) = R^{-5} [A n_\mu \delta_{\rho\sigma} + B (n_\rho \delta_{\sigma\mu} + n_\sigma \delta_{\rho\mu}) + D n_\mu n_\rho n_\sigma]. \quad (7.48)$$

Trace gives $4A + 2B + D = 0$. Vector divergence gives the coefficients $-2A + 2B$ and $-2D - 12B$ of, respectively, $R^{-6} \delta_{\rho\sigma}$ and $R^{-6} n_\rho n_\sigma$. Thus $A = B = \gamma$ and $D = -6\gamma$, giving

$$W(\ell_\infty J_\mu(y) t_{\rho\sigma}(z)) = \gamma R^{-5} (n_\mu \delta_{\rho\sigma} + n_\rho \delta_{\sigma\mu} + n_\sigma \delta_{\rho\mu} - 6 n_\mu n_\rho n_\sigma). \quad (7.49)$$

This also obeys tensor divergence. Take the special vector field centered at y , $X_\alpha^\sigma = 2(z - y)_\alpha(z - y)^\sigma - R^2\delta_\alpha^\sigma$. The contraction of its angular part with the tensor in (7.49) is

$$n^\rho(2n_\alpha n^\sigma - \delta_\alpha^\sigma)(n_\mu \delta_{\rho\sigma} + n_\rho \delta_{\sigma\mu} + n_\sigma \delta_{\rho\mu} - 6n_\mu n_\rho n_\sigma) = -\delta_{\alpha\mu} - 2n_\alpha n_\mu. \quad (7.50)$$

Its sphere average is $-3\delta_{\alpha\mu}/2$, since $\langle n_\alpha n_\mu \rangle = \delta_{\alpha\mu}/4$. The field contributes R^2 and the area R^3 , canceling R^{-5} . The flux is therefore $-3\gamma|S^3|\delta_{\alpha\mu}/2$. It is zero by (7.23), because a centered special transformation annihilates the primary J . Hence $\gamma = 0$.

Theorem 7.11 (Return of the scalar wave). *The formal scalar vacuum wave equals the accessible scalar:*

$$l = S, \quad \ell = 0. \quad (7.51)$$

Proof. Expand $W(\ell_\infty V(y)T(z)) = 0$, which is a case of (7.18), using (7.16). The Jt term has just vanished; the terms with derivatives of L vanish by (7.25) and (7.47). What remains is $\partial_y W(\ell_\infty L(y)t(z)) = 0$. Translation covariance and homogeneity of degree -4 force that function to be zero. Its centered dilation flux around L , using (7.23), now gives

$$0 = 2c W(\ell_\infty L). \quad (7.52)$$

The energy-slice calculation in Proposition 6.4 gave $D(e) = B(e, e) = b$ after excluding the massless contribution. Thus the constant two-point density of S with $l - S$ vanishes: the scalar wave channels are orthogonal, not merely their labels on one time slice. It follows that $W(\ell_\infty L)$ equals the scalar two-point coefficient $W(\ell_\infty \ell)$, which is the nonnegative radial norm $\|\ell\|^2$ and is strictly positive for a nonzero scalar wave. Since $c \neq 0$, (7.52) makes that norm zero. Therefore $l = S$. $\square$

8 Real-axis reconstruction and physical conformal currents

By Theorem 7.11, the extracted scalar wave is the entire trace polarization: $S = l$. In this Section we write L for that accessible entry. Until the joint domain is constructed below, this notation denotes its vacuum wave and its analytic products, rather than an operator on arbitrary states. In particular, the vacuum-wave division by M^2 is never extended to another leg. The ingredients from Section 6 are bounded local approximations, analytic vector products of every finite length, their reflected Hilbert splittings, and gluing at spacelike ties. The leg equation is

$$\square_i W(\dots, L(x_i), \dots) = W(\dots, \Theta(x_i), \dots). \quad (8.1)$$

The remaining task is to establish joint tempered boundaries, their ordered spectra, and affiliation with the original local algebras.

The construction has three stages. First, we obtain real-axis boundaries for pure scalar products and then for every mixed ordering. Second, their Hilbert splittings provide a common invariant domain, and bounded approximants give affiliation in the fixed net of Assumption 2.1. Only after these steps do we regard L as a physical field and use its improvement to define the local conformal Ward derivations.

The first stage proves the following joint boundary statement. Its scalar and vector conclusions are both needed: scalar positivity defines insertion operators, while vector continuity identifies their domain with actual vectors in the original Hilbert space.

Theorem 8.1. *For every finite ordered list of L and original physical fields, the mixed analytic scalar product is the Fourier–Laplace continuation of a tempered Wightman distribution, with the ordered spectrum, adjunction, locality, and similarity covariance, with dilations acting on the original field labels. The analytic vector products have tempered Hilbert-valued boundaries with the vector spectrum. Their reflected splittings hold for joint tests and give all joint Wightman positivity inequalities in the original Hilbert space.*

We first prove this for homogeneous original slots. Pure scalar vectors give the initial mixed orderings; an adjacent-interchange argument then reaches every ordering. After constructing the vector boundaries and their splittings, finite multilinearity restores the arbitrary original field labels under the same joint test kernels.

8.1 Uniformity conventions and pure scalar products

All assertions below concern a fixed finite list of fields and compact real coordinate sets. For a test supported in a compact set K , put

$$p_{K,m}(f) = \max_{|\alpha| \leq m} \sup_K |\partial^\alpha f|. \quad (8.2)$$

A distribution estimate is uniform on the unit ball of one such seminorm; the finite integer m is allowed to increase when a prescribed accuracy increases. For boxes of diameter ε , the same convention uses rescaled tests and hence introduces only powers of ε^{-1} . Imaginary directions range over compact subsets of open ordered cones. Constants can depend on those compact sets, the field list, and the requested derivative order, but never on the dyadic approximation parameter or on $0 < \varepsilon \leq \varepsilon_0$.

For clarity we record the quantitative content of the bounded approximation result. For the product of a fixed number of approximants, there are fixed $A, Q < \infty$ such that its holomorphic norm is at most $C\varepsilon^{-A}R^Q$. On a separated real placement patch its dyadic difference satisfies, for every k ,

$$\|F_{2R} - F_R\| \leq C_k \varepsilon^{-B_k} R^{-k}, \quad \|F_R\| \leq C\varepsilon^{-A} R^Q. \quad (8.3)$$

The first estimate follows by telescoping and moving the differing bounded factor to the vacuum; the finite seminorms for it depend on k . The exponent Q in the second estimate does not. Both estimates include the adjoint waves. These are precisely the shrinking-box estimates of Proposition 6.6; they impose no energy bound on an unknown field operator.

The two exponents in (8.3) play different roles. Increasing the desired accuracy changes k and the required test seminorm, while the polynomial norm exponent Q remains fixed. This is what permits a summable dyadic estimate after a positive fraction of the accuracy has been lost in analytic continuation.

We will obtain spectral support from scaling and local boundary regularity. The exact homogeneous case suffices for pure products. Later we will also need the version in which a scaling error already has the required spectral support.

Lemma 8.2 (Scaling and tube spectrum). *Let $C \subset \mathbb{R}^N$ be an open convex cone and set*

$$K = C^* = \{\xi : \xi \cdot y \geq 0 \text{ for every } y \in C\}.$$

Use Fourier waves $e^{ix \cdot \xi}$. Let u be a tempered distribution with values in a Hilbert space, locally near the origin the distributional boundary of a holomorphic function on a wedge with imaginary directions in C . Suppose that, for a real d and unitaries $V(\lambda)$, $\lambda > 0$, every scaling defect

$$\lambda^d u(\lambda x) - V(\lambda)u(x)$$

has Fourier support in K . Then $\text{supp } \hat{u} \subset K$. In particular this holds for exact scaling covariance, with zero defect. For scalar distributions one may take $V(\lambda) = 1$. The wedge-boundary hypothesis may be replaced by rapid decay of localized Fourier transforms near the origin in every direction outside K .

Proof. A distributional wedge boundary has no analytic, hence no smooth, wavefront at the origin in a direction outside K . This follows by shifting a localized Fourier test a small distance in a strict direction of C with negative frequency pairing. The usual distributional boundary bounds on smaller approach wedges justify the shift. The argument applies in Hilbert norm, or after scalarization uniformly on the unit ball.

Fix a smooth Fourier test supported in a compact subset of $\mathbb{R}^N \setminus K$. Its pairing with $\hat{u}$ after dilation to frequency Λ decreases faster than every power of Λ . To pass from the local wavefront estimate to this unlocalized pairing, insert a cutoff equal to one near the origin. The local part has the asserted rapid decay. Outside the cutoff, the rescaled Schwartz inverse Fourier test and all its derivatives decrease rapidly in every Schwartz seminorm, so temperedness controls the remainder. A finite covering treats all directions in the fixed frequency support.

The scaling defect has zero pairing with that test and with every one of its positive dilates, since K is a cone. The remaining scaling identity makes the norm of the dilated pairing a fixed power of Λ times the norm of the original pairing; $V(\lambda)$ does not change that norm. Rapid decay forces the original pairing to vanish. Every compact Fourier test outside K is allowed, proving the assertion. $\square$

Proposition 8.3. *Every pure L analytic vector product has a tempered Hilbert-valued boundary with the ordinary ordered vector spectrum. Its Fourier–Laplace continuation is the accessible analytic product.*

Proof. First, the product has scalar-primary covariance under sufficiently small simultaneous special transformations. The geometry in Lemma 7.1 applies to every fixed finite family of source and destination patches: its proof uses only finitely many strict wedge inequalities. Choose the conjugate null axis m separated from the finitely many flow directions, take $m \cdot x$ sufficiently negative, and then add a sufficiently large positive multiple of m to make all the required opposite inequalities strict. A small double cone O_- about that point is spacelike to every containing wedge and to the destination patches. Local vacuum cyclicity makes $\{C\Omega : C \in \mathcal{M}(O_-)\}$ total.

For a fixed finite product length, reparametrize the source tests by a small null flow and its inverse primary factors. Write $F_g^{(n)}$ for the limit of the resulting simultaneously conjugated bounded products and $F^{(n)}$ for the accessible destination product. The former limit exists because the source product converges and the common conjugating unitary is bounded. For $C \in \mathcal{M}(O_-)$, move its first bounded factor past C and take the vacuum and adjoint-vacuum limits. The canonical one-wave action and induction on the length give

$$\begin{aligned} \langle C\Omega, F_g^{(n)} \rangle &= \langle L(\bar{f}_1)\Omega, C^* F_g^{(n-1)} \rangle \\ &= \langle L(\bar{f}_1)\Omega, C^* F^{(n-1)} \rangle = \langle C\Omega, F^{(n)} \rangle. \end{aligned} \tag{8.4}$$

Every transformed one-vacuum wave is already the accessible L wave, so this induction uses no finite-length interpolation. Totality proves the covariance identity on the real spacelike patch, and boundary uniqueness proves it in the tube. Compositions of small null flows give small special transformations. For any fixed real compact set and strict approach subcone, choose the transformations small both there and on a finite interior path to the initial patch. Their unitary

factors have norm one. Thus the identity is uniform near that compact set; localization under arbitrary large conformal transformations is not required.

Thus covariance has been obtained separately for every finite pure product length. The next step uses it to transport the known boundary estimates for products of Θ into enough differential estimates to recover the original pure L product.

Here is an algebraic device that converts this covariance into a boundary estimate. On the ambient null cone use

$$X = (X^+, X^-, X^\mu), \quad q(X) = X^\mu X_\mu - X^+ X^-, \quad X = \lambda(1, x^2, x).$$

Here F is the Hilbert-valued analytic vector product. Its lift is Hilbert-valued and homogeneous of degree $w = -2$ in each slot; the differential identities below may be tested against arbitrary Hilbert vectors, with estimates uniform on their unit ball. On degree w homogeneous functions, define the flat Thomas derivative

$$\mathcal{D}_A(w) = (2 + 2w)\partial_A - X_A \partial_{\text{amb}}^2. \quad (8.5)$$

Successive derivatives use the new degree in their slot. The product rule, $\partial_{\text{amb}}^2 q = 12$, and Euler's identity show that these derivatives preserve the quadratic ideal, commute successively, and have zero successive trace. They therefore act intrinsically on the cone. The same calculation gives

$$X^A \mathcal{D}_A(w) = w(2 + 2w) \pmod{q}. \quad (8.6)$$

These are the flat ambient identities underlying the Thomas operator; see also [21]. None of the factors in (8.6) vanishes at $w = -2, -3, \dots$

In particular, contraction can undo each successive derivative at the degrees encountered here. The weight in (8.5) must be lowered after each differentiation in that slot; using a fixed weight throughout would not give these intrinsic commuting operators.

Take the null direction a_∞ representing chart infinity, and extend the lifted function independently of X^- . Then $a_\infty \cdot \mathcal{D}$ is a nonzero smooth chart factor times $\square_x$. Consequently

$$G_a F := \prod_{i=1}^n (a \cdot \mathcal{D}_i) F \quad (8.7)$$

has polynomial boundary growth for $a = a_\infty$, by the ordinary all- Θ product and (8.1). Primary transport proves the same statement for a in an open real null-cone patch about a_∞ . Finite derivatives preserve these bounds after shrinking the approach subcone, by Cauchy's inequalities.

For this transport, write $\tilde{F}$ for the homogeneous lift. For each fixed sufficiently small transformation g , its local covariance is $\tilde{F}(gX) = U_g \tilde{F}(X)$, with the primary chart factors absorbed in the lift. The flat ambient chain rule gives $G_{ga} \tilde{F}(gX) = U_g G_a \tilde{F}(X)$. The unitary U_g is fixed as the insertion coordinates vary, so the coordinate derivatives act only on $\tilde{F}$. Taking $a = a_\infty$ transfers the known bound to the null direction ga_∞ . The shifted homogeneous weights supply smooth chart factors bounded on the chosen compact set. Scalarization replaces a testing vector by its unitary image and therefore keeps the Hilbert norm estimate uniform.

We give the finite algebraic argument needed to recover F . In constant complex variables ξ_{iA} , let I be the ideal generated by ξ_i^2 and by $\prod_i (a \cdot \xi_i)$ for all a in that patch. If a common zero had every $\xi_i \neq 0$, the patch would be covered by the finitely many proper hyperplane sections $a \cdot \xi_i = 0$. The real null patch is Zariski dense in the irreducible complex null cone and spans the ambient space, so this is impossible. Thus every common zero has $\xi_i = 0$ for some i .

For each of the finitely many monomials $m = \prod_i \xi_{iA_i}$, the Nullstellensatz gives $m^{e_m} \in I$. Let e dominate their exponents. Every monomial of degree at least $6(e - 1) + 1$ in each slot is divisible

by one of the m^e and belongs to I . Noetherianity gives finite generator representations, and separate homogeneity in each slot gives multihomogeneous coefficient representations. Substitute the successively commuting operators (8.5). Trace terms vanish; the remaining terms are finite derivatives of (8.7). Hence sufficiently high derivative tensors of F have polynomial growth. Contracting their indices with the X_i and using (8.6) repeatedly recovers F with a nonzero constant factor. The chart factors are bounded on the compact set. This proves moderate growth of F .

The algebraic recovery used only a finite number of null directions and derivatives for this fixed product length. Its constants may depend on that length; no estimate uniform over all lengths is needed for constructing the polynomial field domain.

The moderate-growth tube boundary theorem, applied weakly and then uniformly in unit Hilbert vectors, gives a Hilbert-valued distribution locally. Its exact similarity covariance globalizes this boundary and makes it tempered: rescaling dyadic annuli to a fixed compact set changes a finite distribution seminorm by only a power of the radius, and the dilation unitary leaves its norm unchanged. Lemma 8.2 gives its ordered vector spectrum. Fourier–Laplace uniqueness then identifies its continuation with the original product. $\square$

8.2 Boundary values and a distribution criterion

We specify the analytic facts used in the mixed-product argument so that no growth assertion is concealed in a boundary-value notation. The point of the distinction is that holomorphy alone will initially give boundary hyperfunctions for the coefficients of a causal wave ambiguity. We will prove a quantitative Gaussian estimate before using any of those coefficients as an ordinary distribution. The boundary bf of a holomorphic function on a local convex wedge exists as a hyperfunction without a growth assumption. Restriction to a smaller approach cone is compatible with this boundary map; the map is injective for the local germ on a connected single wedge and agrees with ordinary distributional boundary values whenever those exist. We fix the Euclidean coordinate volume form when identifying scalar hyperfunctions with the corresponding top-form convention. Compactly supported hyperfunctions are analytic functionals carried by their support. Thus, for each complex neighborhood K_γ of the carrier, a representative u satisfies

$$|\langle u, h \rangle| \leq C_\gamma \sup_{K_\gamma} |h|, \quad h \text{ holomorphic near } K_\gamma. \quad (8.8)$$

Komatsu constructs the injective boundary map on a Stein complex neighborhood with an open convex imaginary cone [27, Section 1]; equations (1.1) and (1.5) on pp. 253–254 give injectivity and compatibility with smaller cones. Small complex boxes are Stein, so restriction to a smaller real patch gives the local germ statement used here. The comparison on p. 257 identifies polynomial-growth boundaries with distributional ones. Martineau’s Theorems 1–2 give the analytic-functional realization [32]; the carrier estimate is in Section 1(a)–(b) and Proposition 1 of that paper. These constructions alone give no polynomial growth estimate. In the analytic-functional notation used below, the canonical distribution embedding is described in [24, Section 4, equation (4.3)].

Use real Euclidean coordinates for the Gaussian, with their complex-bilinear extension, and the Fourier wave convention $e^{ix \cdot \xi}$. For a compact analytic-functional representative put

$$\mathcal{T}_\Lambda u(w, \eta) = \left\langle u(z), e^{i\Lambda(w-z) \cdot \eta - \Lambda(w-z)^2/2} \right\rangle. \quad (8.9)$$

Only real centers w and real frequencies η are used for the estimates below.

Lemma 8.4. *Suppose that, near a real point w_0 , a hyperfunction u has a compact analytic-functional representative such that*

$$|\mathcal{T}_\Lambda u(w, \eta)| \leq C(1 + \Lambda)^N \quad (w \in U, |\eta| = r_*, \Lambda \geq 1) \quad (8.10)$$

for some neighborhood U of w_0 and some $r_ > 0$. Then u is a distribution near w_0 .*

If $u = bf$ from a wedge, its transform near w_0 can be computed by integrating f on a sufficiently small fixed interior translate of a local real box, with an exponentially small error. The box and height can be chosen uniformly for w in a smaller box and η in a fixed compact set.

Proof. The first assertion is the distribution case of the generalized FBI criterion of Hoepfner–Ragognette [24, Theorem 6.3 and Section 8.1]. In their notation take the positive quadratic polynomial $p(v) = |v|^2/(2r_*)$ and $\xi = \Lambda\eta$, $|\eta| = r_*$. Then $|\xi|p(w - z) = \Lambda(w - z)^2/2$, so their transform is (8.9) up to an irrelevant fixed normalization. Their local inverse decomposition separates a distribution, obtained from the polynomially bounded part, from an analytic remainder. Finite covering of the frequency sphere makes the polynomial estimate uniform. The compact representative is immaterial because changing it away from w_0 changes this estimate by an exponentially small term.

For the second assertion, use the fixed-height integration representation of a wedge boundary value [24, Remark 4.1], using a rounded box with smooth boundary. The difference is an analytic functional carried in an arbitrarily thin complex neighborhood of the boundary of the chosen real box. This is also obtained from the product Cauchy representation: entries that extend across one coordinate cancel under contour displacement. Axes strictly inside the wedge give the requisite product half-domain description. Let the distance between the smaller box of centers and that boundary be $a > 0$. For $z = x + iy$, $|\eta| \leq M$, the real exponent in the Gaussian is

$$\Lambda \left(y \cdot \eta + \frac{|y|^2}{2} - \frac{|w - x|^2}{2} \right). \quad (8.11)$$

Choose all side contours with $|y| \leq t$ and then the carrier neighborhood with width γ so small that $(t + \gamma)M + (t + \gamma)^2/2 < a^2/4$. Equation (8.8) bounds their contributions by $Ce^{-c\Lambda}$. This is the localization estimate of [24, Proposition 4.5]. Heights and carrier neighborhoods are fixed before $\Lambda \rightarrow \infty$; no integrability of f down to the real edge has been assumed. $\square$

A related fact for a tempered distribution will be useful. Rapid Gaussian decay near the origin in a frequency cone implies rapid decay when its Fourier transform is tested at large frequencies strictly inside that cone. One can use the smooth wavefront characterization, or directly integrate the Gaussian inversion against that frequency test. Away from the origin, integrations by parts in the frequency variable gain arbitrary powers; a derivative of the Gaussian costs at most a square-root power beyond the fixed distribution order. This gives the alternative Fourier-decay hypothesis in Lemma 8.2.

8.3 A quantitative boundary estimate at one merged face

Until the finite-multilinearity passage below, every original-field slot in the mixed-product argument is homogeneous, including every spectator slot. Original derivative slots carry their shifted dimensions; derivatives of L will follow by differentiating its reconstructed products. We fix an arbitrary finite list and an arbitrary ordering; neither its length nor the set of lists is bounded. The constants and distribution orders may depend on this list. We shall first complete both scalar and Hilbert-valued reconstruction for these homogeneous lists.

The scalar induction is simultaneous over every finite homogeneous old-field list and is indexed only by the number of L entries. Its zero stage is the original Wightman family. At the next stage, orderings with every L on the right are tempered: the analytic splitting (6.14) pairs an ordinary homogeneous old bra with the pure vector boundary of Proposition 8.3. This continuous bilinear pairing on the conjugate bra Hilbert space and the ket Hilbert space gives a joint Schwartz distribution by the kernel theorem. The reversed bra vector cone, the ket vector cone, and translation invariance supply its scalar ordered cones.

It remains to move L entries through old entries. In each adjacent interchange, applying $\square$ in the L slot replaces it by the homogeneous old field Θ . Both source orderings therefore belong to the preceding induction stage, however many old spectators are present. We prove the interchange step below before iterating it.

Let W_1 be a mixed scalar ordering already known to be tempered, and obtain W_2 by interchanging an adjacent $L(x)$ and homogeneous original physical field $A(y)$. Write $r = x - y$. The variable z consists of y and all spectator coordinates in their merged order. Holding z in a compact subset of its strict tube keeps every gap except the new relative gap bounded away from the boundary. Replacing this L by Θ gives two already known tempered sources, with the homogeneous dimension-four field Θ in that slot, $S_i = \square_r W_i$.

In these coordinates the momentum at the merged y slot is the sum of the two adjacent momenta; all other spectator momenta are unchanged. Write C_z for the strict merged approach cone and $K_z = C_z^*$ for its closed spectral cone, in the Fourier convention of Lemma 8.2. A strict relative direction v_r of the target order, sufficiently small compared with a fixed merged direction v_z , makes (v_r, v_z) a strict full target direction. These are different cones: operations in r preserve K_z , while the missing relative spectral condition will require a further argument.

Here the relative gap tends to the boundary while the merged gaps remain strict. The estimate must therefore keep a positive analytic propagation exponent as the relative gap shrinks. Once this face estimate is available, the wave equation will carry it from a thin real-time strip to every compact relative-coordinate set.

Lemma 8.5. *For compact real r -sets, compact strict merged z -sets, and compact sets of relative imaginary directions v of the target order, there are $C, N < \infty$ such that*

$$|W_2(r + i\varepsilon v; z)| \leq C\varepsilon^{-N}, \quad 0 < \varepsilon \leq \varepsilon_0. \quad (8.12)$$

The same statement holds for every fixed finite derivative order. Consequently W_2 has an ordinary distributional r -boundary $F_0(r; z)$, holomorphic as a distribution-valued function on the strict merged tube.

Proof. We first work with $|r^0| \leq C_0\varepsilon$ and bounded $\mathbf{r}$. Shift $\mathbf{r}$ outward by at most $C_1\varepsilon$ so that the resulting r_* has spacelike clearance at least $4c\varepsilon$. Put the real merged placements z_* sufficiently far apart on a common time slice. Their spacelike margins, including separation from both members of the pair, can be chosen of order one. The tests for the two nearby slots have diameter $c\varepsilon$; all tests may be taken of that size. The bounds (8.3) hold on the resulting separated patch.

Here F_R denotes the scalar vacuum expectation of the bounded product (6.13). On the vector tube, vacuum invariance removes its initial factor $U(z_1)$. The resulting expression depends only on consecutive gaps and extends holomorphically, with the same polynomial norm bound, to the scalar tube where those gaps are strict. Thus this scalar face estimate does not require the first-gap condition that remains necessary for vector products.

Write $\eta = \text{Im } z$. Consider the complex line

$$\zeta \longmapsto (r_* + \varepsilon\zeta v, z_* + \zeta\eta), \quad \text{Im } \zeta > 0. \quad (8.13)$$

For sufficiently small ε its direction is in the full target cone. There is a real interval $[-a, a]$ on which the approximant supports remain spacelike, with $a > 0$ independent of ε : the relative displacement is of size ε and the corresponding clearance is also of that size, whereas all merged displacements and margins are of order one. The harmonic measure of this interval at $\zeta = i$ is $\alpha = 2 \arctan(a)/\pi > 0$. The two-constants theorem applied to $\log |F_{2R} - F_R|$ gives

$$|F_{2R} - F_R| \leq C_k \varepsilon^{-\alpha B_k - (1-\alpha)A} R^{-\alpha k + (1-\alpha)Q}.$$

Slightly reducing the clearances gives this estimate on a complex r -ball of radius $c_1 \varepsilon$ and a z -ball of fixed radius.

Returning the spatial shift takes a bounded number of ε -balls. At fixed complex r , moving the merged variables to any prescribed merged compact takes a bounded number of fixed-radius z -balls. Moving y moves $x = y + r$ with it, so the small internal gap does not restrict those z -radii. More precisely, take the compact convex hull, inside the strict merged tube, of the initial placements and the prescribed compact set. Its gaps have a fixed positive margin; the two exterior gaps adjacent to the pair keep that margin when the internal displacement is $O(\varepsilon)$. A finite cover therefore supplies radii and a number of balls independent of ε . Take overlapping balls with radii $\rho, 2\rho, 4\rho$ inside the tube. Each three-circle step retains exponent $1/2$. Thus a fixed $\sigma > 0$, independent of ε , gives

$$|F_{2R} - F_R| \leq C_k \varepsilon^{-E_k} R^{-\sigma k + (1-\sigma)Q}, \quad E_k = \sigma \max(B_k, A) + (1-\sigma)A. \quad (8.14)$$

Choose one finite k with $\sigma k - (1-\sigma)Q > 1$ and then fix its finite test seminorm. Summing dyadic differences gives a fixed inverse power of ε . If localization starts at a dyadic $R_0(\varepsilon)$ bounded by a power of ε^{-1} , its initial norm contributes another finite power. The construction of Proposition 6.6 permits this choice. The accuracy order is never increased while ε tends to zero.

This order of choices is essential: the continuation geometry fixes σ , then one chooses k , and only then lets the relative gap shrink. The resulting E_k is a fixed finite exponent in the face bound.

For completeness, unsmearing this estimate does not involve point evaluation of a bounded approximant. The limit already has a holomorphic kernel by Proposition 6.7. Choose a smooth radial bump ρ on $\mathbb{C}^4$ of integral one, supported in a sufficiently small ball. Holomorphic mean value in each of p slots gives, writing Z for the full slot coordinates,

$$\begin{aligned} W_2(Z) &= \int_{(\mathbb{C}^4)^p} \prod_i \varepsilon^{-8} \rho(w_i/\varepsilon) W_2(Z + w) dw \\ &= \int db W_2(f_{\varepsilon, b_1}, \dots, f_{\varepsilon, b_p}; Z + i\varepsilon b). \end{aligned} \quad (8.15)$$

Here $f_{\varepsilon, b}(u) = \varepsilon^{-4} \rho(u/\varepsilon + ib)$, and b ranges over a fixed compact set and $p_{K, m}(f_{\varepsilon, b}) \leq C_m \varepsilon^{-4-m}$. The finite seminorm estimate just proved therefore bounds the last integral by a power. Cauchy's inequalities in smaller ε -balls also bound all fixed derivatives. This proves (8.12) initially in the thin real-time strip.

Finally fix z and the imaginary displacement $i\varepsilon v$. The function $u_\varepsilon(t, \mathbf{r}) = W_2((t, \mathbf{r}) + i\varepsilon v; z)$ satisfies the ordinary inhomogeneous wave equation in the real variables. Its initial value, spatial derivatives, and first time derivative have the bounds just obtained. Its source is the Fourier–Laplace continuation of the inductively tempered S_2 , so it has a power bound of its own at this relative gap. Kirchhoff's formula on $|t| \leq T$, $|\mathbf{r}| \leq B$ gives

$$\|u_\varepsilon\|_{T, B} \leq \|u_\varepsilon(0)\|_{B+T} + T \|\nabla u_\varepsilon(0)\|_{B+T} + T \|\partial_t u_\varepsilon(0)\|_{B+T} + \frac{1}{2} T^2 \|S_2\|_{T, B+T}, \quad (8.16)$$

with an immaterial fixed norm constant. This proves the bound on every real compact set. Derivatives obey the same estimate. Moderate-growth boundary values now give F_0 ; the estimates uniform on z -compacts give holomorphy and locally uniform finite distribution order in r . $\square$

8.4 Causal ambiguity and recovery on the wrong massless branch

The source difference $S_2 - S_1$ vanishes for real spacelike r by inductive locality. Split it into distributions supported in $J^+(0)$ and $J^-(0)$ by a partition away from $r = 0$ and finite Taylor subtraction at that point. Subtraction acts only in r , so all merged z -spectra are preserved. Convolve these two parts with the retarded and advanced fundamental solutions, respectively. The cone-supported convolutions are proper on bounded sets and define tempered distributions; their Schwartz seminorms involve only finitely many seminorms of the sources. Adding their sum to W_1 gives a tempered distribution T_0 satisfying

$$\square_r T_0 = S_2, \quad \text{supp}_r(T_0 - W_1) \subset J(0), \quad (8.17)$$

with the merged spectra. Let the same symbol denote its Fourier–Laplace continuation in z , valued in r -distributions.

At fixed strict z , the difference $H_* = F_0 - T_0$ solves the homogeneous wave equation and is supported in $J(0)$; spacelike tie gluing proves the support assertion. A homogeneous wave distribution has distributional Cauchy data. At time zero those data are supported at the spatial origin, so each is a finite sum of derivatives of a delta function. Cauchy uniqueness consequently gives

$$H_*(r; z) = \sum_{j=1}^m C_j(z) P_j(\partial_r) \Delta_0(r). \quad (8.18)$$

The P_j may be chosen homogeneous and linearly independent modulo the wave quadratic, with Δ_0 the massless commutator function. One fixed finite list works on the connected merged tube. Indeed, on a relatively compact open subset the distribution order is uniform, so sufficiently high spatial monomials annihilate both Cauchy data. These annihilators continue holomorphically to the whole tube. Cauchy restriction is continuous on homogeneous wave solutions, and fixed dual tests recover the finitely many coefficients; hence the C_j are holomorphic.

The unknown part of the boundary has now been reduced to finitely many scalar coefficient functions on the merged tube. The next lemma proves their distribution regularity by testing on the massless branch excluded by the target ordering. The causal solution T_0 is already tempered and will supply the polynomial bound in that test.

Lemma 8.6. *Every coefficient C_j in (8.18) has a local ordinary distribution boundary in the merged variables.*

Proof. Fix a real merged center. Initially bC_j is only a hyperfunction. Choose a small real box about the center, a smaller box of centers w , and a strict merged direction v_z . Lemma 8.4 represents the Gaussian testing by a fixed translate $it_0 v_z$, up to exponentially small side errors.

Choose a strict relative direction v_r for the target ordering, small enough compared with the merged gaps that (v_r, v_z) is a strict full target direction. Choose finitely many nonzero null covectors k_a on the wrong branch, so $k_a \cdot v_r < 0$. The sample points range over a radially open patch of that branch; they are not constrained to a fixed energy shell. The radius variation will be used when separating coefficient polynomials of different degrees; choosing directions only on one fixed shell would not give that independence argument. Then choose $r_* > 0$ sufficiently small that, uniformly for all samples and $|\eta| = r_*$,

$$k_a \cdot v_r + \eta \cdot v_z \leq -c_0 < 0. \quad (8.19)$$

Test (8.18) in r with $\chi(r)e^{-i\Lambda r \cdot k_a - \Lambda r^2/2}$, where $\chi = 1$ near zero, and in z with the Gaussian (8.9) on the fixed translated box.

For the F_0 term, deform the relative contour from zero to it_0v_r , leaving z at height it_0v_z . Lemma 8.5 justifies the initial endpoint. On the final central contour the real exponent, apart from negative spatial Gaussians, is at most

$$\Lambda \left[-t_0c_0 + \frac{1}{2}t_0^2(|v_r|^2 + |v_z|^2) \right].$$

Choose t_0 so small that this is $\leq -c\Lambda$. The continued W_2 is bounded on that fixed interior compact, so this term is exponentially small.

We specify the cutoff error, since the relative contour starts at a distributional face. Let N be an inverse-height growth exponent from (8.12), increased to cover the derivatives used below. Replace $\chi(x)$ on the moving contour by its almost-analytic Taylor lift

$$\chi^{[M]}(x + iy) = \sum_{|\alpha| \leq M} (iy)^\alpha \partial^\alpha \chi(x) / \alpha!, \quad M > N + 1.$$

Its antiholomorphic derivative is $O(|y|^M)$, supported in the transition region. There $|x|$ is separated from zero. By decreasing t_0 before taking the large-parameter limit, (8.11) is $\leq -c'\Lambda$ throughout the transition, even after all linear phase costs. Stokes errors are bounded by a polynomial in Λ times $e^{-c'\Lambda} \int_0^{t_0} t^{M-N} dt$. The same construction applies to any smooth central cutoff in z . For the merely hyperfunctional coefficient boundary we instead use the fixed-height analytic-functional remainder of Lemma 8.4; we do not multiply that boundary by a smooth cutoff.

The T_0 term is polynomially bounded in Λ : move its merged contour back to the real boundary and use its tempered seminorm. Derivatives of the Gaussian and cutoffs cost only powers of Λ . Its contour errors obey the same estimate, now with the known finite distribution order of T_0 .

It remains to check that these tests recover every coefficient. Put $d_j = \deg P_j$ and

$$M_{aj}(\Lambda) = \left\langle P_j(\partial) \Delta_0, \chi(r) e^{-i\Lambda r \cdot k_a - \Lambda r^2/2} \right\rangle.$$

The Fourier transform of Δ_0 is a nonzero normalization times $\text{sgn}(p^0)\delta(p^2)$. Gaussian Fourier integration, with $p = \Lambda k_a + \sqrt{\Lambda}q$, gives

$$\Lambda^{3/2-d_j} M_{aj}(\Lambda) \longrightarrow c(k_a) P_j(ik_a), \quad c(k_a) \neq 0. \quad (8.20)$$

Indeed dp contributes Λ^2 , the delta function $\Lambda^{-3/2}$, and the four-dimensional Gaussian Fourier factor Λ^{-2} . The remaining integral is a nonzero Gaussian integral on $2k_a \cdot q = 0$, with the fixed sign of the branch. The cutoff error is exponentially small by the temperedness of $P_j(\partial)\Delta_0$. The nonzero factor $c(k_a)$ can depend on the row.

The evaluation matrix $P_j(ik_a)$ has full column rank for a suitable finite sample set. Otherwise a nonzero linear combination would vanish on an open real patch of one null branch, hence on the irreducible complex quadratic cone, contradicting independence modulo the wave quadratic. Radial openness matters for polynomials of different degrees. Selecting an invertible square submatrix, (8.20) bounds its inverse by powers of Λ . The tested identity therefore gives

$$|\mathcal{T}_\Lambda(bC_j)(w, \eta)| \leq C(1 + \Lambda)^N \quad (|\eta| = r_*),$$

uniformly on the smaller box. Lemma 8.4 proves that every bC_j is an ordinary distribution there. This conclusion is local in the merged real variables. Growth at infinity and the merged spectral support still require the scaling argument below. $\square$

8.5 Global growth, spectra, and completion of mixed products

Local distribution boundaries do not by themselves imply global temperateness. Here exact similarities provide the missing control. The comparison solution T_0 need not share the exact homogeneity of W_2 . Its scaling defect will instead appear as an explicitly tempered error, which is enough both for dyadic growth estimates and for excluding frequencies outside the merged spectral cone. If W_2 has n_L entries L and homogeneous original entries of dimensions $\Delta_1, \dots, \Delta_k$, put

$$d_W = 2n_L + \sum_{i=1}^k \Delta_i.$$

Here the old dimensions already include any derivative shifts. The accessible analytic product has exact similarity covariance for this homogeneous list, by covariance of the one-wave data and uniqueness of the multilinear accessible products. Its distributional r -boundary F_0 has the same exact scaling. Put $\delta_j = 2 + \deg P_j$, so $P_j(\partial)\Delta_0(\lambda r) = \lambda^{-\delta_j} P_j(\partial)\Delta_0(r)$. Choose fixed compact r -tests φ_j dual to the finite list $P_j(\partial)\Delta_0$. Exact scaling of F_0 and (8.18) show that

$$\lambda^{d_W - \delta_j} C_j(\lambda z) - C_j(z) = \left\langle T_0(r; z) - \lambda^{d_W} T_0(\lambda r; \lambda z), \varphi_j(r) \right\rangle_r. \quad (8.21)$$

The right side has a tempered boundary with merged spectral support. For $\lambda \geq 1$ its Schwartz seminorm constants grow at most polynomially in λ : scaling and the fixed compact extraction tests act continuously on Schwartz space with such bounds.

Cover the complement of a compact z -ball by dyadic annuli and rescale each annulus to a fixed compact set. There the local coefficient distribution has a fixed finite order. Formula (8.21) bounds its value on every outer annulus by a polynomial in the annulus radius and finitely many Schwartz seminorms. Summing against a Schwartz test proves that each C_j is tempered. This argument needs no exact homogeneity of T_0 .

The scaling defect in (8.21) has Fourier support in K_z . Apply Lemma 8.2 to the now tempered coefficient boundary, with $d = d_W - \delta_j$ and $V(\lambda) = 1$. Its local wedge boundary is supplied by Lemma 8.6. We obtain $\text{supp } \hat{C}_j \subset K_z$. Fourier–Laplace continuation and injectivity identify the original C_j with the corresponding tempered continuation; in particular it now has ordinary polynomial bounds at ordered approaches.

It follows that F_0 is the merged continuation of the globally tempered distribution

$$D_0(r, z) = T_0(r, z) + \sum_j C_j(z) P_j(\partial_r) \Delta_0(r). \quad (8.22)$$

It is exactly homogeneous, because its merged continuation F_0 is. We next establish the full target spectrum, rather than just the merged one. For a covector strictly outside the full target cone, choose a strict target imaginary direction (v_r, v_z) with negative pairing. Local Gaussian testing near $(r, z) = 0$ first moves z to a small height $it_0 v_z$, using the already proved merged temperateness. At that face, $D_0 = F_0$; move r to $it_0 v_r$ using Lemma 8.5. Intermediate fractions of the latter move remain in the target tube. The final contour is an interior W_2 contour and has exponential Gaussian decay. The side errors are those already estimated in the proof of Lemma 8.6. This supplies the local off-cone Fourier decay used in Lemma 8.2; exact homogeneity of D_0 therefore gives its full target spectral support. Its full Fourier–Laplace continuation agrees with W_2 , since they have the same r -boundary at fixed strict merged damping.

One adjacent interchange is therefore complete: it preserves temperateness, recovers the full ordered spectrum, and retains spacelike locality. Iterate this step in the simultaneous induction stated above. At fixed number of L entries, begin with all of them on the right and move them

to the prescribed positions. Every source has one fewer L and one additional old entry Θ , so it belongs to the preceding induction stage. Repeat this for every finite ordering of the old entries. Tie gluing, followed by the ordinary merged spectral boundary with a spacelike relative test, then proves locality also for adjacent old-field interchanges, with the prescribed grading. Adjunction and exact similarity covariance for each homogeneous list follow from those identities of the analytic accessible products and uniqueness of their boundaries.

We next complete the Hilbert-valued assertion for the same homogeneous lists. The squared norm of a mixed analytic vector product is its reflected scalar split. A strict vector approach gives a strict scalar split approach: the central gap is twice the first vector damping, and the other gaps are the reflected vector gaps. The scalar polynomial bound therefore gives a polynomial norm bound for the vector product. The Hilbert-valued moderate-growth theorem supplies its local boundary. Its exact similarity covariance for this homogeneous list, with a unitary dilation factor on the output, globalizes the boundary by dyadic annuli and gives the vector support by Lemma 8.2. Thus this boundary is a tempered Hilbert-valued distribution with the full ordered vector spectrum. Taking both strong distributional boundaries in the reflected split, first for factor tests and then for joint tests by continuity, gives

$$W(B^*A) = \langle \Psi(B), \Psi(A) \rangle$$

for any two homogeneous words A, B with their tests. The concatenated cross list is itself a list of homogeneous slots, even when the two words have different total dimensions.

It remains to pass from homogeneous slots to arbitrary original fields. Write each such field in the fixed list as $A_i = \sum_{\Delta \in S_i} A_{i,\Delta}$, with S_i finite, as in Assumption 2.1. Let $\mathbf{A}$ denote the complete ordered list, including its L slots, and let $\mathbf{A}_\Delta$ replace each old A_i by A_{i,Δ_i} . If there are no old slots, the indexing product below is a singleton. Multilinearity of the accessible analytic products already gives the corresponding finite expansion in the tube. For an already constructed homogeneous list $\mathbf{B}$, write $W_{\mathbf{B}}$ and $\Psi_{\mathbf{B}}$ for its scalar and vector boundary maps. For every joint Schwartz test h , in particular every compact joint test, define the maps for $\mathbf{A}$ by

$$\begin{aligned} W_{\mathbf{A}}(h) &= \sum_{\Delta \in \prod_i S_i} W_{\mathbf{A}_\Delta}(h), \\ \Psi_{\mathbf{A}}(h) &= \sum_{\Delta \in \prod_i S_i} \Psi_{\mathbf{A}_\Delta}(h). \end{aligned} \tag{8.23}$$

The same joint kernel h occurs in every term. This is a finite expansion of field labels, not an assertion that h has finite tensor rank. Each summand is an already constructed tempered distribution, and every summand has the same cone for the chosen ordering. Their finite sum therefore has the required temperateness and support and has the specified analytic product as its Fourier–Laplace continuation. Adjunction and locality pass through the same finite expansion. The exact scaling laws of the summands give similarity covariance with dilations acting on the original field labels; a nonhomogeneous list is not assigned one total dimension. Orders and constants may be chosen by taking a maximum of the finitely many seminorm orders and a sum of the corresponding constants, without any uniform estimate over dimensions or lengths. Finite distributional derivatives in any slot preserve temperateness and cone support and have the shifted homogeneous covariance laws term by term.

The reflected identity also passes through this expansion with all cross terms. For arbitrary finite linear combinations P, Q of finite ordered words with joint tests, expand their original labels finitely and apply the homogeneous split to each pair of component words. Here P^*Q uses reversed adjunction on each bra word and the tensor product of its conjugated, reversed test with the ket

test. The result is

$$W(P^*Q) = \langle \Psi(P), \Psi(Q) \rangle. \quad (8.24)$$

In particular, the full joint Wightman positivity inequalities are the Gram inequalities for these vectors in the original $\mathcal{H}$. This uses the cross-tuple splittings, not a claim of termwise positivity for the individual summands.

The homogeneous induction, vector construction, and finite identities (8.23) and (8.24) complete the proof of Theorem 8.1.

8.6 The joint domain and affiliation with the given net

Let $\mathcal{D}$ be the span of the reconstructed vacuum products with joint Schwartz test kernels, including compact joint kernels and finite sums of factorized smearings.

The passage from positive vacuum distributions to an invariant insertion domain is the Wightman reconstruction mechanism, in the test-algebra formulation of Borchers [4, 40]. Here the vacuum vectors are already in the given $\mathcal{H}$; after this domain construction we must still prove affiliation in the fixed net.

For an old vacuum word represented by such a kernel, expand its field labels as in (8.23) under the same joint kernel. Old product multilinearity and the old-only identification in Proposition 6.7 give equality of the old word vector with this finite sum of homogeneous reconstructed word vectors. This exact label expansion makes no finite-rank assertion about the joint kernel. Since $\mathcal{D}_{\text{old}}$ is exactly the algebraic span of these original vacuum-word vectors, it follows that $\mathcal{D}_{\text{old}} \subset \mathcal{D}$. In particular, $\mathcal{D}$ is dense in the original $\mathcal{H}$. This inclusion concerns the specified vacuum-generated domain, not arbitrary additional vectors of a separately chosen invariant domain.

Regard a test polynomial as a finite algebraic sum of finite ordered words with the allowed kernels. Multiplication concatenates the words and tensors their kernels, and adjunction reverses the entries and conjugates and reverses the kernel. All resulting lengths remain finite, with no fixed upper bound. Left insertion defines joint fields on $\mathcal{D}$. To check that it is well-defined, let a whole polynomial A represent a zero vector. Equation (8.24) gives $W(A^*A) = 0$. Positivity and Cauchy–Schwarz applied to A and B^*BA , for any test polynomial B , give

$$|W(A^*B^*BA)|^2 \leq W(A^*A)W((B^*BA)^*(B^*BA)) = 0.$$

Every term exists by the all-finite-length construction, and $W(A^*B^*BA) = \|\Psi(BA)\|^2$. Thus zero states form a left ideal. This applies to the whole polynomial, even when its zero vector results from cancellation between different scaling components. Adjoint insertion gives $A^\dagger(\bar{f})|_{\mathcal{D}} \subset A(f)^*$, where the right side is the Hilbert-space adjoint. Its dense domain therefore makes each smeared field closable. For an old field inserted into a vector of $\mathcal{D}_{\text{old}}$, the same old-only identification applied to the longer word, with the inserted test tensored with the original joint kernel, proves agreement with the old insertion. For compact old smearings, Section 2 identified that insertion and its adjoint with the restrictions of the original affiliated realizations. Thus the old fields and adjoints agree with their original polynomial restrictions on $\mathcal{D}_{\text{old}}$, and (8.1) becomes the operator identity

$$\square L = \Theta \quad \text{on } \mathcal{D}. \quad (8.25)$$

Translation covariance, similarity covariance, and locality are operator identities on this common invariant domain. No additional Hilbert-space sector has been introduced.

This gives the common operator domain but has not yet supplied a closed realization in a prescribed local algebra. For that purpose we prove exchange with every bounded element of that

algebra's commutant and then enlarge the domain by those commutant actions. The distinction matters because weak field locality on $\mathcal{D}$ alone does not establish affiliation.

We will use the same extension argument for the new scalar, the old fields on the enlarged domain, and compact joint polynomials. It is convenient to prove it once for an adjoint pair.

Lemma 8.7 (Extension by commutant actions). *Let $\mathcal{D}$ be dense in $\mathcal{H}$, and let $A, A^\sharp$ be operators on $\mathcal{D}$. Fix a localization set K and the original algebras $\mathcal{M}(O)$ for open neighborhoods O of K . Suppose that for every such O and every $C \in \mathcal{M}(O)'$,*

$$\langle \phi, CA\psi \rangle = \langle A^\sharp \phi, C\psi \rangle, \quad \phi, \psi \in \mathcal{D},$$

and that the corresponding identity also holds with $A, A^\sharp$ interchanged. Then A has a closed extension affiliated with every $\mathcal{M}(O)$, whose adjoint extends $A^\sharp$. The reverse adjoint pair has the same property. If $A^\sharp = A$, the extension is symmetric.

Proof. Let $\mathcal{D}_K$ be the span of $C_1 \cdots C_m \psi$, with $\psi \in \mathcal{D}$ and each C_j in the commutant of an allowed neighborhood algebra, including $m = 0$. Every finite collection of these neighborhoods has a common smaller open neighborhood of K . By isotony all its commutant coefficients belong to the commutant of this smaller algebra. Products and adjoints of the coefficients therefore obey the assumed exchange identities.

For a finite sum with product coefficients C_i , prescribe

$$\tilde{A} \sum_i C_i \psi_i = \sum_i C_i A \psi_i, \quad \tilde{A}^\sharp \sum_i C_i \psi_i = \sum_i C_i A^\sharp \psi_i.$$

If the input sum is zero, pairing the proposed first image with $\phi \in \mathcal{D}$ gives $\langle A^\sharp \phi, \sum_i C_i \psi_i \rangle = 0$. Density proves well-definedness; the reverse exchange proves it for the second prescription. For two sums use exchange with $D_j^* C_i$ in a common smaller algebra. This gives

$$\tilde{A}^\sharp \subset \tilde{A}^*, \quad \tilde{A} \subset (\tilde{A}^\sharp)^*.$$

Both operators are thus closable. Their common domain is invariant under every unitary in every allowed commutant, and both operators commute with that unitary and its inverse. Their closed graphs retain these invariances, proving affiliation simultaneously with all the neighborhood algebras. The adjoint inclusions survive closure. In the symmetric case the two prescriptions coincide, so their closure is symmetric as well. $\square$

Proposition 8.8. *For every real test f with compact support in an open region O , $L(f)$ has a closed symmetric extension affiliated with the original algebra $\mathcal{M}(O)$. These realizations can be made compatible for all neighborhoods of $\text{supp } f$.*

Proof. We first prove exchange with every bounded operator of the actual commutant, not merely with opposite-region fields. If $C \in \mathcal{M}(O)'$, then

$$\langle \phi, CL(f)\psi \rangle = \langle L(f)\phi, C\psi \rangle, \quad \phi, \psi \in \mathcal{D}. \quad (8.26)$$

First take ϕ, ψ to be finite sums of words with factorized compactly supported tests, and by linearity treat one word on each side. Translate their slots into real mutually spacelike patches, all spacelike from $\text{supp } f$. Subdivide f into local patches and choose its self-adjoint bounded approximants in O . The simultaneous approximating products converge there in Hilbert norm. Exchange holds at the bounded level because each approximant to $L(f)$ commutes with C . The norm limit proves (8.26) on that real open placement set.

Continue separately in the translated polynomial slots on either side. A fixed $L(f)$ on the left of a polynomial leaves every suffix spectral cone of that polynomial intact; on a bra the corresponding opposite cones hold. The bounded operator C between the vectors does not affect these two separate vector analyticity statements. To see the common domain explicitly, use the positive vector tube of (6.12) for each polynomial before taking the bra adjoint. For a polynomial ψ of length n , both $\psi(b)$ and $L(f)\psi(b)$ are holomorphic on that tube. In the latter product let q_1 be its total momentum and $q_2, \dots, q_{n+1}$ its suffix momenta. Fixing the first, L , coordinate real removes its imaginary damping, but the strict polynomial gaps still damp all q_j with $j \geq 2$. The real smearing contributes $\hat{f}(q_1 - q_2)$, whose rapid decay controls the remaining momentum since $|q_1| \leq |q_1 - q_2| + |q_2|$. The tempered vector distribution therefore has this smeared face, with all finite derivative bounds. The same applies to ϕ and $L(f)\phi$. After conjugating the bra variables, the common scalar domain is the product of the negative bra tube and the positive ket tube. Boundedness of C suffices for these pairings; neither commutation with translations nor invariance of $\mathcal{D}$ under C is used. Their scalar pairings therefore obey boundary uniqueness, which extends the identity to arbitrary placements and smearings. For a compact joint kernel, first use a finite partition into product patches and then approximate each piece by finite sums of factor tests in the required finite smooth seminorms. Theorem 8.1 gives Hilbert-norm convergence both for the polynomial vector and for the vector with the additional $L(f)$ slot. Boundedness of C therefore passes the identity to the joint kernel. This test-topology limit neither asserts finite tensor rank nor uses invariance of $\mathcal{D}$ under C . Compact joint tests are dense in the joint Schwartz space, so the same Hilbert-norm continuity extends exchange to all joint Schwartz kernels in $\mathcal{D}$.

Apply Lemma 8.7 with $A = A^\sharp = L(f)$ and $K = \text{supp } f$. The exchange identity holds for every neighborhood, so the lemma gives one compatible closed symmetric affiliated extension. Essential self-adjointness is neither asserted nor required. $\square$

For a complex test $f = f_1 + if_2$, linearity of the exchange identities for the real tests gives the adjoint pair $A = L(f)$, $A^\sharp = L(\bar{f})$ on $\mathcal{D}$. Lemma 8.7 supplies a compatible closed affiliated realization with the required adjoint. This uses neither self-adjointness nor strong commutation of the real and imaginary parts.

We likewise need compatible realizations of the original entries on the reconstructed domain before using polar tails on its vectors. For an original field A and compact test f , use the bounded approximants of Proposition 6.6, whose vacuum and adjoint-vacuum convergence was established before the mixed reconstruction. Fix an open O containing $\text{supp } f$ and choose these approximants $A_R(f)$ in $\mathcal{M}(O)$. Initially take the vectors to be words with factorized compactly supported tests, as above. At separated real placements the simultaneous product limits give convergence of $A_R(f)\psi_R$ and $A_R(f)^*\phi_R$. Bounded commutation with $C \in \mathcal{M}(O)'$ therefore gives

$$\langle \phi, CA(f)\psi \rangle = \langle A^\dagger(\bar{f})\phi, C\psi \rangle, \quad \phi, \psi \in \mathcal{D}. \quad (8.27)$$

Initially this holds on the separated placement set. The same separate-vector-tube argument, now with the first slot $A(f)$ or $A^\dagger(\bar{f})$ smeared over real coordinates, and the same joint-test continuity extend it to all the stated vectors. Only the already proved mixed vector spectra and the boundedness of C enter this continuation.

Apply Lemma 8.7 to the pair $A(f), A^\dagger(\bar{f})$ on $\mathcal{D}$, using (8.27) and its adjoint version in every neighborhood. This gives compatible affiliated extensions of the reconstructed restrictions, with adjoints extending the corresponding insertions. On $\mathcal{D}_{\text{old}}$ these restrictions agree with the original chosen realizations by the old-domain argument above. We do not require the newly reconstructed vectors to lie in those previously chosen closed realizations, or identify their full graphs with the extensions just constructed.

Only now use polar truncation for these realizations on polynomial states. If $\hat{A}$ is the chosen closed realization, then $\hat{A}$ and $\hat{A}^*$ agree on $\mathcal{D}$ with the corresponding insertions, which preserve $\mathcal{D}$. Induction gives

$$\mathcal{D} \subset \text{Dom}(\hat{A}^* \hat{A})^m \cap \text{Dom}(\hat{A} \hat{A}^*)^m \quad (m = 1, 2, \dots),$$

with the powers there equal to the alternating joint polynomial words. Their norms are finite by Theorem 8.1, and spectral calculus gives the polar-tail estimates on every polynomial vector. For factorized polynomials, the natural-product and polar-cutoff argument of Section 2 applies to these compatible realizations. For compact joint kernels, take finite factor-test approximations in the required test seminorms and pass both the vacuum and adjoint-vacuum vectors by joint Hilbert-distribution continuity. The closed wedge Tomita graph then passes its reality relation from the factorized affiliated polynomial realizations to Hermitian joint polynomial vectors. These are the wedge reality properties needed for the charge estimates of Section 5.

Finally, affiliation is only one part of the physical-membership criterion in Assumption 2.1. The other part is also present: $L = S$ is the explicit time-line extraction from actual modular transforms of the physical virial field in arbitrarily small diamonds. Its separated products are the limits of those physical bounded operations, and their joint continuation is uniquely fixed by them. Thus the field satisfies the stated intrinsic membership criterion in the given net. In a gauge formulation all these operations take place in the chosen physical, gauge-invariant algebras.

All construction requirements have now been checked: localization in arbitrarily small original-net neighborhoods; joint field distributions at every finite length on the common invariant domain $\mathcal{D} \supset \mathcal{D}_{\text{old}}$, with the required adjoints, covariance, spectrum, and locality; and compatible affiliated smeared realizations in the original net. The notation L henceforth denotes the resulting physical Hermitian scalar of dimension two on $\mathcal{D}$.

8.7 Improvement and the local Ward algebra

Define on $\mathcal{D}$

$$t_{\mu\nu} = T_{\mu\nu} + \frac{1}{3}(\partial_\mu \partial_\nu - \eta_{\mu\nu} \square)L. \quad (8.28)$$

It is a physical symmetric conserved tensor, and (8.25) gives $t^\mu{}_\mu = 0$. For an affine Poincaré Killing vector X , its charge difference from T is the divergence of the antisymmetric superpotential

$$\frac{1}{3}\partial^\lambda (X_\lambda \partial_\mu L - X_\mu \partial_\lambda L + (\partial_\lambda X_\mu)L). \quad (8.29)$$

Its compact commutator flux vanishes, so translation and Lorentz charges agree. If the initial stress tensor required the weight replacement of Section 2, also include the local difference between that selected tensor and the initial one. That difference is symmetric and conserved and has zero affine charges by the weight-replacement argument. This proves the improvement criterion. We now check the special-current Ward algebra on the actual reconstructed domain.

We first construct the local Ward action on the represented polynomial algebra generated by the original fields and L ; this is distinct from the vector space $\mathcal{F}$ of point-field labels. Proposition 8.11 will extend the action to every further jointly admitted finite family. For an open O , let $\mathfrak{P}_c(O)$ consist of finite sums of finite ordered products of original fields, L , and their covariant derivatives, smeared by compact smooth joint kernels supported in O^n , and identified when their restrictions to $\mathcal{D}$ agree. The joint-domain construction makes concatenation and reversed adjunction well-defined on this quotient and leaves $\mathcal{D}$ invariant. Write $P^\dagger$ for the reversed field-adjoint polynomial, to distinguish it from the Hilbert adjoint. Every finite length and every admitted compact joint kernel are included; the kernel need not have finite tensor rank.

These compact polynomial insertions have compatible affiliated realizations in the original net. For a factorized word, take the natural-domain product of the single-field realizations constructed above, and take natural sums for finite polynomials. Their domains contain $\mathcal{D}$, and their adjoints contain the reversed adjoint polynomials there, so they are closable. Every commutant unitary and its inverse preserve their natural graphs; their closures are therefore affiliated. This uses no commutation among the unbounded factors. In particular, $t(f)$ is the finite-sum case of this argument.

For a compact joint kernel, partition into product patches inside O^n and approximate in the smooth test topology by finite sums of tensor tests. If P_j are the resulting factorized polynomials, then Theorem 8.1, with any fixed $\mathcal{D}$ word appended, gives

$$P_j\psi \longrightarrow P\psi, \quad P_j^\dagger\phi \longrightarrow P^\dagger\phi \quad (\phi, \psi \in \mathcal{D})$$

in Hilbert norm. Affiliation of the factorized closures gives exchange with commutant unitaries, hence with every $C \in \mathcal{M}(O)'$ by linear combinations of unitaries. Boundedness of C passes it to the limit:

$$\langle \phi, CP\psi \rangle = \langle P^\dagger\phi, C\psi \rangle, \quad \phi, \psi \in \mathcal{D}.$$

The reversed adjoint polynomial obeys the reverse exchange identity. Lemma 8.7, with $A = P$ and $A^\sharp = P^\dagger$, therefore supplies compatible affiliated extensions for every neighborhood of the compact localization. This is an extension with the required adjoint, not an assertion about the minimal closure or a limit of the factorized closures. Its polar truncations are bounded original-net operations converging on $\mathcal{D}$, with adjoint convergence there. All the commutants here are those of the original net.

For a real conformal Killing field X , set $j_{X,\mu} = t_{\mu\nu}X^\nu$ and $\sigma_X = (\partial \cdot X)/4$. Conservation and tracelessness give $\partial^\mu j_{X,\mu} = 0$. Define

$$\delta_X A = i[q_X, A] \tag{8.30}$$

as a compact commutator flux enclosing the insertion. This is an operator insertion on $\mathcal{D}$, without assuming the existence of a global unbounded charge on every state. More explicitly, smear the two fields with a compact kernel in their relative coordinate and the target test in their merged coordinate. Caps carry the same unit-flux normalization and orientation. In relative coordinates the commutator current is divergence-free and supported in causal positions. The difference of two normalized enclosing caps is a compact gradient there, up to spacelike outer cutoff terms. Conservation and locality therefore allow exact deformation or shrinking while keeping the target enclosed. The resulting expression is a finite sum of joint smeared products; combining the two momenta preserves the ordered cones for the remaining slots. Polynomial coefficients of X give finite derivatives of test kernels and do not spoil these properties.

For a joint target compactly localized in O , partition into product patches and choose the relative caps within their localization margins. A flux about a word is the finite sum of the fluxes about its slots, so these representatives belong to $\mathfrak{P}_c(O)$. To check the quotient by zero operators, choose one absolute normalized compact cap enclosing the finite union of target projections for the polynomial representatives. It may lie outside O . Deformation identifies every slotwise cap with this cap. The algebraic commutator rule identifies the sum with $i[Q_X, P]$ on factor tests, and the same joint Hilbert continuity extends the identity to the joint kernel. Since Q_X and P preserve $\mathcal{D}$, this vanishes if $P|_{\mathcal{D}} = 0$. Thus the local derivations are well-defined on $\mathfrak{P}_c(O)$ and preserve it; their iterates are finite joint polynomials. The preceding exchange argument supplies their compatible original-net affiliated extensions.

The complete vacuum flux vanishes on compact joint products by Lemma 5.4 and its subsequent dyadic charge estimate. That proof applies here because the new compact polynomial vectors have

the wedge reality properties just proved, while the current vacuum wave is the same purely timelike, translation-smooth wave used there. Its mass-band estimate is uniform; the summable low-energy and high-energy dyadic bounds remain unchanged. Consequently

$$\sum_{i=1}^n W(A_1, \dots, \delta_X A_i, \dots, A_n) = 0. \quad (8.31)$$

The derivation preserves adjunction. Its translation, Lorentz, and dilation parts on the original fields are the physical actions: (8.29) handles the first two, and Proposition 5.8 removes the possible conserved virial charge in the third. Identities first proved on an old vacuum leg extend to $\mathcal{D}$ by locality, old polynomial density, and ordered boundary uniqueness.

We need the transformation of the current itself to identify the Lie algebra. The two required primary identities are

$$\delta_X L = X^\lambda \partial_\lambda L + 2\sigma_X L, \quad (8.32)$$

$$\delta_X t_{\mu\nu} = X^\lambda \partial_\lambda t_{\mu\nu} + (\partial_\mu X^\lambda) t_{\lambda\nu} + (\partial_\nu X^\lambda) t_{\mu\lambda} + 2\sigma_X t_{\mu\nu}. \quad (8.33)$$

We explain their return from the previously established pair representation. All short analytic products now agree with the joint tempered ones: they agree on separated vacuum-wave products, then everywhere by continuation. Thus the Wick sphere/cap comparison of Proposition 7.8 applies directly to physical joint pair boundaries. For a translation and target T , the Lorentzian cap difference is the physical translation Ward identity. Its sphere version has the common nonzero normalization fixed there. The translation sphere identity for L was already proved, and differentiating it at a fixed enclosing surface gives the same identity for the derivative terms in (8.28). It therefore holds with target t .

Conjugate these translation identities in the positive pair representation of Proposition 7.7, whose primary targets include L and t . Transporting the surrounding sphere and its holomorphic Killing coefficient gives the sphere identities for every conformal Killing field. Return to Lorentzian commutators with the same averaged cylinders as in Proposition 7.8. The time caps converge by the joint tempered boundary theorem; the sides have a fixed spacelike margin and vanish with their thickness. The Wick pullback of X is holomorphic on these compact surfaces, so the identical argument works for its degree-zero, degree-one, and degree-two coefficients. The normalization is the translation normalization in every case. This proves (8.32)–(8.33) on the vacuum. Their discrepancies are relatively local insertions. Move such an insertion to an extreme position on a spacelike open set; old polynomial density and ordered uniqueness then show that its vacuum vanishing implies vanishing on $\mathcal{D}$. The two displayed identities are therefore operator insertion identities on that domain.

Proposition 8.9. *The compact-flux Ward derivations realize the conformal Lie algebra with the active-action convention*

$$[\delta_X, \delta_Y] = -\delta_{[X, Y]}. \quad (8.34)$$

They obey (8.31) and extend the physical affine Ward derivations on the original field algebra.

Proof. Coordinates multiplying fields are held fixed when taking a Ward variation. Using (8.33), the conformal Killing equation, and conservation, direct differentiation gives

$$\delta_X j_{Y, \mu} = \partial^\lambda (X_\lambda j_{Y, \mu} - X_\mu j_{Y, \lambda}) - j_{[X, Y], \mu}. \quad (8.35)$$

For example, expanding the divergence gives $4\sigma_X j_{Y, \mu} + X \cdot \partial j_{Y, \mu} - (\partial^\lambda X_\mu) j_{Y, \lambda}$; subtracting $j_{[X, Y]}$ cancels the derivative of Y , and $\partial_\lambda X_\mu + \partial_\mu X_\lambda = 2\sigma_X \eta_{\lambda\mu}$ yields exactly the two index terms and the remaining $2\sigma_X$ in (8.33).

Choose one compact flux $Q_Y(F)$ implementing δ_Y on an insertion A and its local X variation. Use a larger compact flux for δ_X on $Q_Y(F)$ and these insertions. The derivation rule on the joint domain gives

$$[\delta_X, \delta_Y]A = i[\delta_X Q_Y(F), A].$$

The first term on the right of (8.35) is an antisymmetric divergence. Its commutator flux about the compact insertion is zero: integration by parts leaves only the outer boundary, where locality permits the cutoff to be removed. The second term therefore gives $-i[Q_{[X,Y]}(F), A] = -\delta_{[X,Y]}A$. All products and iterated insertions in this calculation are defined by compact joint kernels on $\mathcal{D}$. Equation (8.31) and the affine identification were proved above. $\square$

8.8 Centered fields and compatible families

A Ward variation with polynomial coordinate coefficients is an insertion, but need not itself be a translation-covariant point-field label. We now remove that coordinate part and verify the intrinsic membership criterion for each finite family needed in a finite Ward calculation. Write

$$\begin{aligned} p_c(z) &= c, & d(z) &= z, & m_{b,c}(z) &= (b \cdot z)c - (c \cdot z)b, \\ k_b(z) &= 2(b \cdot z)z - z^2 b, & b, c &\in \mathbb{R}^{1,3}. \end{aligned}$$

Fix a homogeneous finite Lorentz multiplet A of dimension Δ among the original fields, L , and their covariant derivatives. Let N_A be the dimension of its component space. Define the centered insertion

$$B_{b,A}(x) = \delta_{k_b^{[x]}} A(x), \quad k_b^{[x]}(z) = k_b(z - x).$$

For a target test f , this denotes i times the difference of joint products with relative kernel $f(x)h^\mu(r)k_b^\nu(r)$, $r = y - x$, where h is a normalized enclosing cap. It involves no pointwise product or unsmeared surface restriction. The cap deformation just proved makes the definition independent of its admissible normalized choice.

Define its iterates on the same polynomial domain by

$$\begin{aligned} B_A^{(0)} &= A, \\ B_{b_r \dots b_1, A}^{(r)}(x) &= \delta_{k_{b_r}^{[x]}} B_{b_{r-1} \dots b_1, A}^{(r-1)}(x), \quad r \geq 1. \end{aligned}$$

Every fixed expression is a finite compact joint polynomial because δ_X preserves the local polynomial algebra.

Proposition 8.10 (Finite families of centered fields). *Let A range over a fixed finite collection of homogeneous original fields, L , and their covariant derivatives. For every fixed finite collection of depths, the corresponding centered fields $B_A^{(r)}$ have all mutual and original-field joint products on $\mathcal{D}$, with the required adjoints, locality, covariance, ordered spectra, and compatible original-net affiliated realizations. If A has real dimension Δ and N_A components, depth r has dimension $\Delta - r$ and at most $4^r N_A$ components before field identities. These finite families satisfy the intrinsic membership criterion of Assumption 2.1. Every finite Ward word on the original fields and L is a finite sum of polynomial-coordinate multiples of derivatives of such centered fields.*

Proof. For a similarity $g(z) = \lambda \Lambda z + a$, $\lambda > 0$, write $U(g)$ for the already given Poincaré/dilation implementer. Changing variables in the smooth current test for the dimension-four t sends X to $g_* X = \lambda \Lambda X \circ g^{-1}$ and an absolute normalized cap test h_0 to $(h_0)_g = \lambda^{-1} \Lambda h_0 \circ g^{-1}$, in the tensor

convention of their contraction. For a relative cap, the translation cancels in this formula. Since $g_*k_b^{[x]} = \lambda^{-1}k_{\Lambda b}^{[gx]}$, cap independence gives

$$U(g)B_{b,A}(x)U(g)^{-1} = \lambda^{\Delta-1}\rho_A(\Lambda)B_{\Lambda b,A}(gx),$$

where ρ_A is the Lorentz component matrix of A . Thus $B_{b,A}$ is a possibly zero translation-covariant homogeneous insertion of real grade $\Delta - 1$, in a finite Lorentz family with one additional vector index. Only the existing similarity covariance and the proved cap deformation have been used.

Let $s_{b,c}$ denote the constant component matrix in the affine action

$$\delta_{p_c}A = c \cdot \partial A, \quad \delta_d A = (x \cdot \partial + \Delta)A, \quad \delta_{m_{b,c}}A = (m_{b,c}(x) \cdot \partial + s_{b,c})A.$$

Expanding $k_b(z - x) = k_b(z) - 2(b \cdot x)d(z) - 2m_{b,x}(z) + p_{k_b(x)}(z)$ in the centered flux gives

$$\delta_{k_b}A(x) = (k_b(x) \cdot \partial + 2\Delta(b \cdot x) + 2s_{b,x})A(x) + B_{b,A}(x). \quad (8.36)$$

The polynomial-coordinate term is an insertion, in general not a translation-covariant point-field label; its coefficients multiply the target test after smearing. For example, (8.32) gives $B_{b,L} = 0$.

The local affine Ward actions on B also follow from the proved Lie relation. Indeed $[p_c, k_b] = 2(b \cdot c)d + 2m_{b,c}$ and $[d, k_b] = k_b$. At a fixed enclosing cap target differentiation commutes with the commutator, so (8.34) gives

$$\begin{aligned} (\partial_c - \delta_{p_c})\delta_{k_b}A &= 2(b \cdot c)\delta_d A + 2\delta_{m_{b,c}}A, \\ \delta_d \delta_{k_b}A &= (x \cdot \partial + \Delta - 1)\delta_{k_b}A. \end{aligned}$$

The polynomial term in (8.36) has the same two defects. Subtraction gives $\delta_{p_c}B = c \cdot \partial B$ and $\delta_d B = (x \cdot \partial + \Delta - 1)B$. The Lorentz bracket $[m_M, k_b] = -k_{Mb}$ for $m_M(z) = Mz$ similarly gives the tensor-product component action. Thus these affine identifications are consequences, not an assumed conformal closure.

For any fixed finite collection of centered insertions, substituting their relative kernels into a joint product expands it into finitely many of the already constructed products. This is a continuous map of compact smooth or Schwartz test spaces: the relative supports are compact and the coefficients are polynomial. It supplies all joint products at every finite length on $\mathcal{D}$, including adjoints and all mixed orderings. Merging each block retains the ordered suffix cones. For real b and h , reversing the two products gives $B_{b,A}^\dagger = B_{b,A^\dagger}$. Positivity holds in the original Hilbert space, and locality follows by shrinking the relative supports on spacelike product patches; a compact test on the spacelike configuration set is first partitioned into such patches. The preceding exchange argument supplies compatible original-net affiliated realizations and their adjoints for these finite homogeneous families.

At each depth, choose a shrinking representative of the previous depth inside an inner neighborhood, and its enclosing cap inside the prescribed target neighborhood. The previous depth's locality makes its current commutator causally supported, so conservation gives the same normalized-cap deformation and shrinking property at the next depth. Every nested expression is still a finite joint polynomial in the original fields and L . The joint continuity and full-commutant construction above therefore give all mutual products and compatible affiliation for any fixed finite collection of depths. The covariance calculation gives grade $\Delta - r$ and at most $4^r N_A$ components before field identities. Equation (8.36), the affine actions just proved, and finite Leibniz rules express every fixed finite Ward word as a finite sum of polynomial-coordinate multiples of finite derivatives of these families. For an original $A = \sum_{\Delta \in S_A} A_\Delta$, do this separately for the finitely many components, using the same target joint kernel in every component term and assigning no single weight to their sum. Thus each

finite Ward calculation uses homogeneous insertions in a jointly constructed finite family on $\mathcal{D}$. This does not identify them with original labels in $\mathcal{F}$, assert global closure under centered descendants, or construct a conformal action on the weakly closed bounded net. $\square$

The same currents also act on other fields already admitted by the intrinsic criterion. This requires a joint reconstruction with L ; separate admission of two families is not by itself a statement about their mutual products.

Proposition 8.11 (Extension to compatible physical families). *Let $\mathcal{G}$ be a finite homogeneous finite-component family jointly admitted with the original fields under Assumption 2.1. The scalar L has all joint products with this family and the original fields, with compatible adjoints and original-net affiliated realizations on a common invariant domain $\mathcal{D}_{\mathcal{G}} \supset \mathcal{D}$ in the original Hilbert space. On the resulting represented polynomial algebra, the same currents j_X give unbroken local conformal Ward derivations, extending the physical Poincaré and dilation actions on $\mathcal{G}$ as well as on the original fields. Every fixed finite family of their centered Ward descendants satisfies the same intrinsic membership criterion. These constructions agree when one compatible family is contained in another.*

Proof. Regard the original fields and $\mathcal{G}$ as the spectator fields in the boundary reconstruction of this section. Their joint products, common invariant realization, adjoints, and original-net affiliations are given by the admission criterion. Each fixed homogeneous list has tempered vacuum distributions by the scaling argument of Section 2; finite sums treat the original labels. Their affiliated polar truncations obey Proposition 6.6, since the required moments are alternating words in these already given joint products. Thus Proposition 6.7 supplies their analytic products with the same accessible L and the same reflected splittings. No current Ward identity on $\mathcal{G}$ is needed for this step.

The pure L construction is unchanged. In the simultaneous mixed induction, replacing L by Θ lowers the number of L entries and adds an original homogeneous spectator. The merged-face estimate, causal ambiguity, Gaussian coefficient recovery, and scaling argument therefore apply with $\mathcal{G}$ among the fixed spectator labels. They prove Theorem 8.1 for every such enlarged list. Its positivity and full-commutant exchange then give the invariant joint domain $\mathcal{D}_{\mathcal{G}}$ and the compatible affiliated realizations exactly as above. Products involving only the original fields and L agree with the previous ones: their analytic products are the same approximation-independent limits, and their tempered boundaries are unique. Consequently $\mathcal{D} \subset \mathcal{D}_{\mathcal{G}}$ and all insertion actions agree there. Old-and- $\mathcal{G}$ products likewise retain their given vacuum-generated restrictions, without requiring agreement of chosen closed realizations on an independently larger domain. The same reasoning proves agreement for nested compatible families. If derivatives of the added fields are included as labels, their dimensions lie in the finite union of progressions $\Delta + \mathbb{Z}_{\geq 0}$; each fixed grade receives only finitely many derivative components. This preserves finite scaling support and the stipulated locally finite, bounded-below grade convention.

Compact commutator fluxes now define δ_X on the enlarged polynomial algebra. Conservation, locality, and joint continuity give cap deformation, the derivation rule, and preservation of the quotient by zero operators just as before. Its compact polynomial vectors have wedge reality by their affiliated realizations and the joint graph limits. Lemma 5.4 and its summable dyadic estimate therefore give vanishing complete vacuum flux and the Ward identity (8.31) on the enlarged domain.

We check the physical affine actions on the added fields, rather than assume them. Let X be a translation, Lorentz, or dilation generator, and let G_X be its given self-adjoint generator, with the convention that the physical variation is $i[G_X, \cdot]$. For a compact smearing F of a member of $\mathcal{G}$ and an old compact polynomial C , the vacuum Ward identity gives

$$\langle C\Omega, \delta_X F\Omega \rangle = -\langle (\delta_X C)\Omega, F\Omega \rangle = i\langle G_X C\Omega, F\Omega \rangle = \langle C\Omega, iG_X F\Omega \rangle.$$

The middle identity uses the already proved affine action on old polynomials. Smooth compact smearing and the given similarity covariance place $F\Omega$ and $C\Omega$ in the generator domains; the vacuum is invariant. Density of old polynomial vectors thus identifies the vacuum action with the physical affine variation of F , including its component and weight terms.

The discrepancy between these two variations is relatively local and closable on $\mathcal{D}_G$: it is a finite compact joint insertion with a densely defined reversed adjoint. It annihilates the vacuum. Choose an old double cone spacelike to its compact localization. Locality makes it annihilate every old polynomial vacuum vector from that cone, a dense set by Reeh–Schlieder. The closure of its graph therefore contains $\mathcal{H} \times \{0\}$, and being an operator graph it can contain no nonzero output. The discrepancy vanishes on the whole joint domain. This proves the required affine identifications without assuming new current Ward identities.

The primary identities for L and t likewise persist on the enlarged domain: their relatively local discrepancies vanish on the original dense domain, and the same adjoint and closed-graph argument applies. The current-variation computation and compact nested flux proof of Proposition 8.9 consequently give the conformal Lie relation on every enlarged polynomial insertion. Finally the centered construction of Proposition 8.10 uses only these Ward identities, similarity covariance, cap shrinking, and finite joint products. Repeating it on $\mathcal{D}_G$ proves its conclusion for every fixed finite family of added targets and finite descendant depths. $\square$

Thus the improvement (8.28) lies in the original physical theory, has the original translation and Lorentz charges, and is traceless. Its special-conformal currents are physical conserved currents with unbroken local Ward identities. The conclusion concerns flat spacetime and does not impose a curved-background Weyl identity.

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